Advanced Combinatorics Notes

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Part I

Extremal Set Theory

Extremal set theory is the study of extremal problems for finite sets. Notable examples include:

- (Intersecting Families) What is the maximum size a collection of sets $\mathcal{F}$ can have if $F \cap F' \neq \emptyset$ for all $F, F' \in \mathcal{F}$?

- (Antichains) What is the maximum size a collection of sets $\mathcal{F}$ can have if $F \not\subseteq F'$ for all distinct $F, F' \in \mathcal{F}$?

These problems reflect the most basic operations and relationships that one can have between pairs of sets, namely intersections (equivalently unions) and containment. Of course, the answer to both questions is "infinite" if one allows arbitrary sets, and as such we typically impose conditions on $\mathcal{F}$ to obtain interesting problems.

Main Definitions. The central objects of study in extremal set theory are *set systems*, which are just collections of sets. We will typically denote set systems by $\mathcal{F}$. For example, $\mathcal{F} = \{\{1,2,3\},\{\square\},\{*,9\}\}$ is a set system. We will refer to the sets in a set system as either *sets*, *edges*, or *hyperedges* and we use notation like $F \in \mathcal{F}$ to denote them; for instance both $\{1,2,3\}$ and $\{\square\}$ are edges of the previous example. The elements of an edge of a set system will be referred to as *vertices* or *elements* and denoted as lower case letters like i, v, or x depending on context; for instance both 3 and $*$ are vertices of the previous example. We say that a set V is a *ground set* or *vertex set* for a set system $\mathcal{F}$ if every vertex of $\mathcal{F}$ lies in V; for instance $\{1,2,3,9,\square,*\}$ is a ground set for the previous example, as is $\mathbb{N} \cup \{\square, *\}$.

Most of the problems we consider will involve bounding set systems with ground sets of a given size n, for which it is often convenient to consider the ground set to be the first n integers. For a positive integer n we write $[n] := \{1,2,\ldots,n\}$ (the symbol $[n]$ is typically read as "bracket n"). We let $2^{[n]}$ denote the set of all subsets of $[n]$, and let $\binom{[n]}{r}$ denote the set of all subsets of $[n]$ of size r. Almost all of the set systems we consider will either be of the form $\mathcal{F} \subseteq 2^{[n]}$ or $\mathcal{F} \subseteq \binom{[n]}{r}$ for some integers n, r, and in this latter case we say that $\mathcal{F}$ is r*-uniform* or simply *uniform* if we do not specify the value of r.

Perspectives on Set Systems. There are several ways to visualize set systems with different perspective leading to different problems to consider. Perhaps the simplest is the *hypergraph perspective*: we draw a point for each element of the ground set of $\mathcal{F}$, and then for each $F \in \mathcal{F}$ we draw a blob which contains exactly the elements of F insert picture.

Another way is the *Boolean lattice* persepective. For this, imagine (but do not draw) $n+1$ horizontal lines stacked on top of each other. Along the ith imaginary line, make $\binom{n}{i-1}$ dots and label them by each subset of $[n]$ of size $i-1$. Then between lines i and $i+1$, draw a line between any two dots labeled by sets A, B with $A \subseteq B$. The picture we have just draw is something called the Hasse diagram for the Boolean lattice, which is a special kind of poset. With this, we can view any set system $\mathcal{F} \subseteq 2^{[n]}$ as a subset of the dots of the Boolean lattice, namely the dots which are labeled by an element of $\mathcal{F}$. Drawing things in this way has the feature that one can visually determine if two elements of $\mathcal{F}$ are contained in one another: you just have to check if there is a vertical path in the Boolean lattice from one set to another.

The dots and lines we drew for the Boolean lattice can be interpreted as a graph, and it is not difficult to see that this graph in fact the n-dimensional hypercube, that is, the graph whose vertex set is bit strings of length n where two strings are adjacent if they differ in exactly 1 vertex. Indeed, this correspondence can be seen be replacing each set A which we used to label the dots of the Boolean lattice and replacing it with its charcteristic vector, which is the bit string w with $w_i = 1$ if and only if $i \in A$. In this way we can view set systems as sets of vertices of the hypercube graph, giving us the perspective of the *Boolean cube* for thinking about how to visualize set systems.

These three perspectives for visualizing set systems (hypergraphs, elements of a poset, and vertices of a graph) each lend themselves to different types of problems, and we will see examples of each of these in the coming chapters.

Comparing with Extremal Hypergraph Theory. Readers familiar with hypergraphs may recognize that set systems are simply hypergraphs, and as such technically the area of extremal set theory should be the same as the area of extremal hypergraph theory. In practice though there are significant culture differences between these two areas. For example, extremal hypergraph theorists tend to consider a problem solved if it is solved for large n, while extremal set theorists often care quite a bit about solving the problem for all n. Similarly hypergraph theoristis tend to look at generalizations of graph theory problems which are less natural to state in set theoretic language (e.g. how many edges can a hypergraph without containing a "cycle of length 6"?) wheras the problems considered by extremal set theorists tend to be trivial for graphs (e.g. how many edges can a hypergraph have if every two edges intersect?).

Other Resources. Much of these notes are inspired by Frankl and Tokushige's book *Extremal Problems for Finite Sets* which gives a much deeper dive into the topic than we do here, as well as excellent notes by Liu. The reader may also enjoy Junka's *Extremal Combinatorics* especially if they are interested in connections with theoretical computer science.

Asymptotic Notation. Eventually in the text it will be convenient for us to use the following asymptotic notation which we record here for ease of reference.

Let $f(n), g(n)$ be two functions.

- We write $f(n) = O(g(n))$ if there exists a constant $C > 0$ such that $f(n) \le Cg(n)$ for all n.
- We write $f(n) = \Omega(g(n))$ if there exists a constant $c > 0$ such that $f(n) \le cg(n)$ for all n.
- We write $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. In this case we say that f, g have the same *order of magnitude*.
- We write $f(n) = o(g(n))$ if $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 0$. In particular, writing $f(n) = o(1)$ means $\lim_{n\to\infty} f(n) = 0$.

1 Antichains and Posets

1.1 Sperner's Theorem

What is generally considered to be the first paper in extremal set theory is Sperner's Theorem from 1928 on antichains, which is the main topic of this chapter.

Definition 1. A set system $\mathcal{F}$ is an *antichain* if it does not contain $F, F' \in \mathcal{F}$ with $F \subsetneq F'$.

That is, no two elements of an antichain contain one another. The extremal question we want to ask is: how large can an antichain $\mathcal{F} \subseteq 2^{[n]}$ be?

One simple construction is to take $\mathcal{F}$ to be all the sets of size k for some integer k, which is trivially an antichain of size $\binom{n}{k}$. This size $\binom{n}{k}$ is maximized when $k = \lfloor n/2 \rfloor$ or $k = \lceil n/2 \rceil$, i.e. when $\mathcal{F}$ is a "middle layer of the Boolean lattice." Sperner's Theorem states that this simple construction is the best one can do.

Theorem 1.1 (Sperner's Theorem)**.** *If $\mathcal{F} \subseteq 2^{[n]}$ is an antichain, then $|\mathcal{F}| \le \binom{n}{\lfloor n/2 \rfloor}$.*

We will see a few proofs of Sperner's Theorem in these notes, beginning with a relatively simple combinatorial argument.

Proof. Our proof will work by showing that $2^{[n]}$ has a "symmetric chain decomposition." We say that a collection of sets $\{F_k, F_{k+1}, \ldots, F_{n-k}\} \subseteq 2^{[n]}$ is a *symmetric chain* if $F_k \subseteq F_{k+1} \subseteq \cdots \subseteq F_{n-k}$ and if $|F_i| = i$ for all i. From the Boolean lattice perspective, a symmetric chain is a vertical line which is symmetric about the middle of the Boolean lattice. A *symmetric chain decomposition* is a partition of $2^{[n]}$ into disjoint symmetric chains.

Claim 1.2. *To prove the theorem, it suffices to show $2^{[n]}$ has a symmetric chain decomposition.*

Proof. Say there was a decomposition into non-empty symmetric chains $\mathcal{C}_1, \ldots, \mathcal{C}_t$ and consider any antichain $\mathcal{F} \subseteq 2^{[n]}$. Observe that $|\mathcal{F} \cap \mathcal{C}_i| \le 1$ for all i, since if $\mathcal{F}$ contained two elements of a chain then it would contain two elements which are comparable, contradicting $\mathcal{F}$ being an antichain. Since the $\mathcal{C}_i$ partition $2^{[n]}$ it follows that

$$|\mathcal{F}| = \sum_i |\mathcal{F} \cap \mathcal{C}_i| \le \sum_i 1 = t.$$

On the other hand, we claim that we must have $t = \binom{n}{\lfloor n/2 \rfloor}$. Indeed, by definition every non-empty symmetric chain must contain exactly 1 element from $\binom{[n]}{\lfloor n/2 \rfloor}$, and since the $\mathcal{C}_i$ partition $2^{[n]}$ there must be exactly $\binom{n}{\lfloor n/2 \rfloor}$ of them. This implies $t = \binom{n}{\lfloor n/2 \rfloor}$, giving the bound. $\square$

Playing around with small n suggests that $2^{[n]}$ should indeed have a symmetric chain decomposition for all n, but it is not immediate how to construction such a decomposition in a systematic way. We resolve this by considering a clever inductive construction.

For $n = 0$ the single chain $\{\emptyset\}$ forms a symmetric chain decomposition. Assume inductively that $\mathcal{C}_1, \ldots, \mathcal{C}_t$ is a symmetric chain decomposition for $2^{[n-1]}$. For each $\mathcal{C}_i = \{F_k, \ldots, F_{n-1-k}\}$

we define $\mathcal{C}_i' = \{F_k, \ldots, F_{n-1-k}, F_{n-1-k} \cup \{n\}\}$, and if we further have $|\mathcal{C}_i| \neq 1$ then we also define $\mathcal{C}_i'' = \{F_k \cup \{n\}, \ldots, F_{n-2-k} \cup \{n\}\}$. It is not difficult to check that $\mathcal{C}_i', \mathcal{C}_i''$ are symmetric chains of $2^{[n]}$ since $\mathcal{C}_i$ was for $2^{[n-1]}$. Moreover, each element $F \in 2^{[n-1]}$ appears in exactly one of the $\mathcal{C}_i'$ sets (namely the one with $F \in \mathcal{C}_i$) and no $\mathcal{C}_i''$ set, and similarly each element $F \in 2^{[n]}$ containing n is contained in exactly one $\mathcal{C}_i'$ or $\mathcal{C}_i''$ set (a $\mathcal{C}_i'$ set if $F \setminus \{n\}$ was a top element of some $\mathcal{C}_i$ and otherwise the $\mathcal{C}_i''$ with $F \setminus \{n\} \in \mathcal{C}_i$). We conclude that this is a symmetric chain decomposition, giving the result. □

Sperner's Theorem shows up in a variety of mathematical contexts. For example, it can be used to solve a seemingly unrelated probability problem.

Recall that a *Rademacher random variable* is a random variable X satisfying $\Pr[X = 1] = \Pr[X = -1] = \frac{1}{2}$. An old problem of Littlewood and Offord asks: if $a_1, \ldots, a_n$ are non-zero real numbers and if $X_1, \ldots, X_n$ are independent Rademacher random variables, then how large is

$$\sup_{r \in \mathbb{R}} \Pr[\sum_i a_i X_i = r]?$$

That is, how likely is the likeliest outcome to be. Problems like this show up in random matrix theory and can be viewed as an anticoncentration problem.

The exact answer to this problem depends on the choice of the a_i. For instance, if $a_i = 2^i$ then this supremum is 2^{-n} which is the lowest this probability could be. If instead $a_i = 1$ for all i and n is even then

$$\Pr[\sum a_i X_i = 0] = \binom{n}{n/2} 2^{-n} = \Theta(n^{-1/2}),$$

and if n is odd then $\Pr[\sum a_i X_i = 1] = \binom{n}{\lfloor n/2 \rfloor} 2^{-n}$. It turns out that this is the largest this probability can be.

Theorem 1.3 (Erdős)**.** *If $a_1, \ldots, a_n$ are non-zero real numbers and if $X_1, \ldots, X_n$ are independent Rademacher random variables, then*

$$\sup_{r \in \mathbb{R}} \Pr[\sum_i a_i X_i = r] \leq \binom{n}{\lfloor n/2 \rfloor} 2^{-n}.$$

Proof. Proving this is equivalent to proving that the number of assignments of the X_i variables with $\sum a_i X_i = r$ is at most $\binom{n}{\lfloor n/2 \rfloor}$. As this bound suggests, we will show this by associating each assignment to a subset of $2^{[n]}$ and then show that this collection of subsets is an antichain. To do this, we note by the symmetry of the problem we may assume $a_i > 0$ for all i.

Fix some real number r. Let $\mathcal{F} \subseteq 2^{[n]}$ be the collection of sets F with the property

$$\sum_{i \in F} a_i - \sum_{j \notin F} a_j = r,$$

and observe $\Pr[\sum a_i X_i = r] = |\mathcal{F}| 2^{-n}$ since $\sum a_i X_i = r$ happens if and only if $\{i : X_i = 1\} \in \mathcal{F}$.

We claim that $\mathcal{F}$ is an antichain. Indeed, assume for contradiction that there existed $F \subsetneq F'$ both in $\mathcal{F}$. Crucially, because we assumed each a_i is positive we find

$$r = \sum_{i \in F} a_i - \sum_{j \notin F} a_j < \sum_{i \in F'} a_i - \sum_{j \notin F'} = r,$$

a contradiction. We conclude that $\mathcal{F}$ is an antichain and hence by Sperner's Theorem,

$$\Pr[\sum a_i X_i = r] = |\mathcal{F}|2^{-n} \leq \binom{n}{\lfloor n/2 \rfloor} 2^{-n}.$$

$\square$

1.2 Posets and Dilworth's Theorem

The term antichain comes from the study of posets. A *partially ordered set* (often referred to as a *poset* for short) is a pair $P = (X, \leq)$ where X is a set of *elements* of the poset and $\leq$ is a partial order on the element of X, which means it is a binary relation satisfying the following:

- (Reflexivity) $x \leq x$ for all $x \in X$.
- (Antisymmetry) If $x \leq y$ and $y \leq x$ then $x = y$.
- (Transitivity) if $x \leq y$ and $y \leq z$ then $x \leq z$.

For a partial order we write $x < y$ whenever $x \leq y$ and $x \neq y$. We emphasize that a partial order (unlike a total order) does not require all pairs of elements of X to be comparable to one another.

For example, the main poset that we care about in these notes is the *Boolean lattice* which is the poset $P_n = (2^{[n]}, \subseteq)$; that is the elements of the poset are subsets of $[n]$ and the partial order is defined by subset containment which one can easily check defines a partial order. There are many other examples one could consider, such as the divisibility poset $D_n = ([n], |)$ whose elements are the integers 1 through n and the partial order is defined by one element being a divisor of another.

We say two elements x, y of a poset are *comparable* if either $x \leq y$ or $y \leq x$, and we say x, y are *incomprable* otherwise. A *chain* of a poset is a set of elements $\{x_1, \ldots, x_\ell\}$ such that any two elements are comparable (equivalently, one can reorder the elements so that $x_i \leq x_{i+1}$ for all i), and an *antichain* is a set of elements $\{x_1, \ldots, x_a\}$ such that any two elements are incomparable. Note that an antichain for the Boolean lattice is a collection of sets such that neither is contained in the other, thereby recovering our previous notion of what an antichain is. One can thus ask the broader question: given a poset P, what is the size of its largest antichain? We call this quantity the *width* of the poset P (since for the Boolean lattice this is the width of the middle layer) and we denote this by $w(P)$.

Looking to our proof of Sperner's Theorem for inspiration, we see that if a poset P has a "symmetric chain decomposition" with t chains, then $w(P) \leq t$. What "symmetric" means for a general poset is not so clear, but in fact a closer look shows that this symmetric condition is not really needed to get a bound.

To this end, we say that a collection of disjoint chains $\mathcal{C}_1, \ldots, \mathcal{C}_t$ of a poset P is a *chain decomposition of size t* if every element of P appears in exactly one $\mathcal{C}_i$, and it is straightforward to show that the existence of such a chain decomposition implies $w(P) \leq t$. There are many possible chain decompositions for P, such as the trivial chain decomposition into chains of one element $\{x\}$, and to get an effective bound we want to consider one of smallest possible size. Surprisingly, this simple approach gives the best possible bound for every poset.

Theorem 1.4 (Dilworth's Theorem)**.** *For every finite poset P, the maximum size of an antichain is equal to the minimum size of a chain decomposition.*

Dilworth's Theorem is an example of a "max-min" theorem where the maximum of some combinatorial quantity is equal to the minimum of another combinatorial quantity. A number of results take this form, such as Hall's Theorem and König's Theorem from graph theory, and some of the many proof of Dilworth's Theorem work by making use of such results. The proof we present is a clever combinatorial argument.

Proof. That the size of any chain decomposition upper bounds $w(P)$ is straightforward to show. What remains then is to argue that the maximum equals the minimum, i.e. that P contains a chain decomposition of size (at most) $w(P)$. As with Sperner's Theorem, we prove this by a careful inductive argument.

The result is trivially true if P has only one element, so assume we have inductively proven the result up to some size and let P be a poset of this size. Consider a maximal element m of P, that is, m is an element such that there exists no x with $m < x$; such a maximal element exists because P is finite. Consider the poset $P - m$ obtained by removing m. If $w(P - m) < w(P)$ then inductively one can find a chain decomposition of $P - m$ using at most $w(P - m) \le w(P) - 1$ elements, which together with the trivial chain $\{m\}$ gives a chain decomposition of P of size at most $w(P)$, proving the result.

As such, we may assume $w(P - 1) = w(P) := t$. Inductively this means there exists a chain decomposition $\mathcal{C}_1, \ldots, \mathcal{C}_t$ of $P - m$. Observe that every antichain of size t of $P - m$ (of which at least one exists since $w(P - m) = t$) must intersect each $\mathcal{C}_i$ in exactly one vertex. To this end, for each i we let $a_i \in \mathcal{C}_i$ denote the maximal element of the chain with the property that it is contained in some antichain of size t in $P - m$.

Claim 1.5. *The set $\{a_1, \ldots, a_t\}$ is an antichain of $P - m$.*

Proof. Assume for contradiction that we had, say, $a_i < a_j$ for some i, j. By definition there exists some antichain A_j of size t which contains a_j, and also A_j must intersect $\mathcal{C}_i$ at some element a'_i. If $a'_i \le a_i$ then we have $a'_i < a_j$, a contradiction to A_j being an antichain. Otherwise since $\mathcal{C}_i$ is a chain we must have $a'_i > a_i$, a contradiction to a_i being the maximal element with respect to being contained in an antichain of size t. □

Now because $w(P) = w(P - m)$, it must be that $\{m, a_1, \ldots, a_t\}$ is not an antichain. The claim implies that m is comparable to some a_i, and since m is maximal we must have $a_i < m$. The idea now is to try and use the chain

$$\mathcal{C}' = \{m\} \cup \{c \in \mathcal{C}_i : c \le a_i\}$$

in our decomposition, where we emphasize that this is a chain by our assumption $a_i < m$. To this end, consider the subposet $P - \mathcal{C}'$.

We claim that $w(P-\mathcal{C}') < t = w(P)$. Indeed, we noted that every antichain A of $P-m \supseteq P-\mathcal{C}'$ of size t uses an element from each $\mathcal{C}_j$, and in particular it uses some element $a_i' \in \mathcal{C}_i$. By definition of a_i we must have $a_i' \le a_i$ and hence $a_i' \in \mathcal{C}$. It follows then that any antichain of size t of $P-m$ is not contained in $P-\mathcal{C}'$, so $w(P-\mathcal{C}') \le t-1$.

From this last claim we can find a chain decomposition of $P-\mathcal{C}'$ of size at most $t-1$, which together with $\mathcal{C}'$ gives a chain decomposition of P of size at most t, giving the result. □

Dilworth's Theorem is a powerful tool in poset theory and beyond, with it being particularly useful in the context of Ramsey-theoretic problems. Most of these applications come from the following observation, where here we define the *height* $h(P)$ of a poset P to be the length of a longest chain.

Corollary 1.6. *If P is a poset on n elements with height $h(P)$ and width $w(P)$, then*

$$h(P)w(P) \ge n.$$

In particular, P has a chain or an antichain of size at least $\sqrt{n}$.

Proof. By Dilworth's Theorem there exists a chain decomposition $\mathcal{C}_1, \ldots, \mathcal{C}_{w(P)}$ for P, meaning

$$n = \sum_i |\mathcal{C}_i| \le h(P)w(P).$$

□

Corollary 1.6 is analogous to the graph Ramsey statement which says that every n-vertex graph has either a clique or an independent set of size at least $\Theta(\log n)$. The $\log n$ term is best possible for arbitrary graphs, but Corollary 1.6 suggests that significantly stronger bounds can be obtained for graphs with additional structure. Indeed, we have the following.

Corollary 1.7. *If G is an n-vertex comparability graph, meaning there exists a poset $P = (V(G), \le)$ with $x \sim y$ if and only if $x < y$; then G has a clique or independent set of size at least $\sqrt{n}$.*

This follows from Corollary 1.6 together with the straightforward observation that chains in P are exactly cliques in G and antichains are exactly independent sets. One place where additional structure like that in Corollary 1.7 tends to pop up is in geometric settings. For example, we have the following.

Proposition 1.8. *If $I_1, \ldots, I_n$ are intervals in $\mathbb{R}$, then there exist at least $\sqrt{n}$ of these intervals which are either pairwise intersecting or pairwise disjoint.*

This bound is best possible, as can be seen by considering $I_1, \ldots, I_n$ to consist of $\sqrt{n}$ copies of some $\sqrt{n}$ disjoint intervals.

Proof. Define a graph G on $I_1, \ldots, I_n$ where $I_i \sim I_j$ if and only if I_i and I_j are disjoint (this is known as a *disjointness graph*). The proposition then is equivalent to showing G has either a clique or independent set of size at least $\sqrt{n}$, and by Corollary 1.7 it suffices to construct a partial order on $I_1, \ldots, I_n$ which encodes the edges of G. One natural ordering is to have $I_i < I_j$ if I_i lies entirely to the left of I_j, which one can easily check is a partial ordering. In this case having $I_i < I_j$ or $I_j < I_i$ guarantees that these two intervals are disjoint from each other, while I_i, I_j being incomparable implies they intersect. It follows that G is the comparability graph for $P = (V(G), \leq)$, proving the result by Corollary 1.6. $\square$

In practice it can often be too restrictive to consider graphs which are exactly comparability graphs. By loosening this condition slightly we can still obtain (weaker) polynomial bounds.

Proposition 1.9. *Let G be an n-vertex graph. If there are posets $P_1 = (V(G), \leq_1), \ldots, P_d = (V(G), \leq_d)$ such that $x \sim y$ if and only if there exists some i with either $x <_i y$ or $y <_i x$, then G contains either a clique or an independent set of size at least $n^{\frac{1}{d+1}}$.*

We leave the proof of this as an exercise and explain briefly how to use this to give a polynomial bound for a higher dimensional analog of Proposition 1.8.

Theorem 1.10 (Larman-Matoušěk-Pach-Törőcsik)**.** *If $C_1, \ldots, C_n$ are convex sets in $\mathbb{R}^2$, then there exist $n^{1/5}$ of them which are either pairwise intersecting or pairwise disjoint.*

Sketch of Proof. By considering the disjointness graph of $C_1, \ldots, C_n$ together with Proposition 1.9, we see that it suffices to construct four partial orderings such that any two disjoint C_i, C_j are comparable for at least one of these orders. This can be done with a bit of work. $\square$

There are a number of followup questions one can ask at this point. The first is whether the bound of $n^{1/5}$ can be improved. As far as we are aware this is still the best bound known for this problem, with the best known construction showing that the bound can not exceed roughly $n^{\log(2)/\log(5)} \approx n^{.43}$ (the construction essentially comes from the fact that the iterated 5-cycle graph can be realized as the disjointness graph of convex sets in $\mathbb{R}^2$).

Another immediate question is if one can generalize Theorem 1.10 to higher dimensions. Surprisingly the answer is no: an old result of Tietze shows that every graph can be realized as the intersection graph of convex sets in $\mathbb{R}^3$, which in particular means one can not find a clique or independent set of size larger than the Ramsey bound $\Theta(\log n)$. One can, however, say something more if we consider restricted classes of convex sets, such as boxes in $\mathbb{R}^d$, though we omit doing so here.

We close by mentioning a dual version of Dilworth's Theorem stated in terms of the height of a poset rather than its width that can be useful in certain contexts.

Theorem 1.11 (Mirsky's Theorem)**.** *If $P = (X, \leq)$ is a finite poset of height $h(P)$, then there exists a partition of X into $h(P)$ antichains.*

While Mirsky's Theorem and Dilworth's Theorem are essentially duals of each other, as far as we know there is no elementary way of deriving one from the other. Mirsky's Theorem turns out to be substantially easier to prove than Dilworth's and we leave this as an exercise.

1.3 The LYM Inequality

An important refinement of Sperner's Theorem is the LYM inequality, which is named after Lubell, Yamamoto, and Meshalkin who all independently proved this same inequality.

Theorem 1.12 (The LYM Inequality). *If $\mathcal{F} \subseteq 2^{[n]}$ is an antichain, then*

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} \leq 1.$$

The proof and intuition for this theorem will be deferred to next chapter as part of a broader discussion of Katona's circle method. Until then, we discuss a few of the many applications of this result to problems on antichains.

As an initial observation, we note that the LYM inequality together with the fact that $\binom{n}{k} \leq \binom{n}{\lfloor n/2 \rfloor}$ immediately implies that every antichain $\mathcal{F}$ satisfies

$$\frac{|\mathcal{F}|}{\binom{n}{\lfloor n/2 \rfloor}} \leq \sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} \leq 1,$$

and as such the LYM inequality gives a new proof of Sperner's Theorem. With more work one can use the LYM inequality to obtain a nice generalization of Sperner's Theorem.

Theorem 1.13 (Erdős). *If $\mathcal{F} \subseteq 2^{[n]}$ contains no chain of size larger than t, then*

$$|\mathcal{F}| \leq \max_{T \in \binom{\{0,1,\ldots,n\}}{t}} \sum_{r \in T} \binom{n}{r}.$$

For example, when $t = 1$ this says if $\mathcal{F}$ has no chain of size 2 (meaning it is an antichain), then $|\mathcal{F}| \leq \max_r \binom{n}{r} = \binom{n}{\lfloor n/2 \rfloor}$, so the $t = 1$ case is exactly Sperner's Theorem. The bound for general t is best possible by considering $\mathcal{F}$ to be the t largest layers of the Boolean lattice which trivially has no chain of size larger than t.

Proof. By Mirsky's Theorem, any $\mathcal{F}$ as in the theorem is equal to the union of t antichains $\mathcal{F}_1, \ldots, \mathcal{F}_t$. By the LYM inequality it follows that

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} = \sum_{i=1}^{t} \sum_{F \in \mathcal{F}_i} \frac{1}{\binom{n}{|F|}} \leq t.$$

It is not difficult to show that this inequality can only hold if $|\mathcal{F}|$ is at most the size of the t largest layers of $2^{[n]}$, proving the result. □

As this proof and the LYM inequality demonstrates, it can sometimes be useful to measure the size of a family not by its cardinality but by the function

$$\lambda(\mathcal{F}) := \sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}},$$

which is commonly referred to as the *Lubell measure* of $\mathcal{F}$. For example, the proof of Theorem 1.13 implicitly shows that any $\mathcal{F}$ with no chain of size larger than t has $\lambda(\mathcal{F}) \leq t$. In general it can be of use to consider extremal problems which measure the "size" of $\mathcal{F}$ by various functions of the size of its elements; we will see another example of this later in Theorem 2.7.

1.4 Exercises

Each chapter will end with a set of exercises. Following the notation of Stanley, we will add numbers after each exercise to indicate the problem's rough level of difficulty as follows:

- [1] problems are elementary and routine requiring little to no thought,
- [2] problems have simple solutions (though that does not necessarily mean it is easy to find!),
- [3] problems tend to have involved solutions,
- [4] problems have extremely difficult solutions (to the extent that such questions should never be used in a classroom setting),
- [5] problems are unsolved.

Additionally, plus and minus symbols may be used to indicate higher or lower levels of difficulty for the problem. For example, a [2+] problem might have a simple solution that's pretty challenging to find, while a [3-] problem might have an involved solution that's actually not too hard to work out. Ultimately, all of the ratings that I give are only rough estimates and the reader may find a given [3] problem easier to solve than a [2-] depending on the circumstances.

1. Let $\mathcal{F}$ be an r-uniform set system with the property that $F' \cap F'' \not\subseteq F$ for any distinct $F, F', F'' \in \mathcal{F}$. Prove that $|\mathcal{F}| \leq 1 + \binom{r}{\lfloor r/2 \rfloor}$ (Hint: let $F'' \in \mathcal{F}$ be arbitrary and consider the set system $\{F \cap F'' : F \in \mathcal{F} \setminus \{F''\}\}$) [2-].

2. Let $a_1, \ldots, a_n$ be real numbers with $|a_i| \geq 1$ for all i and $X_1, \ldots, X_n$ independent Rademacher random variables.

   (a) Prove that
   $$\sup_{r \in \mathbb{R}} \Pr[\sum_i a_i X_i \in [r, r+2)] \leq \binom{n}{\lfloor n/2 \rfloor} 2^{-n}.$$
   That is, the random signed sum is unlikely to land in any given half interval of length at most 2 [2-].

   (b) More generally prove for any integer $t \geq 1$ that
   $$\sup_{r \in \mathbb{R}} \Pr[\sum_i a_i X_i \in [r, r+2t)] \leq \max_{T \subseteq \binom{\{0,1,\ldots,n\}}{t}} \sum_{r \in t} \binom{n}{r} 2^{-n},$$
   where to be clear this upper bound is equal to 2^{-n} times the sum of the t largest binomial coefficients [2].

* * *

3. Verify that set containment $F \subseteq F'$ and divisibility of integers $x|y$ define partial orders on $2^{[n]}$ and $[n]$, respectively [1].

4. (1-dimensional Helly Theorem) Prove that if $I_1, \ldots, I_k$ are a collection of intervals of $\mathbb{R}$ with the property that $I_i \cap I_j \neq \emptyset$ for all i, j, then $\bigcap_i I_i \neq \emptyset$. In particular, one can strengthen Proposition 1.8 slightly by replacing the intervals pairwise intersecting with having a point in common [2-].

5. Prove Proposition 1.9 (Hint: try proving the weaker lower bound of n^{-2^d} to get a feel for the proof idea) [2].

6. Finish the proof of Theorem 1.10 by constructing four partial orders with the desired properties [3-].

7. A famous Ramsey result of Erdős and Szekeres says that if $(a_1, \ldots, a_n)$ are a sequence of distinct real numbers, then there exists either an increasing subsequence of length at least $\sqrt{n}$ or a decreasing subsequence of length at least $\sqrt{n}$. That is, there exist indices $1 \leq i_1 < \cdots < i_t \leq n$ such that $t \geq \sqrt{n}$ and such that either $a_{i_1} < \cdots < a_{i_t}$ or $a_{i_1} > \cdots > a_{i_t}$.

   (a) Prove this using Dilworth's Theorem [2].

   (b) Prove this without using Dilworth's Theorem (Hint: for each j define i_j to be the length of the longest increasing subsequence which ends at a_j and d_j to be the length of the longest decreasing subsequence which ends at a_j. Can we have $(i_j, d_j) = (i_{j'}, d_{j'})$ for $j < j'$?) [2+].

8. Prove Mirsky's Theorem: if $P = (X, \leq)$ is a finite poset of height $h(P)$, then there exists a partition of X into $h(P)$ antichains (Hint: prove that the set of maximal elements M of P form an antichain) [2].

9. Prove that if $\mathcal{F} \subseteq 2^{[n]}$ is an antichain with $|F| \leq r$ for all $F \in \mathcal{F}$ then $|\mathcal{F}| \leq \binom{n}{r}$ [2-].

2 Intersecting Families and Katona's Circle

2.1 Katona's Circle Method

In this chapter we explore a general technique known as Katona's circle method which is based on the following core idea.

Mantra 1 (Katona Circle Philosophy)**.** If an extremal set theory problem is difficult, try solving it for "simple" set systems and then reduce your problem to simple set systems by an averaging argument.

To illustrate this method, let us try and prove Sperner's Theorem using this approach. For this, we ultimately want to find the largest antichain in $2^{[n]}$, but such set systems can be rather complicated. To remedy this, we will look at $\mathcal{F}$ not contained in $2^{[n]}$ but rather some simpler family.

One (very clever) choice for a simpler family is the collection $\mathcal{A}_n$ consisting of all $A \subseteq [n]$ which are "cyclic intervals", meaning $A = \{i, i+1, \ldots, i+j\}$ for some $0 \le i, j \le n$ where these elements are taken modulo n. Thus for example $\{n-1, n, n+1\} = \{n-1, n, 1\} \in \mathcal{A}_n$. The intuition for this collection comes from imagining placing the numbers 1 through n consecutively around a circle, in which case $\mathcal{A}_n$ consists exactly of the consecutive arcs of this circle draw a picture. In this setting the antichain problem is very simple to solve.

Lemma 2.1. *If $\mathcal{F} \subseteq \mathcal{A}_n$ is an antichain, then $|\mathcal{F}| \le n$.*

Proof. Observe that if $\emptyset \in \mathcal{F}$ or $[n] \in \mathcal{F}$ then $|\mathcal{F}| = 1$ for $\mathcal{F}$ an antichain, so we may assume from now on that $\emptyset, [n] \in \mathcal{F}$. This implies that for every $F \in \mathcal{F} \subseteq \mathcal{A}_n \setminus \{\emptyset, [n]\}$ there exists a unique integer i_F with $i_F \in F$ and $i_F - 1 \notin F$ (where again $i_F - 1$ is written modulo n). Observe that if $F, F' \in \mathcal{F}$ were distinct elements with $i_F = i_{F'} :=$, then this would imply $F = \{i, i+1, \ldots, i+j\}$ and $F' = \{i, i+1, \ldots, i+j'\}$ for some $j \ne j'$ which would imply one of F, F' contains the other, a contradiction to $\mathcal{F}$ being an antichain.

We conclude that for each $i \in [n]$ that there is at most one $F \in \mathcal{F}$ with $i = i_F$, and from this it follows that $|\mathcal{F}| \le n$. $\square$

Crucially, we can upgrade this result for families on a circle to arbitrary families and thereby recover the LYM inequality.

Proof of the LYM inequality. Recall that we wish to show that if $\mathcal{F} \subseteq 2^{[n]}$ is an antichain, then

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} \le 1.$$

We will do this by "averaging" over all the ways of embedding $\mathcal{F}$ into $\mathcal{A}_n$ where we already know how to solve the problem by Lemma 2.1.

For each bijection $\pi : [n] \to [n]$ (which we think of as a way of arranging the numbers 1 through n along a circle) and set $F = \{x_1, \ldots, x_r\}$, let $\pi(F) = \{\pi(x_1), \ldots, \pi(x_r)\}$. Let $\mathcal{P}$ denote the set of all pairs (π, F) such that $\pi : [n] \to [n]$ is a bijection and $F \in \mathcal{F}$ satisfies $\pi(F) \in \mathcal{A}_n$ (that is, F is a consecutive arc when the circle is labeled according to π). We aim to count $|\mathcal{P}|$ in two ways by observing two facts.

Claim 2.2. *Each $F \in \mathcal{F}$ is in exactly $n \cdot |F|!(n - |F|)!$ pairs of $\mathcal{P}$.*

Proof. If $(\pi, F) \in \mathcal{P}$ then this means the image of F under π is an arc of the circle. There are n possible choices for what its image could be, and given this there are $|F|!$ ways for π to map F to its image and $(n - |F|)!$ to map the remaining elements, giving the claim. $\square$

Claim 2.3. *Each bijection π is in at most n pairs of $\mathcal{P}$.*

Proof. Consider the family $\mathcal{F}' = \{\pi(F) : (\pi, F) \in \mathcal{P}\}$. By definition $|\mathcal{F}'|$ is the number of pairs π is in, $\mathcal{F}' \subseteq \mathcal{A}_n$, and $\mathcal{F}'$ is an antichain because $\mathcal{F}$ is. It follows from Lemma 2.1 that $|\mathcal{F}'| \leq n$, giving the claim. $\square$

Combining these two claims gives

$$\sum_{F \in \mathcal{F}} n \cdot |F|!(n - |F|)! \leq |\mathcal{P}| \leq n \cdot n!,$$

and dividing both sides by $n \cdot n!$ exactly gives the stated inequality. $\square$

This double counting argument can be rewritten in the language of probability.

Probabilistic Proof of LYM Inequality. Let $\mathcal{F} \subseteq 2^{[n]}$ be an antichain and $\pi : [n] \to [n]$ a bijection chosen uniformly at random. Let X denote the (random) number of $F \in \mathcal{F}$ such that $\pi(F)$ is an arc of the circle. By Lemma 2.1 we deterministically have $X \leq n$, so in particular $\mathbb{E}[X] \leq n$. On the other hand, since $\pi(F)$ is a uniform random subset of $[n]$ of size $|F|$ and since exactly n such sets are an arc of the circle we find $\Pr[\pi(F) \in \mathcal{A}] = \frac{n}{\binom{n}{|F|}}$, and hence by linearity of expectation

$$\mathbb{E}[X] = \sum_{F \in \mathcal{F}} \frac{n}{\binom{n}{|F|}},$$

which combined with $\mathbb{E}[X] \leq n$ gives the result. $\square$

The idea of using probabilistic thinking to solve combinatorial problems is at the heart of the probabilistic method, which is perhaps the most important tool in extremal combinatorics. Because GSU has an entire course dedicated to this topic we will refrain from discussing this further except where absolutely necessary.

We explore a few more applications of the Katona circle method in the coming subsections. For an even more detailed discussion on the topic, we refer the reader to the survey by Frankl dedicated to this method.

2.2 Intersecting Families

We next use Katona's circle method to study another fundamental object in extremal set theory, namely that of an intersecting family.

Definition 2. A set system $\mathcal{F}$ is said to be an *intersecting family* or simply *intersecting* if $F \cap F' \neq \emptyset$ for all $F, F' \in \mathcal{F}$.

We want to determine how large such a set system can be. If we consider arbitrary set systems this is somewhat boring.

Proposition 2.4. *Every intersecting family $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| \leq 2^{n-1}$ and this bound is best possible.*

Proof. Observe that for every $F \subseteq [n]$, we must either have F not in $\mathcal{F}$ or its complement $[n] \setminus F$ not in $\mathcal{F}$ in order for $\mathcal{F}$ to be intersecting. As such, any intersecting family can contain at most half the elements of $2^{[n]}$, giving the bound of $|\mathcal{F}| \leq 2^{n-1}$. This bound can be achieved by a large number of constructions: the simplest is to take $\mathcal{F}$ to be the family of sets containing 1; if n is odd then another construction is to take all sets of size at least $(n+1)/2$. In fact, in the exercises it is shown that *every* maximal intersecting family $\mathcal{F} \subseteq 2^{[n]}$ has size 2^{n-1}. □

This argument, while cute, is too simple to justify a large amount of time studying intersecting families. The saving grace is that the problem becomes significantly more interesting when we look at the setting of uniform families.

Theorem 2.5 (Erdős-Ko-Rado)**.** *If $\mathcal{F} \subseteq \binom{[n]}{r}$ is intersecting and $n \geq 2r$, then $|\mathcal{F}| \leq \binom{n-1}{r-1}$.*

The hypothesis $n \geq 2r$ is needed for this result to hold because for $n < 2r$ the entire family $\mathcal{F} = \binom{[n]}{r}$ is trivially intersecting. The upper bound $\binom{n-1}{r-1}$ is best possible when $\mathcal{F}$ is a *star*, which means there exists some element $i \in [n]$ such that $\mathcal{F}$ consists of every element of $\binom{[n]}{r}$ which contains i. In fact, one can prove that stars are the unique constructions which achieve the bound $\binom{n-1}{r-1}$ whenever $n > 2r$.

Historically the Erdős-Ko-Rado Theorem was the second result ever proven[1] in extremal set theory and is considered by many to be the fundamental theorem in the field. Its central importance is due to a number of factors:

(i) There are many ways of interpreting the problem (such as a hypergraph Turán problem of avoiding two disjoint edges, or as finding the independence number of the Kneser graph).

(ii) There are many ways to try to extend or generalize the problem (as we discuss at the end of the chapter).

(iii) There are many ways to prove the result. In fact, it is not too much of an exaggeration to say that a method is only useful in extremal set theory if it can be used to prove some version of the Erdős-Ko-Rado Theorem.

[1]The result was proven in 1938, but due to other events going on around Europe at the time they delayed publication until 1961, making it the third paper published in extremal set theory.

We will see a number of proofs for the Erdős-Ko-Rado Theorem and its variants throughout these notes. We begin with what is perhaps the slickest proof using Katona's circle method. For this, we let $\mathcal{A}_{r,n} \subseteq 2^{[n]}$ be the sets of the form $A_s = \{s, s+1, \ldots, s+r-1\}$ for $1 \leq s \leq n$ where these numbers are written modulo n. In other words, we can view $\mathcal{A}_{r,n}$ as consisting of the arcs of length r of the circle which has the numbers 1 through n written cyclically along it. As with the LYM inequality, we begin our proof by solving the intersecting family problem restricted to this simpler family $\mathcal{A}_{r,n}$.

Lemma 2.6. *If $\mathcal{F} \subseteq \mathcal{A}_{r,n}$ is intersecting and $n \geq 2r$, then $|\mathcal{F}| \leq r$.*

Proof. If $\mathcal{F} = \emptyset$ then we are done, so assume there is some $A_s = \{s, \ldots, s+r-1\} \in \mathcal{F}$. Because $n \geq 2r$, the only other sets $A_{s+j} \in \mathcal{A}_{r,n}$ which can intersect A_s are those with $-r < j < r$ (for example, $A_{s+r} = \{s+r, \ldots, s+2r-1\}$ is disjoint since $n \geq 2r$ implies $2r-1 \not\equiv 0 \mod n$). This immediately gives the weaker bound $|\mathcal{F}| \leq 2r-1$. To strengthen this, we observe that we can not simultaneously have $A_{s+j} \in \mathcal{F}$ for all $-r < j < r$. Indeed, partition these elements into pairs $\{A_{s+j}, A_{s+j+r}\}$ for all $-r-1 \leq j \leq 1$ together with the singleton $\{A_s\}$. Observe that the two elements in each pair are disjoint because $n \geq 2r$, and as such $\mathcal{F}$ can contain at most one element from each of these $r-1$ pairs together with A_s, in total giving the bound of r. □

We can now translate Lemma 2.6 for circular arcs into our original problem by considering a clever double counting argument.

Proof I of the Erdős-Ko-Rado Theorem. Let $n \geq 2r$ and $\mathcal{F} \subseteq \binom{[n]}{r}$ intersecting. Let $\mathcal{P}$ denote the set of all pairs (π, F) where $\pi : [n] \to [n]$ is a bijection and $F = \{x_1, \ldots, x_r\} \in \mathcal{F}$ is such that $\pi(F) = \{\pi(x_1), \ldots, \pi(x_r)\} \in \mathcal{A}_{r,n}$. That is, $\mathcal{P}$ consists of pairs (π, F) where π can be thought of as a way to arrange the elements of $[n]$ around a circle such that the elements of F are a consecutive arc of the circle. We estimate $|\mathcal{P}|$ in two ways.

First, we observe that every $F \in \mathcal{F}$ is contained in exactly $n \cdot r!(n-r)!$ pairs $(\pi, F) \in \mathcal{P}$ since each such π can be identified by choosing the image of F in at most n ways (since its image is a consecutive interval of length r on the circle), after which there are $r!(n-r)!$ ways to choose how π maps F onto this image. This implies that

$$|\mathcal{P}| = n \cdot r!(n-r)!|\mathcal{F}|.$$

On the other hand, Lemma 2.6 implies that each π can be in at most r pairs since the family $\{\pi(F) : (\pi, F) \in \mathcal{P}\}$ is a set of intersecting arcs on the circle. This implies

$$|\mathcal{P}| \leq n! \cdot r,$$

which combined with our other inequality implies

$$|\mathcal{F}| \leq \frac{r}{n}\binom{n}{r} = \binom{n-1}{r-1}.$$

□

Again this counting argument can be interpreted through the lens of probability which we spell out to partially motivate our next proof.

Proof II of the Erdős-Ko-Rado Theorem. Let $\mathcal{F} \subseteq \binom{[n]}{r}$ be intersecting, let $\binom{\pi:[n]}{[n]}$ be a bijection chosen uniformly at random, and let X denote the (random) number of $F \in \mathcal{F}$ such that $\pi(F)$ is an arc of the circle. By Lemma 2.6 we deterministically have $X \le r$, so in particular $\mathbb{E}[X] \le r$. Moreover, one can show using linearity of expectation that $\mathbb{E}[X] = n|\mathcal{F}|\binom{n}{r}^{-1}$, and combining this with our upper bound on $\mathbb{E}[X]$ gives the desired result. □

At this point we have solved the two most basic problems concerning intersecting families by determining the largest size of such a family in either $2^{[n]}$ or $\binom{[n]}{r}$, but there are many more problems one could consider. For example, we might try to adjust the intersecting family problem for $2^{[n]}$ in some way to make it more interesting.

Partially motivated by our discussion around the LYM inequality, we will do this by measuring the size of $\mathcal{F} \subseteq 2^{[n]}$ by something other than their cardinality. One way to do this is by considering the Lubell measure for $\mathcal{F}$, and this turns out to be an easy exercise to solve. Another popular way to measure size is through a certain product measure.

Definition 3. For a real number $0 \le p \le 1$ and a family $\mathcal{F} \subseteq 2^{[n]}$ we define the *p-biased measure* of $\mathcal{F}$ by

$$\mu_p(\mathcal{F}) = \sum_{F \in \mathcal{F}} p^{|F|}(1-p)^{n-|F|}.$$

In other words, if $S_p \subseteq [n]$ is a random set obtained by keeping each element independently and with probability p, then $\mu_p(\mathcal{F})$ is the probability that $S_p \in \mathcal{F}$.

Note that $\mu_{1/2}(\mathcal{F}) = |\mathcal{F}|2^{-n}$, so bounding $\mu_{1/2}(\mathcal{F})$ is equivalent to bounding the size of $\mathcal{F}$. In a similar spirit, $\mu_p(\mathcal{F})$ for other values of p can be interpreted as measuring the "size" of $\mathcal{F}$ but with a bias towards favoring small or large sets depending on if p is less than or greater than $\frac{1}{2}$. Using this new measure of size allows us to prove a more interesting variant of the intersecting family problem for $2^{[n]}$.

Theorem 2.7. *If $\mathcal{F} \subseteq 2^{[n]}$ is intersecting, then $\mu_p(\mathcal{F}) \le p$ for all $p \le \frac{1}{2}$.*

The bound $\mu_p(\mathcal{F}) \le p$ is best possible by considering all sets which contain a given element i, and in fact one can show for $p < \frac{1}{2}$ that these are the unique extremal constructions. For $\frac{1}{2} < p < 1$ it turns out that $\mu_p(\mathcal{F}) > p$ so our hypothesis in the theorem is necessary; we leave this as an exercise.

Proof. There are again many proofs of this result, we give a proof based on Katona's circle method due to Filmus. We have already proven the result when $p = \frac{1}{2}$ with Proposition 2.4, so for some slight ease of notation we only prove the result for $p < \frac{1}{2}$.

Consider the unit circle C and do two random things: place points $x_1, \ldots, x_n$ independently and uniformly at random on C and let $I = (x, x+p)$ be an interval of the circle chosen uniformly at random. For a choice of $\vec{x} = (x_1, \ldots, x_n)$ and I, let $F = F_{\vec{x},I} = \{i : x_i \in I\}$. Observe that F has the same distribution as if we included each element of $[n]$ in this set independently with probability p, and hence for any family $\mathcal{F}$ we have $\Pr[F \in \mathcal{F}] = \mu_p(\mathcal{F})$. It remains to effectively upper bound this probability for $p < \frac{1}{2}$ and $\mathcal{F}$ an intersecting family, for which the following will be crucial.

Claim 2.8. *We have*

$$\Pr[F \in \mathcal{F} | \vec{x}] \le p,$$

That is, regardless of what $\vec{x}$ is conditioned to be, the probability that $F = F_{\vec{x},I} \in \mathcal{F}$ is small.

Proof. fix $\vec{x}$ arbitrarily. If there does not exist any interval I with $F_{\vec{x},I} \in \mathcal{F}$ then this conditional probability is 0, so we may assume there exists some interval $I' = (x', x'+p)$ such that $F_{\vec{x},I'} \in \mathcal{F}$. Now observe that if $I = (x, x+p)$ is any interval with $F_{\vec{x},I} \in \mathcal{F}$ then we must have $I \cap I' \neq \emptyset$ since $\mathcal{F}$ is intersecting and hence in particular there must be some $x_i \in I \cap I'$. Having $I \cap I' \neq \emptyset$ is only possible if $|x - x'| \le p$, i.e. if $x \in [x' - p, x' + p]$, which for $p < \frac{1}{2}$ is a proper subset of C.

The observation $x \in [x' - p, x' + p]$ already implies the non-trivial result $\Pr[F \in \mathcal{F} | \vec{x}] \le 2p$ since this is the probability that the random interval $I = (x, x+p)$ has x lying in the appropriate interval. To improve this by a factor of 2 we use the same trick in the original Katona circle proof that one can pair arcs in such a way that at most one of these corresponds to an element of $\mathcal{F}$. Notably, for each $x \in [x' - p, x']$ at most one of the arcs $I_1 = (x, x+p)$ and $I_2 = (x+p, x+2p)$ can have $F_{\vec{x},I_j} \in \mathcal{F}$ since necessarily $F_{\vec{x},I_1} \cap F_{\vec{x},I_2} = \emptyset$. It follows that conditional on $I = (x, x+p)$ having $x \in [x' - p, x' + p]$ that there is at most a $\frac{1}{2}$ probability of $F_{\vec{x},I} \in \mathcal{F}$, and hence in total

$$\Pr[F \in \mathcal{F} | \vec{x}] \le 2p \cdot \frac{1}{2} = p.$$

□

From this claim it follows that

$$\mu_p(\mathcal{F}) = \Pr[F \in \mathcal{F}] = \mathbb{E}_{\vec{x}} \Pr[F \in \mathcal{F} | \vec{x}] \le p,$$

proving the result. □

2.3 The Bollobás Set Pairs Inequality

Theorem 2.7 shows one variant of the intersecting family problem, namely by changing how we measure the size of our family. We now consider a problem where we change what exactly we mean by "intersecting." The exact condition we consider is somewhat unnatural but turns out to appear often in practice.

Definition 4. We say that a pair of set systems $\mathcal{A} = \{A_1, \ldots, A_m\}$, $\mathcal{B} = \{B_1, \ldots, B_m\}$ is *Bollobás intersecting* if $A_i \cap B_j \neq \emptyset$ for all $i \neq j$ and $A_i \cap B_i = \emptyset$.

We emphasize that it would be far more natural to define a pair of families to be "intersecting" if $A_i \cap B_j \neq \emptyset$ for all i, j rather than for all $i \neq j$. This more natural notion of intersecting is also studied in the literature (under the name of *cross-intersecting*), but the notion of Bollobás intersecting turns out to be more widely applicable as we will shall shortly see. To begin, we establish the main theorem regarding how large Bollobás intersecting pairs can be.

Theorem 2.9 (Bollobás Set Pairs Inequality)**.** *If $\mathcal{A} = \{A_1, \ldots, A_m\}, \mathcal{B} = \{B_1, \ldots, B_m\}$ are Bollobás intersecting, then*

$$\sum_i \frac{1}{\binom{|A_i|+|B_i|}{|A_i|}} \le 1.$$

Observe that this bound is independent of the ground sets of $\mathcal{A}, \mathcal{B}$, which is very different behavior compared to the results we have seen up to this point.

Proof. Our proof uses a variant of the Katona circle argument which puts $[n]$ on a line rather than a circle. To this end, let X denote the common ground set of $\mathcal{A}$ and $\mathcal{B}$ and consider the set $\mathcal{P}$ of pairs (π, i) where $\pi : X \to X$ is a bijection satisfying that $\pi(x) < \pi(y)$ for all $x \in A_i$ and $y \in B_i$. Observe that the number of pairs (π, i) that a given index i is in is exactly

$$\binom{|X|}{|A_i| + |B_i|} |A_i|!|B_i|!(|X| - |A_i| - |B_i|)! = \frac{|X|!}{\binom{|A_i|+|B_i|}{|A_i|}},$$

since we can choose the image of $A_i \cup B_i$ in $\binom{|X|}{|A_i|+|B_i|}$, and given this we have $|A_i|!|B_i|!(|X| - |A_i| - |B_i|)!$ ways to choose π so that all of A appears before all of B (both these facts implicitly use that A_i, B_i are disjoint by hypothesis, as otherwise no such π would exist). In total this implies

$$|\mathcal{P}| = |X|! \sum_i \frac{1}{\binom{|A_i|+|B_i|}{|A_i|}}.$$

On the other hand, we claim that each π is in at most one pair. Indeed, assume for contradiction that $(\pi, i), (\pi, j) \in \mathcal{P}$ for some $i \neq j$. By hypothesis there exists some $x \in A_i \cap B_j$ and $y \in A_j \cap B_i$, which by definition of $(\pi, i) \in \mathcal{P}$ means $\pi(x) < \pi(y)$ and by definition of $(\pi, j) \in \mathcal{P}$ means $\pi(y) < \pi(x)$, a contradiction. This implies

$$|\mathcal{P}| \le |X|!,$$

which combined with the equality above gives the desired result. $\square$

The following simpler corollary of the Bollobás set pairs inequality is often more convenient to work with.

Corollary 2.10. *If $\mathcal{A}, \mathcal{B}$ are Bollobás intersecting and if every $A_i \in \mathcal{A}$ has $|A_i| \le a$ and every $B_i \in \mathcal{B}$ has $|B_i| \le b$, then*

$$|\mathcal{A}| = |\mathcal{B}| \le \binom{a+b}{a}.$$

This bound is best possible when $\mathcal{A} = \binom{[a+b]}{a}$ and $\mathcal{B}$ has $B_i = [a+b] \setminus A_i$ for all i.

The statement of the Bollobás set pairs inequality should remind the reader of the LYM inequality. And indeed, one can quickly derive the LYM inequality from this, showing that the Bollobás set pairs inequality is at least as strong as this result.

Proof of LYM Inequality. For $\mathcal{F} = \{F_1, \ldots, F_m\} \subseteq 2^{[n]}$ an antichain, define $\mathcal{A}, \mathcal{B}$ by $A_i = F_i$ and $B_i = [n] \setminus F_i$. Trivially $A_i \cap B_i = \emptyset$ for all i, and also $A_i \cap B_j = F_i \cap ([n] \setminus F_j) = F_i \setminus F_j \neq \emptyset$ for $i \neq j$ because F_j does not contain F_i by assumption of $\mathcal{F}$ being an antichain. We conclude that $\mathcal{A}, \mathcal{B}$ are Bollobás intersecting, and hence by the Bollobás set pairs inequality

$$\sum_i \frac{1}{\binom{|A_i|+|B_i|}{|A_i|}} = \sum_i \frac{1}{\binom{n}{|F_i|}} \leq 1,$$

which is exactly the LYM inequality. $\square$

Another application of the Bollobás set pairs inequality is to hypergraph Turán-type problems. To this end, for an r-uniform set system $\mathcal{F}$ we say that a set T of $t \geq r$ of its vertices is a copy of $K_t^{(r)}$ if $\binom{T}{r} \subseteq \mathcal{F}$, that is if the set T has all $\binom{t}{r}$ possible of its edges in $\mathcal{F}$. Note that when $r = 2$ this is the same as saying T is a copy of the graph clique K_t. We say $\mathcal{F}$ is $K_t^{(r)}$-free if it contains no copy of $K_t^{(r)}$.

One of the central unsolved problems in extremal hypergraph theory is to determine the maximum number of edges in a $K_t^{(r)}$-free hypergraph on a given number of vertices. One can also naively ask for the minimum number of edges in such a hypergraph, but this is a boring question because the answer is 0. To make this interesting, we consider *maximal* n-vertex $K_t^{(r)}$-free hypergraphs, which are set systems of the form $\mathcal{F} \subseteq \binom{[n]}{r}$ which are $K_t^{(r)}$-free but $\mathcal{F} \cup \{F\}$ for any $F \in \binom{[n]}{r} \setminus \mathcal{F}$ is not $K_t^{(r)}$-free. We then define the *saturation number* $\mathrm{sat}(n, K_t^{(r)})$ to be the minimum number of edges in a maximal n-vertex $K_t^{(r)}$-free hypergraph.

For example, if $r = 2$ and $t = 3$ then we are considering maximal n-vertex triangle-free graphs. A classic result of Mantel's says that the maximum number of edges in such a graph is $\lfloor n^2/4 \rfloor$. The minimum number of edges turns out to come from the star $K_{1,n-1}$, which is maximal triangle-free because adding any edge to the star creates a triangle. Surprisingly, we can determine the exact saturation number for all r, t despite the corresponding maximum extremal problem being unsolved in general.

Theorem 2.11. *For all $t \geq r$ we have*

$$\mathrm{sat}(n, K_t^{(r)}) = \binom{n}{r} - \binom{n-t+r}{r}.$$

Proof. We leave it as an exercise to construct a maximal $K_t^{(r)}$-free hypergraph with $\binom{n}{r} - \binom{n-t+r}{r}$ edges, the existence of which implies $\mathrm{sat}(n, K_t^{(r)}) \leq \binom{n}{r} - \binom{n-t+r}{r}$ (observe here that constructions for saturation problems imply *upper bounds*, not lower bounds like in the more common maximization problems). It remains to prove the lower bound, that is any $\mathcal{F}$ which is a maximal n-vertex $K_t^{(r)}$-free hypergraph has $|\mathcal{F}| \geq \binom{n}{r} - \binom{n-t+r}{r}$. Equivalently, we want to show that the set of missing edges $\mathcal{A} = \binom{[n]}{r} \setminus \mathcal{F}$ satisfies $|\mathcal{A}| \leq \binom{n-t+r}{r}$.

The only real fact we know about each $A_i \in \mathcal{A}$ is that $\mathcal{F} \cup \{A_i\}$ contains some copy T_i of $K_t^{(r)}$. That is, T_i is a set of t vertices for which $\mathcal{F}$ contains every possible edge except for the edge $A_i \subseteq T_i$. Let $B_i = [n] \setminus T_i$ for each $A_i \in \mathcal{A}$. Since $A_i \subseteq T_i$ we have $A_i \cap B_i = \emptyset$ for all i. On the other hand, if $i \neq j$ then we claim that $A_i \not\subseteq T_j$ as otherwise the set T_j would be missing both the edge A_i and the edge A_j, a contradiction to the fact that adding A_j makes T_j a copy of $K_t^{(r)}$. This in turn implies $A_i \cap B_j \neq \emptyset$, and hence $\mathcal{A}$ and $\mathcal{B} = \{B_1, \ldots\}$ are Bollobás intersecting. Since each $A_i \in \mathcal{A}$ has $|A_i| = r$ and each $B_i \in \mathcal{B}$ has $|B_i| = n - t$, we find by the (corollary of the) Bollobás set pairs inequality

$$\binom{n}{r} - |\mathcal{F}| = |\mathcal{A}| \leq \binom{n-t+r}{r},$$

giving the result. □

many extensions and variants of the Bollobás Set Pairs Inequality. Perhaps the most notable of these is a skew version, for which we say We say that a pair of set systems $\mathcal{A} = \{A_1, \ldots, A_m\}, \mathcal{B} = \{B_1, \ldots, B_m\}$ are *skew Bollobás intersecting* if $A_i \cap B_j \neq \emptyset$ for all $i < j$ and $A_i \cap B_i = \emptyset$. Note that we say nothing now about $A_i \cap B_j$ with $i > j$.

Theorem 2.12 (Skew Bollobás Set Pairs Inequality)**.** *If $\mathcal{A}, \mathcal{B}$ are skew Bollobás intersecting and if every $A_i \in \mathcal{A}$ has $|A_i| \leq a$ and every $B_i \in \mathcal{B}$ has $|B_i| \leq b$, then*

$$|\mathcal{A}| = |\mathcal{B}| \leq \binom{a+b}{a}.$$

As far as we are aware, every known proofs of the Skew Bollobás Set Pairs Inequality relies on some form of the linear algebra method, which is a powerful technique we discuss further in Section 5.

Problem 2.13 (Füredi-Gyárfás-Király)**.** *How large can a Bollobás intersecting pair $\mathcal{A}, \mathcal{B}$ be if $|A_i| \leq a$ for each $A_i \in \mathcal{A}$, $|B_i| \leq b$ for each $B_i \in \mathcal{B}$, and if $|A_i \cap B_j| = 1$ for all $i \neq j$?*

The best known upper bound for this problem is of the form $\frac{5}{6}\binom{a+b}{a}$ due to Kostochka, McCourt, and Nahvi who refined an earlier result of Holzman, which is far off from the best known lower bound of roughly $5^{a/2}$ for $a = b$.

2.4 More on Intersecting Families

While we have solved the most basic problem around intersecting families, there are still many more directions one can consider. Here are a few such directions:

- What if we require $\mathcal{F}$ to be *k-wise intersecting*, meaning $F_{i_1} \cap F_{i_2} \cap \cdots \cap F_{i_k} \neq \emptyset$ for any $F_{i_1}, \ldots, F_{i_k} \in \mathcal{F}$?

- What if we require $\mathcal{F}$ to be *t-intersecting*, meaning $|F \cap F'| \geq t$ for all $F, F' \in \mathcal{F}$?

- What if more generally for a set $L \subseteq [n]$ we require $\mathcal{F}$ to be *L-intersecting*, meaning any two elements of $\mathcal{F}$ have $|F \cap F'| \in L$?

- What if instead of requiring $\mathcal{F}$ to be intersecting, i.e. have no two elements which are pairwise disjoint, we require $\mathcal{F}$ to have no m elements which are pairwise disjoint?

- What if we consider two families $\mathcal{F}, \mathcal{G}$ which are *cross-intersecting*, meaning $F \cap G \neq \emptyset$ for any $F \in \mathcal{F}, G \in \mathcal{G}$? How large can $|\mathcal{F}| \cdot |\mathcal{G}|$ or $\min\{|\mathcal{F}|, |\mathcal{G}|\}$ be in these cases?

- What if $\mathcal{F}$ is a *non-trivial* intersecting family, meaning there does not exist some element i in every edge of $\mathcal{F}$?

- What if instead of looking at $\mathcal{F} \subseteq 2^{[n]}$ or $\mathcal{F} \subseteq \binom{[n]}{r}$ we looked at some other collection of sets? For example one could consider "intersecting sets of permutations" or "intersecting sets of graphs."

- What if we consider some mixture of these properties simultaneously?

Some of these problems have known solutions which we will see later on, while others are not understood in full.

Possibly name some of these big ones explicitly, eg Erdos matching, Chvatal conjecture, F-intersecting graph families.

Maybe mention complete intersection theorem somewhere around here.

2.5 Exercises

1. We consider an abstract version of the Katona circle argument: for a permutation $\pi : [n] \to [n]$, we define for $F = \{x_1, \ldots, x_r\} \subseteq [n]$ the set $\pi(F) = \{\pi(x_1), \ldots, \pi(x_r)\}$ and for $\mathcal{F} \subseteq 2^{[n]}$ we define $\pi(\mathcal{F}) = \{\pi(F) : F \in \mathcal{F}\}$. Prove that for any two families $\mathcal{F}, \mathcal{A} \subseteq 2^{[n]}$ that
$$\sum_{F \in \mathcal{F}} \frac{\left|\mathcal{A} \cap \binom{[n]}{|F|}\right|}{\binom{n}{|F|}} \leq \max_{\pi} |\pi(\mathcal{F}) \cap \mathcal{A}|,$$
where the maximum ranges over all permutations $\pi : [n] \to [n]$ [2-].

2. Give an alternative proof of the LYM inequality by using that if $\mathcal{I}$ is the set of initial intervals $\{1, 2, \ldots, i\}$ for $1 \leq i \leq n$ then every antichain of $\mathcal{I}$ has size at most 1 [2-].

* * *

3. Verify for $n < 2r$ that $\binom{[n]}{r}$ is intersecting [1].

4. Classify all intersecting families when $r = 2$ in terms of graph theoretic language [1+].

5. We say that a family $\mathcal{F} \subseteq 2^{[n]}$ is *union* if $F \cup F' \neq [n]$ for all $F, F' \in \mathcal{F}$. Prove that $\mathcal{F}$ is a union family if and only if
$$\overline{\mathcal{F}} := \{[n] \setminus F : F \in \mathcal{F}\}$$
is intersecting. In particular, any problem about intersecting families can be turned into a problem about union families, which in practice can sometimes be easier to solve [1+].

* * *

6. In this problem we investigate maximal intersecting families, i.e. intersecting set systems $\mathcal{F}$ such that $\mathcal{F} \cup \{F\}$ for any $F \notin \mathcal{F}$ with either $F \in 2^{[n]}$ or $F \in \binom{[n]}{r}$ depending on context.

   (a) Prove that every maximal intersecting family $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| = 2^{n-1}$. For this we emphasize that by definition, no intersecting family $\mathcal{F}$ can have $\emptyset \in \mathcal{F}$ since then $F \cap F' = \emptyset$ for $F = F' = \emptyset$ elements in $\mathcal{F}$ [2].

   (b) Prove that there exist maximal intersecting families $\mathcal{F} \subseteq \binom{[n]}{r}$ with $|\mathcal{F}| \leq 4^r$ for all n, r [2].

7. Prove for all $\frac{1}{2} < p < 1$ and $n \geq 3$ that there exists some $\mathcal{F} \subseteq 2^{[n]}$ which is intersecting and which has $\mu_p(\mathcal{F}) > p$ (Hint: there is a pretty natural construction when n is odd; try comparing this to the star construction) [2+].

8. Recall that for a family $\mathcal{F} \subseteq 2^{[n]}$ we define its Lubell measure

$$\lambda(\mathcal{F}) = \sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}}.$$

   Prove that if $\mathcal{F}$ is intersecting then $\lambda(\mathcal{F}) \leq \frac{n+1}{2}$ [2].

9. Prove that for any family $\mathcal{F} \subseteq 2^{[n]}$ we have

$$\int_0^1 \mu_p(\mathcal{F}) dp = \frac{1}{2} \int_0^1 \mu_p(\mathcal{F}) + \mu_{1-p}(\mathcal{F}) dp = \frac{1}{n+1} \lambda(\mathcal{F}).$$

   In particular, any upper bound on $\mu_p(\mathcal{F})$ or $\mu_p(\mathcal{F}) + \mu_{1-p}(\mathcal{F})$ leads to upper bounds on $\lambda(\mathcal{F})$ which can be a useful mode of attack, see for example this paper [2+].

* * *

10. Give a proof of the Bollobás set pairs inequality in terms of probabilistic language [1].

11. For all $t \geq r$, find a construction of a maximal n-vertex $K_t^{(r)}$-free hypergraph $\mathcal{F}$ with $|\mathcal{F}| = \binom{n}{r} - \binom{n-t+r}{r}$ [2].

12. Prove without using the Bollobás set pairs inequality that $\mathrm{sat}(n, K_3) = n - 1$ [2-].

3 Compression Arguments and Isoperimetric Inequalities

One of the most frequently used methods in extremal set theory is that of compression arguments. Abstractly, compression arguments work as follows:

(i) Start with some family $\mathcal{F}_0$ and apply some carefully chosen "compression operations" to turn $\mathcal{F}_0$ into a new family $\mathcal{F}_1$, and then again apply compression operations to $\mathcal{F}_1$ to obtain a new family $\mathcal{F}_2$ and so on.

(ii) Verify that the compression operations are such that $\mathcal{F}_{i+1}$ maintains or improves upon all the desired "properties" of $\mathcal{F}_i$.

(iii) Stop your process when we reach some $\mathcal{F}_k$ which is a "stable structure", in the sense that applying any compression operation to $\mathcal{F}_k$ gives back the family $\mathcal{F}_k$.

(iv) Solve your extremal problem when your family is a "stable structure."

Arguments like this are present throughout many areas of mathematics. Indeed, an illustrative example of this approach is Zykov's proof of Turán's Theorem from graph theory. Here we wish to determine the maximum number of edges an n-vertex K_{r+1}-free graph G can have, and to do this we proceed as follows:

(i) Iteratively if G has two non-adjacent vertices u, v with distinct neighborhoods we delete a vertex with smallest degree and replace it with a clone of the other vertex.

(ii) We verify that this operation preserves the property of being n-vertex K_{r+1}-free and that the number of edges does not decrease.

(iii) By applying a finite number of operations we can transform G into a complete $(r-1)$-partite graph.

(iv) We solve the problem for complete $(r-1)$-partite graphs, which by the steps above is enough to solve the problem in general.

For compression arguments, the main difficulty usually lies in choosing the right "compression operations" to work with in order to fulfill steps (ii) and (iv). For example, the very strong compression argument "replace $\mathcal{F}$ with the conjectured extremal example" makes (iv) trivial to solve, but now verifying (ii) is just as difficult as proving the original problem. Similarly the very weak compression argument "leave $\mathcal{F}$ as it is" trivially satisfies (ii), but now (iv) is just as hard as the base problem. Striking the right balance between the difficulties of (ii) and (iv) is the real heart of this choice of operation.

We begin our investigation of compression arguments with the familiar territory of intersecting families, after which we use these methods to prove some discrete isoperimetric inequalities for set systems.

3.1 Shifting and Intersecting Famlies Revisited

In this section we explore a compression operation known as shifting, which is perhaps the most commonly used compression operation in extremal set theory. Its definition comes from the original proof of the Erdős-Ko-Rado Theorem and is motivated by the idea of trying to shift the elements of some family $\mathcal{F}$ to look more like the extremal example, namely the collection of all r-sets containing some element i. That is, starting with $\mathcal{F}_0 = \mathcal{F}$, we want to iteratively define new families $\mathcal{F}_1, \mathcal{F}_2, \ldots$ such that $|\mathcal{F}_k| = |\mathcal{F}_{k-1}|$, such that each $\mathcal{F}_k$ remains intersecting, and such that each $\mathcal{F}_k$ is "closer" to being a star on some given element i.

Keeping this idea in mind, if we have some $F \in \mathcal{F}$ which does not contain our given element i, then we want to transform F into a different r-element set which does contain i to make it closer to this extremal example. Perhaps the simplest way to do this is to replace F with a set of the form $F \cup \{i\} \setminus \{j\}$ for some $j \in F$. The issue with this replacement is that we may already have $F \cup \{i\} \setminus j$ in $\mathcal{F}$, so doing this operation will change the size of $\mathcal{F}$ which is something we want to avoid. The clever but simple workaround is that we will only do this replacement F such that $F \cup \{i\} \setminus j \notin \mathcal{F}$ and otherwise we leave F as is.

More precisely, for integers i, j, a family of sets $\mathcal{F}$, and $F \in \mathcal{F}$, we define the shifted edge

$$S_{j \to i}(F; \mathcal{F}) = \begin{cases} F \cup \{i\} \setminus \{j\} & j \in F,\ i \notin F,\ F \cup \{i\} \setminus \{j\} \notin \mathcal{F}, \\ F & \textit{otherwise}, \end{cases}$$

and we denote this simply by $S_{j \to i}(F)$ whenever $\mathcal{F}$ is understood. We similarly define the shifted family

$$S_{j \to i}(\mathcal{F}) = \{S_{j \to i}(F) : F \in \mathcal{F}\}.$$

Crucially, this operation preserves all the properties we care about.

Lemma 3.1. *Let $\mathcal{F} \subseteq 2^{[n]}$ be a family and $i, j \in [n]$.*

(a) For every $F \in \mathcal{F}$ we have $|S_{j \to i}(F)| = |F|$; in particular, if $\mathcal{F} \subseteq \binom{[n]}{r}$ then $S_{j \to i}(\mathcal{F}) \subseteq \binom{[n]}{r}$.

(b) We have $|S_{j \to i}(\mathcal{F})| = |\mathcal{F}|$.

(c) If $\mathcal{F}$ is intersecting then so is $S_{j \to i}(\mathcal{F})$.

Proof. Throughout the proof we let $F' := S_{j \to i}(F)$ for ease of notation.

Part (a) is immediate from the definition. For (b), we claim that $F' \neq G'$ for any distinct $F, G \in \mathcal{F}$. This is immediate if $F' = F$ and $G' = G$, so suppose that instead $F' \neq F$, that is $F' = F \cup \{i\} \setminus \{j\}$. By definition of shifted edges, having $F' \neq F$ means $F' \notin \mathcal{F}$, so in particular $G \neq F'$. If $G' = F' \neq G$ then we must have $G' = G \cup \{i\} \setminus \{j\} = F \cup \{i\} \setminus \{j\}$ with $j \in F, G$ and $i \notin F, G$. But this implies $F = G$, a contradiction to us assuming F, G were distinct. This implies

$$|S_{j \to i}(\mathcal{F})| = |\{F' : F \in \mathcal{F}\}| = |\{F : F \in \mathcal{F}\}| = |\mathcal{F}|,$$

proving (b).

For (c), consider any $F, G \in \mathcal{F}$. Because $\mathcal{F}$ is intersecting, we will trivially have that F', G' intersect if $F' = F$ and $G' = G$, and similarly these two edges will intersect if $F' \neq F$ and $G' \neq G$ since in this case we have i in both of these edges. As such we may assume without loss of generality that $F' = F$ and $G' \neq G$, and hence that $G' = G \cup \{i\} \setminus \{j\}$. The fact that $F' = F$ means either that: (i) $j \notin F$, in which case whatever $k \in F \cap G$ which exists by assumption of $\mathcal{F}$ being intersecting also lies in $F' \cap G'$; (ii) $i \in F$, in which case $i \in F' \cap G'$; or (iii) we have $F \cup \{i\} \setminus \{j\} \in \mathcal{F}$. If (iii) holds then by definition of $\mathcal{F}$ being intersecting there must exist some $k \in (F \cup \{i\} \setminus \{j\}) \cap G$, and by definition $k \neq j$ and also by definition of $G' \neq G$ we have $i \notin G$ and hence $k \neq i$ either. This means that in fact $k \in F = F'$ and $k \in G'$, proving that F', G' intersect. This proof logic could probably be cleaned up somewhat. □

We now wish to take some arbitrary family $\mathcal{F}$ and shift it to be as close as possible to, say, a star with every edge containing 1. Certainly we want to apply the operations $\mathcal{S}_{j\to 1}$ for all j, but we might also consider performing $\mathcal{S}_{j\to i}$ with $j > i$ in general to give us even more structure. To this end, we say that a family $\mathcal{F}$ is *shifted* if $\mathcal{F} = S_{j\to i}(\mathcal{F})$ for all $j > i$ and observe the following.

Lemma 3.2. *For every family $\mathcal{F}_1 \subseteq 2^{[n]}$ there is a sequence of integer pairs $(i_1, j_1), \ldots, (i_t, j_t)$ with $i_k < j_k$ for all k such that if we define $\mathcal{F}_{k+1} = S_{j_k \to i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is shifted.*

Proof. For this proof, we observe that for any family $\mathcal{F}$, integers $j > i$, and $F \in \mathcal{F}$ that

$$\sum_{k\in F} k \geq \sum_{k \in S_{j\to i}(F)} k$$

with the inequality being strict whenever $F \neq S_{j\to i}(F)$ simply because either $F = S_{j\to i}(F)$ or we removed j and replaced it with the strictly smaller number i.

Returning to the problem at hand, if some $\mathcal{F}_s$ we have constructed is shifted then we are done. Otherweise by definition there is some $j_1 > i_1$ with $\mathcal{F}_s \neq S_{j_s \to i_s}(\mathcal{F}_s) := \mathcal{F}_{s+1}$. By the observation above and $\mathcal{F}_s \neq \mathcal{F}_{s+1}$ we find

$$\sum_{e\in\mathcal{F}_s}\sum_{k\in e} k > \sum_{e\in\mathcal{F}_s}\sum_{k\in S_{j_s\to i_s}(e)} k = \sum_{e\in\mathcal{F}_s}\sum_{k\in e} k.$$

Because this (finite) sum decreases by at least 1 every time we go from $\mathcal{F}_s$ to $\mathcal{F}_{s+1}$ and because this sum most always be non-negative, we conclude that after finitely many steps this process must terminate, meaning $\mathcal{F}_s$ is shifted, giving the result. □

The two lemmas above establish (ii) and (iii) of our general compression argument paradigm, and as such it remains to do step (iv) of proving Erdős-Ko-Rado for shifted families. We aim to do this by induction on n, which will require us to replace our given family $\mathcal{F} \subseteq \binom{[n]}{r}$ with appropriate subfamilies on a strictly smaller ground set.

One way to try and achieve this is to partition $\mathcal{F}$ into the sets which contain a given element $i \in [n]$ and those which don't, and for this we define

$$\mathcal{F}(i) = \{F : F \in \mathcal{F},\ i \in F\},$$

$$\mathcal{F}(\bar{i}) := \{F : F \in \mathcal{F},\ i \notin F\}.$$

Observe that $\mathcal{F}(\bar{i})$ is a set system on the smaller ground set $[n] \setminus \{i\}$ but this is typically not true for $\mathcal{F}(i)$. However, we observe the key fact that every element in $\mathcal{F}(i)$ contains i, and hence us including i in each element gives us no new information. Motivated by this, we define the *link hypergraph*

$$\mathcal{F}_L(i) := \{F \setminus \{i\} : F \in \mathcal{F}(i)\} = \{F \setminus \{i\} : F \in \mathcal{F},\ i \in F\},$$

which is a set system on the smaller ground set $[n]\setminus\{i\}$. We moreover observe that the properties of this and the other family $\mathcal{F}(\bar{i})$ satisfy a number of reasonable properties.

Lemma 3.3. *Let $\mathcal{F} \subseteq \binom{[n]}{r}$, $i \in [n]$, and $\mathcal{F}(\bar{i}), \mathcal{F}_L(i)$ be the families as defined above.*

(a) We have $\mathcal{F}(\bar{i}) \subseteq \binom{[n]\setminus\{i\}}{r}$, $\mathcal{F}_L(i) \subseteq \binom{[n]\setminus\{i\}}{r-1}$, and

$$|\mathcal{F}| = |\mathcal{F}(\bar{i})| + |\mathcal{F}_L(i)|.$$

(b) If $\mathcal{F}$ is intersecting, then $\mathcal{F}(\bar{i})$ is intersecting.

(c) If $\mathcal{F}$ is intersecting and shifted, then $\mathcal{F}_L(n)$ is intersecting.

Proof. Part (a) is straightforward to verify, and (b) follows from the fact that $\mathcal{F}(\bar{i}) \subseteq \mathcal{F}$ and $\mathcal{F}$ is intersecting. For (c), the result is trivial if $n \leq r$, so we may assume $n \geq r+1$ from now on. Assume for contradiction that there existed $F', G' \in \mathcal{F}_L(n)$ which were disjoint. By definition of this family we must have $F = F' \cup \{n\}$ and $G = G' \cup \{n\}$ in $\mathcal{F}$ which satisfy $F \cap G = \{n\}$ by hypothesis of F', G'. The fact that the family is shifted and $F \in \mathcal{F}$ implies that for every $i < n$ that $F \cup \{i\} \setminus \{n\} \in \mathcal{F}$, as otherwise $S_{n\to i}(\mathcal{F})$ would contain $F \cup \{i\} \setminus \{n\}$ and hence not be equal to $\mathcal{F}$. But now take any $i \in [n] \setminus G$ (which exists since $n \geq r+1$) and observe that $F \cup \{i\} \setminus \{n\}$ is disjoint from G since $F \cap G = \{n\}$, a contradiction to $\mathcal{F}$ being intersecting, giving the result. □

Putting all these pieces gives us a new proof of the Erdős-Ko-Rado Theorem.

Proof III of Erdős-Ko-Rado Theorem. We prove the result by induction on r and then induction on $n \geq 2r$, the base cases $r = 1$ and $n = 2r$ being straightforward to verify. Let $\mathcal{F} \subseteq \binom{[n]}{r}$ be intersecting with $n > 2r$. By Lemma 3.2 we can apply a finite sequence of shift operators to $\mathcal{F}$ to reach a new family $\mathcal{F}'$ which by Lemma 3.1 will also be r-uniform and intersecting with $|\mathcal{F}| = |\mathcal{F}'|$. Moreover, by Lemma 3.3 and the inductive hypothesis we find

$$|\mathcal{F}| = |\mathcal{F}'| = |\mathcal{F}'(\overline{n})| + |\mathcal{F}'_L(n)| \leq \binom{n-2}{r-1} + \binom{n-2}{r-2} = \binom{n-1}{r-1},$$

proving the result. □

We note that the idea in this proof of essentially partitioning $\mathcal{F}$ into the two sets $\mathcal{F}(\overline{n})$ and $\mathcal{F}_L(n)$ based on whether n is in a given element of $\mathcal{F}$ to induct on the size of the ground set is a very common trick in extremal set theory which we will see several more times throughout these notes.

Shifting is typically used for problems involving uniform families, but one can use it to solve non-uniform problems as well. Specifically, we will study *t-intersecting families*, which are families $\mathcal{F}$ such that any $F, F' \in \mathcal{F}$ have $|F \cap F| \geq t$. We are thus led to the problem: how large can $\mathcal{F} \subseteq 2^{[n]}$ be if $\mathcal{F}$ is t-intersecting?

Let us spend a second thinking about which constructions might give us the largest size, and to begin let us think about the $t = 1$ case which we have already solved. While there are many extremal constructions achieving 2^{n-1} for $t = 1$, the two simplest we mentioned were (i) the family containing every set containing 1 and (ii) when n is odd the family containing all sets of size $(n+1)/2$. To make these examples t-intersecting the most natural way to generalize them is via taking (i') the family containing every set containing $\{1, 2, \ldots, t\}$ and (ii') when $n+t$ is even taking the family which contains all sets of size at least $(n+t)/2$. A quick analysis shows that the construction from (i') has size exactly 2^{n-t} while (ii') has size

$$\sum_{k=(n+t)/2}^{n} \binom{n}{k} = 2^{n-1} - O_t(n^{-1/2}2^n),$$

and this suggests that (ii') should be the right answer for $n+t$ even. For $n+t$ odd the best one can naively do is to take all sets of size at least $(n+t+1)/2$. However, this family is in fact $(t+1)$-intersecting and hence we can add more elements before we lose the t-intersecting property. After some thought, one might come up with the idea to take all sets of size $(n+t+1)/2$ together with all sets of size $(n+t-1)/2$ which contain 1. One can check that this is indeed t-intersecting and has size

$$\binom{n-1}{(n+t-1)/2} + \sum_{k=(n+t+1)/2}^{n} \binom{n}{k} = 2 \sum_{k=(n+t-1)/2}^{n} \binom{n-1}{k},$$

with the equality following from Pascal's identity.

We now have a conjectured bound for what to try and prove. One can try and replicate our shifting proof of Erdős-Ko-Rado in this context and the inductive step will work when $n+t$ is odd, essentially because the extremal construction $\mathcal{F}$ in this case is such that the families $\mathcal{F}(\overline{n})$ and $\mathcal{F}_L(n)$ turn out to both be extremal constructions for $n-1$. However, when $n+t$ is even the families $\mathcal{F}(\overline{n})$ and $\mathcal{F}_L(n)$ are not extremal when $\mathcal{F}$ is, and hence there is too much slack in our argument for this approach to work. We can remedy this by applying a small trick.

Mantra 2. If a problem if giving you trouble, try taking its "complement."

In extremal set theory there are two natural notions of the complement of a set system $\mathcal{F} \subseteq 2^{[n]}$, namely $2^{[n]} \setminus \mathcal{F}$ and $\{[n] \setminus F : F \in \mathcal{F}\}$. This latter definition plays nicely with the t-intersecting property through the following observation.

Lemma 3.4. *If $\mathcal{F} \subseteq 2^{[n]}$ is t-intersecting, then the family $\mathcal{F}' = \{[n] \setminus F : F \in \mathcal{F}\}$ has the property that for any $F, F' \in \mathcal{F}'$, we have*

$$|F \cup F'| \leq n - t.$$

Motivated by this, we say a family $\mathcal{F} \subseteq 2^{[n]}$ is *t-union* if $|F \cup F'| \le n - t$ for any $F, F' \in \mathcal{F}$. By this lemma, we see that the size of a largest t-intersecting family on $[n]$ is exactly the size of a largest t-union family on $[n]$, and as such it suffices to solve this latter problem. In particular, it turns out that shifting will work for this modified problem, giving us the following.

Theorem 3.5 (Katona's Union Theorem)**.** *Let $\mathcal{F} \subseteq 2^{[n]}$ be a t-union family with $n > t > 0$, and for convenience let $d = n - t$ so that $F, F' \in \mathcal{F}$ satisfy $|F \cup F'| \le d$. If $d = 2h$ for some integer h then*

$$|\mathcal{F}| \le \sum_{i=0}^{h} \binom{n}{i},$$

and if $d = 2h + 1$ for some integer h then

$$|\mathcal{F}| \le 2 \sum_{i=0}^{h} \binom{n-1}{i}.$$

Sketch of Proof. We prove the result by induction on $d \ge 1$ and then induction on $n \ge d + 1$. The base case of $d = 1$ is straightforward, and the base case $n = d+1$ is simply Proposition 2.4, so assume we have proven the result up to some value d and then up to some value $n \ge d + 2$. One can easily show that shifting preserves $\mathcal{F}$ being t-union, so it suffices to prove the result under the assumption that $\mathcal{F}$ is shifted. As in our proof of the Erdős-Ko-Rado Theorem we let $\mathcal{F}(\overline{n}) = \{F \in \mathcal{F} : n \notin \mathcal{F}\}$ and $\mathcal{F}_L(n) = \{F \setminus \{n\} : F \in \mathcal{F},\ n \in \mathcal{F}\}$.

Since $\mathcal{F}(\overline{n}) \subseteq \mathcal{F}$ we immediately have $|F \cup F'| \le d$ for any $F, F' \in \mathcal{F}(\overline{n})$. On the other hand, we claim that any $F, F' \in \mathcal{F}_L(n)$ has $|F \cup F'| \le d - 2$. Indeed, if their union were at least $d - 1$ then that means $F \cup \{n\}, F' \cup \{n\} \in \mathcal{F}$ would have union at least d and hence size exactly d by hypothesis of $\mathcal{F}$. But now for any $i \notin F \cup F' \cup \{n\}$ (which exists since $n \ge d + 2$) we have by definition of $\mathcal{F}$ being shifted and $F \cup \{n\} \in \mathcal{F}$ that $F \cup \{i\} \in \mathcal{F}$ and the union of this with $F' \cup \{n\}$ has size $d + 1$, a contradiction to our assumption on $\mathcal{F}$.

In total we find that $\mathcal{F}(\overline{n})$ satisfies the conditions of the theorem with the parameters $n - 1$ and d, and that $\mathcal{F}_L(n)$ satisfies the conditions with $n - 1$ and $d - 2$. We can thus inductively upper bound the size of these two sets, which together with the observation $|\mathcal{F}| = |\mathcal{F}(\overline{n})| + |\mathcal{F}_L(n)|$ and some binomial coefficient manipulations gives the desired bound. $\square$

Somewhat surprisingly we can use the non-uniform Theorem 3.5 to give another proof of the uniform Erdős-Ko-Rado Theorem.

Proof IV of Erdős-Ko-Rado. Let $\mathcal{F} \subseteq \binom{[n]}{r}$ with $n \ge 2r$ be intersecting. Let $\mathcal{F}' = \mathcal{F} \cup \{F \subseteq [n] : |F| \le r - 1\}$ and observe that this family satisfies $|F \cup F'| \le 2r - 1$ for any $F, F' \in \mathcal{F}'$ (this trivially holds if a set of size at most $r - 1$ is involved, and otherwise it follows from the intersecting property). By Theorem 3.5 applied with $d = 2r - 1 < n$ we find that

$$|\mathcal{F}| + \sum_{k=0}^{r-1} \binom{n}{k} = |\mathcal{F}'| \le 2 \sum_{k=0}^{r} \binom{n-1}{k} = \binom{n-1}{r} + \sum_{k=0}^{r-1} \binom{n}{k},$$

giving the desired bound. $\square$

We note that this trick of adding in all the elements of size less than r to go from a uniform problem to a non-uniform problem appears somewhat frequently in the field, and we will see another example of this later in this chapter.

Before moving on, it is natural to ask about what can be said about t-intersecting families in the uniform setting. The natural guess is t-intersecting $\mathcal{F} \subseteq \binom{[n]}{r}$ have $|\mathcal{F}| \leq \binom{n-t}{r-t}$ with this being best possible by considering all families which contain $\{1, 2, \ldots, t\}$. This turns out to be true for n sufficiently larger than some constant $n_{r,t}$, but as in the non-uniform case the proof of this for larger t is more complicated. In particular, the shifting arguments we have here generalize to prove the inductive step without issue, but the difficulty now is that the base case is not so easy to prove. This in part due to the fact that for small n there are various different types of constructions that can occur which makes it extra difficult to determine the optimal value of $n_{r,t}$ such that the bound of $\binom{n-t}{r-t}$ holds for all $n \geq n_{r,t}$.

3.2 Shadows and the Kruskal-Katona Theorem

The use of shifting does not apply only to problems about intersecting families; they can be used for any problem involving properties which are well-behaved under shifting. For example, given a set system $\mathcal{F}$ and an integer t we define the *t-shadow*

$$\partial_t(\mathcal{F}) = \{T : |T| = t,\ \exists F \in \mathcal{F}\ T \subseteq F\}.$$

That is, these are the t-element subsets which are contained in some set of $\mathcal{F}$. If $\mathcal{F} = \{F\}$ we will often simply write $\partial_t(F)$ instead of $\partial_t(\{F\})$ for simplicity, and we will also sometimes write $\partial_t \mathcal{F}$ for $\partial_t(\mathcal{F})$. We observe that shadows play nicely with shifting operations.

Lemma 3.6. *For every family $\mathcal{F}$ and integers i, j, t we have $\partial_t(S_{j\to i}(\mathcal{F})) \subseteq S_{j\to i}(\partial_t(\mathcal{F}))$.*

That is, the shadow of the shift of $\mathcal{F}$ is contained in the shift of the shadow of $\mathcal{F}$. We leave the verification of this lemma as an exercise and use it to solve a problem around shadows.

To begin, we observe that Lemma 3.6 and Lemma 3.1(b) imply

$$|\partial_t(S_{j\to i}(\mathcal{F}))| \leq |S_{j\to i}(\partial_t(\mathcal{F}))| = |\partial_t(\mathcal{F})|,$$

which means shifting can only decrease the size of the shadow. This suggests that shifting might be used for problems which involve lower bounding the size of the shadow of a family (since the above shows we can assume such a family is shifted without loss of generality). And indeed, we can use this to prove a version of the Kruskal-Katona Theorem, which like the Erdős-Ko-Rado Theorem is one of the biggest results in extremal set theory. Here for a real number x and an integer r we write $\binom{x}{r} = \frac{x(x-1)\cdots(x-r+1)}{r!}$ and we adopt the convention $\binom{x}{0} = 1$ for all x.

Theorem 3.7 (Lovász's Simplified Kruskal-Katona Theorem)**.** *If $\mathcal{F} \subseteq \binom{[n]}{r}$ satisfies $|\mathcal{F}| \geq \binom{x}{r}$ for some real number $x \geq r$, then for all $0 \leq t \leq r$ we have $|\partial_t(\mathcal{F})| \geq \binom{x}{t}$.*

When x is an integer this bound is best possible by considering $\mathcal{F} = \binom{[x]}{r}$. The "full" Kruskal-Katona Theorem discussed later on gives a bound which is tight for all values of $|\mathcal{F}|$, but this simpler bound which was first stated by Lovász is often preferable to work with in practice.

Proof. We prove the result only for $t = r-1$, from which one can derive the full statement by induction on $r-t$. We prove the $t = r-1$ result by induction on r and then on $|\mathcal{F}|$ with the cases $r = 1$ and $|\mathcal{F}| = 0$ being trivial.

Assume we have proven the result up to some r and consider $\mathcal{F} \subseteq \binom{[n]}{r}$ with $|\mathcal{F}| \geq \binom{x}{r}$ for some $x \geq r$. By Lemma 3.6 we may assume $\mathcal{F}$ is shifted, as shifting $\mathcal{F}$ further can only decrease the size of its shadow. Heuristically $\mathcal{F}$ being shifted means it contains many edges containing 1, and looking in this direction leads us to the following where we recall $\mathcal{F}_L(i) = \{F \setminus \{i\} : i \in F \in \mathcal{F}\}$ is the link of $\mathcal{F}$ at i.

Claim 3.8. *We have* $|\partial_{r-1}(\mathcal{F})| \geq |\mathcal{F}_L(1)| + |\partial_{r-2}(\mathcal{F}_L(1))|$.

Proof. By definition every element of $\mathcal{F}_L(1)$ is an edge of $\mathcal{F}$ minus the element 1 and hence is contained in $\partial_{r-1}(\mathcal{F})$. We similarly observe that if we take any element $T \in \partial_{r-2}(\mathcal{F}_L(1))$ then $T \cup \{1\}$ is also an element of $\partial_{r-1}(\mathcal{F})$ which is moreover disjoint from every element of $\mathcal{F}_L(1)$ since it contains 1, in total giving the stated number of elements. $\square$

From this claim and induction on r we see that the result holds whenever $\mathcal{F}_L(1)$ is sufficiently large. Specifically, if $|\mathcal{F}_L(1)| \geq \binom{x-1}{r-1}$ then this claim and induction implies

$$|\partial_{r-1}(\mathcal{F})| \geq \binom{x-1}{r-1} + \binom{x-1}{r-2} = \binom{x}{r-1},$$

where this equality is true if x is an integer by Pascal's rule, and since this polynomial equation is true for all integers it is in fact true for all x. Because of this, we may assume from now on that $|\mathcal{F}_L(1)| < \binom{x-1}{r-1}$, and the fact that $\mathcal{F}_L(1)$ is small means

$$|\mathcal{F}(\bar{1})| = |\mathcal{F}| - |\mathcal{F}_L(1)| \geq \binom{x}{r} - \binom{x-1}{r-1} = \binom{x-1}{r}.$$

Heuristically it should seem strange that $\mathcal{F}(\bar{1})$ is so large in a shifted family which by definition favors edges containing 1. One concrete way to formalize this intuition is through the following.

Claim 3.9. *We have* $\partial_{r-1}(\mathcal{F}(\bar{1})) \subseteq \mathcal{F}_L(1)$.

Proof. For each $T \in \partial_{r-1}(\mathcal{F}(\bar{1}))$ there exists some j such that $T \cup \{j\} \in \mathcal{F}(\bar{1}) \subseteq \mathcal{F}$, which in particular means $1 \notin T \cup \{j\}$. Because $\mathcal{F}$ is shifted, we must have $T \cup \{1\} \in \mathcal{F}$ (as otherwise we would have $S_{j\to 1}(\mathcal{F}) \neq \mathcal{F}$), and hence $T \in \mathcal{F}_L(1)$ as desired. $\square$

Observe that $\mathcal{F}(\bar{1})$ must be a strict subset of $\mathcal{F}$ if $\mathcal{F}$ is non-empty and shifted (i.e. $\mathcal{F}$ must contain at least one edge containing 1), so by induction on $|\mathcal{F}|$ and our inequality $|\mathcal{F}(\bar{1})| \geq \binom{x-1}{r}$ we find

$$|\mathcal{F}_L(1)| \geq |\partial_{r-1}(\mathcal{F}(\bar{1}))| \geq \binom{x-1}{r-1},$$

a contradiction to us assuming the opposite inequality held above. $\square$

The Kruskal-Katona Theorem (even in its simplified Lovász form) has many applications, one of which is that we can give yet another proof of the Erdős-Ko-Rado Theorem due to Daykin.

Proof IV of Erdős-Ko-Rado. The key observation is that having $F \cap F' \neq \emptyset$ is equivalent to the statement $F \not\subseteq \overline{F}'$ where $\overline{F}' := [n] \setminus F'$. If F, F' are r-sets then $F \not\subseteq \overline{F}'$ is equivalent to saying $F \notin \partial_r \overline{F}'$. In total then if we define $\overline{\mathcal{F}} = \{[n] \setminus F : F \in \mathcal{F}\}$ for $\mathcal{F}$ an intersecting family in $\binom{[n]}{r}$ then we find $\mathcal{F} \cap \partial_r \overline{\mathcal{F}} = \emptyset$.

Now assume for contradiction that $|\mathcal{F}| > \binom{n-1}{r-1}$, which in turn implies $|\overline{\mathcal{F}}| = |\mathcal{F}| > \binom{n-1}{r-1} = \binom{n-1}{n-r}$. Applying the Kruskal-Katona Theorem with $t = r \le n - r$ due to the hypothesis $n \ge 2r$ gives

$$|\mathcal{F}| + |\partial_r \overline{\mathcal{F}}| > \binom{n-1}{r-1} + \binom{n-1}{r} = \binom{n}{r},$$

a contradiction to $\mathcal{F}, \partial_r \overline{\mathcal{F}}$ being disjoint subsets of $\binom{[n]}{r}$. $\square$

As mentioned above, the full version of the Kruskal-Katona Theorem determines exactly how large $\partial_t \mathcal{F}$ must be as a function of $|\mathcal{F}|$. To partially motivate the statement of this result, let us think about the case when $\mathcal{F} \subseteq \binom{[n]}{2}$, that is when $\mathcal{F}$ is a graph. If $|\mathcal{F}| = \binom{m}{2}$ for some integer m and we want to minimize $\partial_1 \mathcal{F}$ (which is simply the set of vertices in an edge of $\mathcal{F}$), then Theorem 3.7 says the best we can do is to take $\mathcal{F}$ to be a clique on $[m]$. If $\binom{m}{2} < |\mathcal{F}| < \binom{m+1}{2}$ then a bit of thought suggests that the best we can do is to have a clique on $[m]$ together with $\binom{m+1}{2} - |\mathcal{F}|$ edges from this clique to, say, the vertex $m+1$. That is, we want to add these extra edges while avoiding using additional vertices as much as possible. Thinking about this a bit more, one could perhaps be led to the idea of trying to iteratively add edges to $\mathcal{F} \subseteq \binom{[n]}{r}$ while making its largest element as small as possible (to avoid adding new vertices as much as possible), and then condition on this making its second largest element as small as possible, and so on. And indeed, this is the correct answer.

More precisely, we define the *colexicographical ordering* or simply *colex ordering* $<_C$ on $\binom{[n]}{r}$ by saying two distinct sets $F = \{a_1 < \cdots < a_r\}$ and $F' = \{b_1 < \cdots < b_r\}$ have $F <_C F'$ if for the largest integer i with $a_i \neq b_i$ we have $a_i < b_i$. That is, $F <_C F'$ if either $a_r < b_r$, or if $a_r = b_r$ and $a_{r-1} < b_{r-1}$, or if $a_r = b_r$ and $a_{r-1} = b_{r-1}$ and $a_{r-2} < b_{r-2}$, and so on. Let $\mathcal{C}(r, s)$ denote the set of the first s elements of $\binom{\mathbb{N}}{r}$, so for example $\mathcal{C}(r, 2) = \{[r], [r-1] \cup \{r+1\}\}$.

Theorem 3.10 (Kruskal-Katona Theorem)**.** *For any $\mathcal{F} \subseteq \binom{[n]}{r}$ we have $|\partial_{r-1}\mathcal{F}| \ge |\partial_{r-1}\mathcal{C}(r, |\mathcal{F}|)|$.*

For example, if $|\mathcal{F}| = \binom{m}{r}$ for an integer m then it is not so difficult to show $|\partial_{r-1}\mathcal{C}(r, |\mathcal{F}|)| = \binom{m}{r-1}$, and thereby we recover this case of Lovász's version. For general $|\mathcal{F}|$ we need to work a bit more to understand what exactly the lower bound of Theorem 3.10 is giving.

To this end, for integers $s, r \ge 1$ it turns out there exist integers $b_r > b_{r-1} > \cdots > b_\ell \ge \ell \ge 1$ such that $s = \sum_{i=\ell}^{r} \binom{b_i}{i}$, and we refer to this sum of binomial coefficients as the *r-cascade representation* of s. Note that the cascade representation of s differs from the more conventional base b-representation of the integer s by writing s not as a linear combination of powers of b, but rather as a sum of binomial coefficients. One can show that if s has r-cascade representation $\sum_{i=\ell}^{r} \binom{b_i}{i}$, then

$$|\partial_{r-1}\mathcal{C}(r, s)| = \sum_{i=\ell}^{r} \binom{b_i}{i-1},$$

which gives a more quantitative way of interpreting Theorem 3.10. In fact, our proof of Lovász's version of Kruskal-Katona almost goes through for this more quantitative version by replacing our binomial coefficients with the appropriate r-cascade representation. The only issue that pops up is when we assume now that $|\mathcal{F}_L(1)| \geq \sum \binom{b_i-1}{i}$ we can only apply our inductive hypothesis if this sum is an r-cascade representation, which can fail to be the case if $b_i = i$ for some i. One can get over this issue with some case analysis; we omit doing so since we will derive the Kruskal-Katona Theorem from an even stronger result in the following subsection.

Before moving on, we note that from the Boolean lattice perspective, $\partial_{r-1}(\mathcal{F})$ can be interpreted as the set of elements lying just below $\mathcal{F}$. From this viewpoint it makes as much sense to consider the elements lying just above $\mathcal{F}$. To this end we define for $\mathcal{F} \subseteq \binom{[n]}{r}$ and $t \geq r$ the *upper shadow* $\partial^t(\mathcal{F}) = \{F' \in \binom{[n]}{t} : \exists F \in \mathcal{F},\ F \subseteq F'\}$. The Kruskal-Katona Theorem has an essentially equivalent formulation in terms of upper shadows which can often be useful to consider.

To this end, we define the The *lexicographical ordering* or simply *lex ordering* $<_L$ on $\binom{\mathbb{N}}{r}$ by saying two distinct sets $F = \{a_1 < \cdots < a_r\}$ and $F' = \{b_1 < \cdots < b_r\}$ have $F <_L F'$ if for the smallest integer i with $a_i \neq b_i$ we have $a_i < b_i$. That is, this ordering prioritizes using smaller elements as much as possible, as compared to colex which avoids using larger elements as much as possible. We let $\mathcal{L}(n, r, s)$ denote the set of the first s elements of $\binom{[n]}{r}$ in lex order.

Corollary 3.11 (Kruskal-Katona for Upper Shadows)**.** *For any $\mathcal{F} \subseteq \binom{[n]}{r}$ with $r < n$ we have $|\partial^{r+1}\mathcal{F}| \geq |\partial^{r+1}\mathcal{L}(n, r, |\mathcal{F}|)|$.*

3.3 Isoperimetric Inequalities and Harper's Inequality

The Kruskal-Katona Theorem can be interpreted as a discrete isoperimetric inequality. Broadly speaking, an isoperimetric inequality is a statement which bounds the size of the "boundary" of an object in terms of the "volume" of the object. For example, the classical isoperimetric inequality states that the perimeter of a closed curve of area A is at least that of the circle with same area A, and in a similar spirit the Kruskal-Katona Theorem states that the shadow of any family is at least that of the initial sequence in colex of the same size.

We consider an isoperimetric inequality for graphs G, and in this setting there are several different ways one can define the concept of "boundary" and "volume." For our purposes we will think of the volume of a set of vertices $V \subseteq V(G)$ as its size and its boundary as being the number of neighbors it has outside of its set, that is

$$N(V) = \{u \notin V : \exists v \in V,\ u \sim v\}.$$

The isoperimetric problem for a graph G then is to determine the smallest possible size of $|N(V)|$ amongst all sets V of a given size s. For this it will sometimes be more convenient to work with the closed neighborhood

$$N[V] = N(V) \cup V,$$

and studying $|N(V)|$ for $|V| = s$ is entirely equivalent to studying $|N[V]|$ for $|V| = s$.

For many graphs, a reasonable guess for the isoperimetric problem is that the answer should be V being a ball, i.e. the set of vertices within some distance d of a given vertex x, since this is exactly the answer to the problem for bodies in $\mathbb{R}^n$. In the special case when G is the

n-dimensional hypercube with its vertices identified by subsets of $[n]$ (which for extremal set theory is the primary graph we care about), balls are essentially equivalent to the collection of all subsets of $[n]$ on at most d elements. This will turn out to be the correct answer when $s = \sum_{i=0}^{d} \binom{n}{i}$ for some d, but what happens for values of s which are not of this particular form? Heuristically if $\sum_{i=0}^{d} \binom{n}{i} < s < \sum_{i=0}^{d+1} \binom{n}{i}$ then to try and be as "close" to a ball as posisble we should take all of the sets on at most d elements and then some subset of the elements on $d+1$ elements. In choosing the elements of size $d+1$ there are perhaps two main ways we could do it: prioritize sets which use a given element i as much as possible, or prioritize sets which avoid a given element i as much as possible. That is, we should use either lex or colex order. After a bit of experimenting one is led to believe that lex is the better ordering.

With all this motivation in mind, we define the *simplicial order* $<_S$ on $2^{\mathbb{N}}$ by having $F = \{a_1, \ldots, a_r\} <_S F' = \{b_1, \ldots, b_{r'}\}$ if either $|F| < |F'|$ or if $|F| = |F'|$ and $F <_L F'$. We let $\mathcal{S}(n, s)$ denote the first s elements of $2^{[n]}$ in simplicial order, and slightly more generally we define $\mathcal{S}(X, s)$ for any $X \subseteq [n]$ to be the first s elements of 2^X in simplicial order. For example, the elements of $2^{[4]}$ in simplicial order are:

$$\emptyset, 1, 2, 3, 4, 12, 13, 14, 23, 24, 34, 123, 124, 134, 234, 1234.$$

Our main result for this subsection shows that initial segments of this simplicial order solves the isoperimetric problem.

Theorem 3.12 (Harper's Theorem)**.** *If G is the n-dimensional hypercube graph with vertex set $2^{[n]}$, then any $V \subseteq V(G)$ has $|N(V)| \geq |N(\mathcal{S}(n, |V|))|$.*

Harper's Theorem is quite powerful, and in particular quickly implies the Kruskal-Katona Theorem.

Sketch of Proof of Kruskal-Katona Theorem. For convenience we prove the upper shadow version of Kruskal-Katona. Let $\mathcal{F} \subseteq \binom{[n]}{r}$ and define $\mathcal{F}' = \mathcal{F} \bigcup_{i=0}^{r-1} \binom{[n]}{i}$. In the language of Harper's Theorem we have $N(\mathcal{F}') = \partial^{r+1}(\mathcal{F}) \cup (\binom{[n]}{r} \setminus \mathcal{F})$, and hence if $s = |\mathcal{F}|$ we have by Harper's Theorem

$$|\partial^{r+1}(\mathcal{F})| + \binom{n}{r} - s = |N(\mathcal{F}')| \geq |N(\mathcal{S}(n, s + \sum_{i=0}^{r-1} \binom{n}{i}))| = |\partial^{r+1}(\mathcal{L}(n, r, s))| + \binom{n}{r} - s,$$

with this last step using that $N(\mathcal{S}(n, s + \sum_{i=0}^{r-1} \binom{n}{i}))$ consists of all the $\binom{n}{r} - s$ elements of size r which $\mathcal{S}(n, s + \sum_{i=0}^{r-1} \binom{n}{i}))$ does not contain together with the upper shadow of the elements of size r which $\mathcal{S}(n, s + \sum_{i=0}^{r-1} \binom{n}{i}))$ does contain, which is exactly the first s elements in the lex ordering. Rearranging gives the desired bound. $\square$

One can also use Harper's Theorem to deduce Katona's Union Theorem in a similar way to how we used Kruskal-Katona to deduce Erdős-Ko-Rado; we leave this as an exercise.

With regards to proving Harper's Theorem, this seems to be a difficult task since, in particular, it is not clear what exactly our theoretical lower bound $|N(\mathcal{S}(n, s))|$ should be. In situations like this the following is useful to keep in mind.

Mantra 3. If a problem has an extremal construction with a "simple" structure but a complicated size, then compression arguments might be useful for proving this is optimal.

Indeed, in situations such as this one can hope to compress a given object $\mathcal{F}$ into a simpler object $\mathcal{F}'$ of the same size which one can compare to the extremal construction without having to do any real explicit computations. For example, in extremal graph theory the number of edges in the Turán graph is somewhat complicated to write down, but by using Zykov symmeterization we can reduce the problem to simply determining the multipartite graph with the maximum number of edges.

Motivated by this, we prove Harper's Theorem by a compression argument. As mentioned at the start of this chapter, one has to carefully balance how powerful our choice of compression operation is, and for this argument it turns out we will want to choose a rather strong one. Indeed, while we do not take the most extreme operation of "replace the family by an extremal example", we will do the next most extreme thing and "split the family in half and replace each half by an extremal example."

Before going into the details, we would like to warn the reader that writing down the precise notation to prove Harper's Theorem is a bit cumbersome compared to the rather simple ideas at the heart of the argument. As such, the proof is probably best seen drawn on a board rather than read in notes like this, and for this we highly recommend a beautiful video lecture by Imre Leader which proves Harper's Theorem with essentially the same approach we describe below.

Back to the ugly details: for $\mathcal{F} \subseteq 2^{[n]}$ and $i \in [n]$, we recall that $\mathcal{F}(i), \mathcal{F}(\bar{i})$ are the subsets of $\mathcal{F}$ which do and do not contain i, respectively, and geometrically we can picture these subfamilies by looking at the two halves of the Boolean cube split based on whether element i is in a set or not. We then define

$$C_i(\mathcal{F}) := \mathcal{S}([n] \setminus \{i\}, |\mathcal{F}(\bar{i})|) \cup \{S \cup \{i\} : S \in \mathcal{S}([n] \setminus \{i\}, |\mathcal{F}(i)|)\}.$$

That is, we replace the elements $\mathcal{F}(\bar{i})$ with an equal sized initial segment of the simplicial order for $[n] \setminus \{i\}$ and we similarly replace the elements of $\mathcal{F}(i)$ with an initial segment of $[n] \setminus \{i\}$ and then add i back to each of these elements, which in particular makes the above a disjoint union of sets.

With our compression operation defined, we next want to verify that it maintains the properties which we care about.

Lemma 3.13. *If $\mathcal{F} \subseteq 2^{[n]}$ and if Harper's Theorem is true for $n-1$, then $|C_i(\mathcal{F})| = |\mathcal{F}|$ and $|N[C_i(\mathcal{F})]| \le |N[\mathcal{F}]|$ for all $i \in [n]$.*

Sketch of Proof. That $|C_i(\mathcal{F})| = |\mathcal{F}|$ is straightforward to verify. For some slight ease in notation we prove this only when $i = n$. Our goal will be to show that the number of elements of $N[C_n(\mathcal{F})]$ which do not contain n is at most the number of elements of $N[\mathcal{F}]$ which do not contain i, and also that the number of elements of $N[C_n(\mathcal{F})]$ which contain n is at most the number of elements of $N[\mathcal{F}]$ which contain n, from which the bound follows. For ease of notation we only do this first part, the other part being entirely analogous.

Observe that $\mathcal{F}(\overline{n}) \subseteq 2^{[n-1]}$, and hence we can consider it as a subset of the $(n-1)$-dimensional hypercube and we let $N'[\mathcal{F}(\overline{n})]$ denote its closed neighborhood within this smaller hypercube.

With this we observe that the set of elements of $N[\mathcal{F}]$ which do not contain n equals

$$N'[\mathcal{F}(\overline{n})] \cup \mathcal{F}_L(n),$$

essentially because an element $F \subseteq [n-1]$ is in $N[\mathcal{F}]$ if and only if it is either a neighbor of something which does not contain n or $F \cup \{n\} \in \mathcal{F}$ which exactly says $F \in \mathcal{F}_L(n)$. One can also interpret this claim geometrically as saying if we look at the half of the hypercube not using n, then either F is in the neighborhood of something in this half or the element above F is present.

For ease of notation let $\mathcal{C} = C_n(\mathcal{F})$. Exactly the same reasoning as above says that the elements of $N[\mathcal{C}]$ not containing n equals

$$N'[\mathcal{C}(\overline{n})] \cup \mathcal{C}_L(n).$$

Observe that by construction,

$$|\mathcal{C}_L(n)| = |\mathcal{S}([n-1], |\mathcal{F}(n)|)| = |\mathcal{F}_L(n)|,$$

and by assumption of Harper's Theorem we have

$$|N'[\mathcal{C}(\overline{n})] = |N'[\mathcal{S}([n-1], |\mathcal{F}(\overline{n})|)]| \leq |N'[\mathcal{F}(\overline{n})]|.$$

The last key thing we observe is the following.

Claim 3.14. *Either $N'[\mathcal{C}(\overline{n})] \subseteq \mathcal{C}_L(n)$ or $\mathcal{C}_L(n) \subseteq \mathcal{C}(\overline{n})$.*

Proof. Observe that $\mathcal{C}_L(n)$ is an initial segment in the simplicial order for $[n-1]$ and that $N'[\mathcal{C}(\overline{n})]$ is the neighborhood of an initial segment in the simplicial order for $[n-1]$. It is straightforward to check that neighborhoods of initial segments continue to be initial segments, and therefore one of the two initial segments $N'[\mathcal{C}(\overline{n})]$ or $\mathcal{C}_L(n)$ must contain the other. $\square$

In total this implies that the size of $N[\mathcal{C}](\overline{n})$, i.e. the number of elements of this neighborhood not containing n satisfies

$$\begin{aligned}|N[\mathcal{C}](\overline{n})| = |N'[\mathcal{C}(\overline{n}) \cup \mathcal{C}_L(n)| &= \max\{|N'[\mathcal{C}(\overline{n})]|, |\mathcal{C}_L(n)|\} \\ &\leq \max\{|N'[\mathcal{F}(\overline{n})]|, |\mathcal{F}_L(n)|\} \leq |N'[\mathcal{F}(\overline{n})] \cup \mathcal{F}_L(n)| = |N[\mathcal{F}](\overline{n})|,\end{aligned}$$

proving the result. $\square$

We next want to prove that this sequence of compression operations eventually terminates to a stable structure. To this end, we say a family $\mathcal{F}$ is *simplicially compressed* if $C_i(\mathcal{F}) = \mathcal{F}$ for all i.

Lemma 3.15. *For every family $\mathcal{F}_1 \subseteq 2^{[n]}$ there is a sequence of integers $i_1, \ldots, i_t$ such that if we define $\mathcal{F}_{k+1} = C_{i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is simplicially compressed.*

Sketch of Proof. Iteratively if there exists some i such that $C_i(\mathcal{F}_k) \neq \mathcal{F}_k$ then we take $i_k = i$. If we let $\sigma(F)$ denote the position of F in the simplicial order of $2^{[n]}$, then one can prove that the quantity

$$\sum_{F \in \mathcal{F}_k} \sigma(F)$$

is decreasing with k, and as such this process must terminate at some point. $\square$

With all the above established it remains only to solve the problem for simplicially compressed families. Because our compression operations are so powerful, there are relatively few cases which we need to consider.

Lemma 3.16. *Let $F_1, F_2, \ldots, F_{2^n}$ be the simplicial ordering of $2^{[n]}$. If $\mathcal{F} \subseteq 2^{[n]}$ is simplicially compressed then there exists some integer s such that either $\mathcal{F} = \mathcal{S}(n,s)$ or $\mathcal{F} = \mathcal{S}(n,s) \setminus \{F_{s-1}\}$ and $F_s = [n] \setminus F_{s-1}$.*

Proof. If $\mathcal{F}$ is an initial segment in the simplicial ordering then we are done. Otherwise this implies that there exists some $F <_S F'$ with $F \notin \mathcal{F}$ and $F' \in \mathcal{F}$.

Claim 3.17. *Any such F, F' satisfy $F' = [n] \setminus F$.*

Proof. Assume for contradiction that there was some $i \in F \cap F'$. The fact that $\mathcal{F}$ is simplicially compressed implies that all the elements of $\mathcal{F}$ containing i form an initial segment in the simplical order, so $F <_S F'$ and $F' \in \mathcal{F}$ would imply $F \in \mathcal{F}$, a contradiction to our assumption. A similar holds for $i \notin F \cup F'$, so we find that every i is in exactly one of the set F, F', giving the claim. □

In total this claim implies that for every $F' \in \mathcal{F}$ the only element smaller in simplicial order than it which can be missing from $\mathcal{F}$ is $[n] \setminus F'$, so if F_s is a largest element in simplicial order in $\mathcal{F}$ then we find that the only earlier element $\mathcal{F}$ can be missing is $F = [n] \setminus F_s$. But now if $F \neq F_{s-1}$ then the claim applied with $F' = F_{s-1}$ would imply $F = [n] \setminus F_{s-1}$, a contradiction to the above, so we must have $F = F_{s-1}$, giving the second case of the lemma. □

We now have all we need to prove Harper's Theorem.

Proof of Harper's Theorem. We prove the result by induction on n, the case $n = 1$ being trivial. Assume we have proven the result up to some value n and consider $\mathcal{F} \subseteq 2^{[n]}$. By Lemmas 3.15 and 3.13 we find that there exists some simplicially stable $\mathcal{F}'$ with $|\mathcal{F}| = |\mathcal{F}'|$ and $|N[\mathcal{F}]| \geq |N[\mathcal{F}']|$, so it suffices to prove $|N[\mathcal{F}']| \geq |N[\mathcal{S}(n, |\mathcal{F}'|)]|$. This is certainly true if $\mathcal{F}'$ is an initial segment in the simplicial order. Otherwise Lemma 3.16 implies that $\mathcal{F}' = \mathcal{S}(n,s) \setminus \{F_{s-1}\}$ and $F_{s-1} = [n] \setminus F_s$, so it remains to do a little bit of case analysis.

First consider the case that $1 \in F_{s-1}$. The simplicial ordering requires F_s to try and keep 1 if it is at all possible, so the fact that $1 \notin F_s = [n] \setminus F_{s-1}$ is only possible if $F_{s-1} = \{1, r, r+1, \ldots, n\}$ for some $r > 2$ and hence $F_s = \{2, 3, \ldots, r-1\}$. Again by the simplicial ordering in this case we must have $r - 2 = |F_s| = |F_{s-1}| = n - r + 2$. In total then this case can only happen if $n = 2r - 4$ for some r and its largest simplicial element is $\{2, 3, \ldots, r-1\}$, meaning we must have

$$\mathcal{F}' = \binom{[n]}{< r-2} \cup \{F \in \binom{[n]}{r-2} : 1 \in F\} \cup \{\{2, 3, \ldots, r-1\}\} \setminus \{\{1, r, r+1, \ldots, n = 2r-4\}\}.$$

This in turn implies that

$$\mathcal{S}(n, |\mathcal{F}'|) = \binom{[n]}{< r-2} \cup \{F \in \binom{[n]}{r-2} : 1 \in F\},$$

and it is straightforward to check that the size of this neighborhood is no larger than that of $\mathcal{F}'$, proving the result in this case.

Now consider the case $1 \notin F_{s-1}$. Again because the next element in the simplicial order F_s contains 1 this situation is only possible if $F_{s-1} = \{r, r+1, \ldots, n\}$ for some r and $F_s = \{1, 2, \ldots, n-r+2\}$ in which case $\mathcal{S}(n, |\mathcal{F}'|)$ is simply $\binom{[n]}{n-r+1}$ and again one can check that this works out. $\square$

3.4 Exercises

1. Prove that if $\mathcal{F} \subseteq 2^{[n]}$ is t-intersecting, then the family $\mathcal{F}' = \{[n] \setminus F : F \in \mathcal{F}\}$ is t-union, i.e. that any $F, F' \in \mathcal{F}'$ have
$$|F \cup F'| \leq n - t$$
[1].

2. Here we explore properties of $\mathcal{F} \subseteq 2^{[n]}$ which are preserved under the shifting operation $S_{j \to i}$ for integers j, i.

   (a) Prove that if $\mathcal{F}$ is t-intersecting then so is $S_{j \to i}(\mathcal{F})$ [1+].

   (b) Prove that if $\mathcal{F}$ is t-union then so is $S_{j \to i}(\mathcal{F})$ [1+].

   (c) Prove that for all integers t we have $\partial_t(S_{j \to i}(\mathcal{F})) \subseteq S_{j \to i}(\partial_t(\mathcal{F}))$ [2].

3. For all integers $r \geq t \geq 2$, find a t-intersecting set system $\mathcal{F} \subseteq \binom{2r+1}{r}$ with $|\mathcal{F}| > \binom{2r+1-t}{r-t}$ [2+].

4. We say that an intersecting family $\mathcal{F} \subseteq \binom{[n]}{r}$ is *non-trivial* if $\mathcal{F}$ is not a subsystem of a star, i.e. if there does not exist an element $i \in [n]$ with $i \in F$ for all $F \in \mathcal{F}$.

   (a) Prove that for any non-trivial intersecting family $\mathcal{F}_1 \subseteq \binom{[n]}{r}$ there exists a sequence of integer pairs $(i_1, j_1), \ldots, (i_t, j_t)$ with $i_k < j_k$ for all k such that if we define $\mathcal{F}_{k+1} = S_{j_k \to i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is shifted and non-trivial intersecting [2+].

   (b) (Hilton-Milner) Prove that if $\mathcal{F} \subseteq \binom{[n]}{r}$ is a non-trivial intersecting family and $n \geq 2r$, then $|\mathcal{F}| \leq \binom{n-1}{r-1} - \binom{n-1-r}{r-1} + 1$. Note that this bound is of the form $O(n^{r-2})$ and hence is substantially smaller than the Erdős-Ko-Rado bound of $\binom{n-1}{r-1} = O(n^{r-1})$ [3-].

   (c) Construct a non-trivial intersecting family $\mathcal{F} \subseteq \binom{[n]}{r}$ when $n \geq 2r$ which satisfies $|\mathcal{F}| = \binom{n-1}{r-1} - \binom{n-1-r}{r-1} + 1$ [2-].

* * *

5. Derive the Kruskal-Katona for Upper Shadows Theorem from the Kruskal-Katona Theorem [2-].

6. Given a graph G, let $\kappa(G)$ denote the total number of cliques of G, i.e. this is the number of K_1's of G plus the number of K_2's of G, and so on. For integers m, n with $m = \binom{k}{2}$ for some integer $k \leq n$, determine the largest possible value of $\kappa(G)$ amongst n-vertex graphs with m edges [2+].

7. In this exercise we show how to use Harper's Theorem to prove Katona's Union Theorem.

   (a) Prove that if $\mathcal{F} \subseteq 2^{[n]}$ satisfies $|\mathcal{F}| = \sum_{i=0}^{d} \binom{n}{i}$ for some $0 \leq d \leq n-1$ then $|N(\mathcal{F})| \geq \sum_{i=0}^{d+1} \binom{n}{i}$ [2-].

   (b) For $\mathcal{F} \subseteq 2^{[n]}$ and an integer t, define $N_{\leq t}(\mathcal{F})$ to be the collection of all sets $F \subseteq [n]$ such that there exists $F' \in \mathcal{F}$ whose symmetric difference with F satisfies $|F \Delta F'| \leq t$; equivalently, $N_{\leq t}(\mathcal{F})$ is obtained by iteratively applying N to $\mathcal{F}$ a total of t times. Prove that if $\mathcal{F} \subseteq 2^{[n]}$ satisfies $|\mathcal{F}| = \sum_{i=0}^{d} \binom{n}{i}$ for some $0 \leq d \leq n-t$ then $|N(\mathcal{F})| \geq \sum_{i=0}^{d+t} \binom{n}{i}$ [1].

   (c) Deduce Katona's union theorem (or equivalently its complementary intersecting version) [2+].

* * *

8. The shifting operator $S_{j \to i}$ is perhaps the most natural and simplest compression operation for problems involving uniform set systems. For the remainder of the exercise we study what is perhaps the most natural and simplest compression operation for non-uniform set systems. Given a family $\mathcal{F} \subseteq 2^{[n]}$ and an element $i \in [n]$, we define for each $F \in \mathcal{F}$ the *squashed edge*

$$s_{-i}(F; \mathcal{F}) = \begin{cases} F \setminus \{i\} & i \in F,\ F \setminus \{i\} \notin \mathcal{F}, \\ F & \textit{otherwise}, \end{cases}$$

   and we define the squashed family

$$s_{-i}(\mathcal{F}) = \{s_{-i}(F; \mathcal{F}) : F \in \mathcal{F}\}.$$

   That is, $s_{-i}(\mathcal{F})$ by iteratively going through each edge and removing i if this does not create a new element and otherwise it keeps it the same. This operation is perhaps most naturally viewed from the perspective of thinking of $\mathcal{F}$ as elements of a Boolean cube: the operation $s_{-i}(\mathcal{F})$ can be viewed as moving the Boolean cube so that the $x_i = 0$ face lies flat on the ground and then literally trying to squash down the elements on the $x_i = 1$ face down, with the elements staying put precisely when the vertex below them is already occupied (i.e. when $e \setminus \{i\} \in \mathcal{F}$).

   To begin our exploration of squashing, we start by establishing what its stable structures are. To this end, we say that a set system $\mathcal{F}$ is a *downset* (sometimes also referred to as *hereditary* or a *simplicial complex*) if for every $F \in \mathcal{F}$ and $F' \subseteq F$ we have $F' \in \mathcal{F}$. Prove that for every family $\mathcal{F}_1 \subseteq 2^{[n]}$ there is some finite sequence of integers $i_1, \ldots, i_t$ such that if we define $\mathcal{F}_{k+1} = s_{-i_k}(\mathcal{F}_k)$ then $\mathcal{F}_{t+1}$ is a downset [2-].

9. Prove that if $\mathcal{F} \subseteq 2^{[n]}$ and $i \in [n]$ then $|N(\mathcal{F})| \geq |N(s_{-i}(\mathcal{F}))|$ [2].

   As an aside, this exercise implies that to prove Harper's Theorem it suffices to do so for downsets, but this is seemingly just as hard as proving the result for arbitrary families. This shows that squashing on its own is too weak of a compression operation to prove Harper's Theorem, whereas our operations C_i were strong enough to do this.

10. We say a family $\mathcal{F}$ has diameter d if $\max_{F,F' \in \mathcal{F}} |F \Delta F'| = d$ where here $F \Delta F'$ denotes the symmetric difference, and we let $\mathrm{diam}(\mathcal{F})$ denote the diameter of $\mathcal{F}$. Prove that $\mathrm{diam}(s_{-i}(\mathcal{F})) \leq \mathrm{diam}(\mathcal{F})$ for all families $\mathcal{F}$ and integers i [2-].

11. Kleitman's Diameter Theorem states that if $\mathcal{F} \subseteq 2^{[n]}$ satisfies $\mathrm{diam}(\mathcal{F}) \leq d$ for some $n > d > 0$ and if $d = 2h$ for some integer h then

$$|\mathcal{F}| \leq \sum_{i=0}^{h} \binom{n}{i},$$

   and if $d = 2h + 1$ for some integer h then

$$|\mathcal{F}| \leq 2 \sum_{i=0}^{h} \binom{n-1}{i}.$$

   (a) Prove Katona's Union Theorem using Kleitman's Diameter Theorem [1+].

   (b) Prove Kleitman's Diameter Theorem using Katona's Union Theorem [2].

4 Sunflowers and the Delta System Method

4.1 Introduction to Sunflowers

The first ever paper in extremal set theory is Sperner's Theorem. The second is a paper of Erdős and Rado from 1960 which proves a sort of Ramsey-esque result saying that large uniform set systems always contain a certain substructure known as a sunflower.

Definition 5. A set system $\mathcal{S} = \{S_1, \ldots, S_p\}$ is a *sunflower with* p *petals* if there exists a set K called the *kernel* such that $S_i \cap S_j = K$ for all $i \neq j$.

For example, in the 2-uniform setting of graphs sunflowers are either stars (if K is a single vertex) or matchings (if K is the empty set). It is an easy exercise to show that graphs with a large number of edges either contain a large star or a large matching, and the sunflower analog of this statement holds for uniform set systems in general.

Theorem 4.1 (Erdős-Rado Sunflower Lemma)**.** *If $\mathcal{F}$ is r-uniform and $|\mathcal{F}| > (p-1)^r r!$ then $\mathcal{F}$ contains a sunflower with p petals.*

Importantly (and somewhat remarkably) the size of the ground set of $\mathcal{F}$ does not appear in this statement, which implies that sunflowers are ubiquitous in large set systems.

Proof. We prove the result by induction on r, the $r = 1$ case being trivial. Assume we have proven the result up to solve value r and let $\mathcal{F}$ be an r-uniform set system with $|\mathcal{F}| > (p-1)^r r!$. If $\mathcal{F}$ contains a matching on p edges (i.e. p disjoint edges), then this defines a sunflower, so we can assume the largest matching of $\mathcal{F}$ has $m \leq p-1$ edges, say with these edges being $F_1, \ldots, F_m$. Crucially, we observe that because this is a largest matching, every $F \in \mathcal{F}$ must intersect one of these F_i edges, and in particular every $F \in \mathcal{F}$ must use at least one vertex in $\bigcup F_i$. Note that $|\bigcup F_i| = mr \leq (p-1)r$, so by the pigeonhole principle there is some $x \in \bigcup F_i$ whose degree in $\mathcal{F}$ is at least $\frac{1}{(p-1)r}|\mathcal{F}| > (p-1)^{r-1}(r-1)!$. This means that the link hypergraph $\mathcal{F}' = \{F - x : F \in \mathcal{F},\ x \in F\}$ is an $(r-1)$-uniform set system with size strictly greater than $(p-1)^{r-1}(r-1)!$, so inductively there exists a sunflower $\{S_1', \ldots, S_p'\}$ for $\mathcal{F}'$, say with kernel K'. But then the sets $S_1' \cup \{x\}, \ldots, S_p' \cup \{x\}$ are edges of $\mathcal{F}$ which have pairwise intersection $K' \cup \{x\}$, so this defines a sunflower with p petals for $\mathcal{F}$. □

Sunflowers show up in a number of combinatorics problems (as we discuss below) as well as in various computer science contexts, and because of this there is a great deal of interest in determining better bounds for when sunflowers are guaranteed to exist. A basic construction which we leave as an exercise shows that Theorem 4.1 is not too far from the truth.

Lemma 4.2. *For all $r, p \geq 1$ there exists an r-uniform set system $\mathcal{F}$ with $|\mathcal{F}| = (p-1)^r$ which does not contain a sunflower with p petals.*

It is generally believed that this latter bound should be closer to the truth.

Conjecture 4.3 (Erdős-Rado Sunflower Conjecture)**.** *For all p there exists some $C = C(p)$ such that any r-uniform set system $\mathcal{F}$ with $|\mathcal{F}| > C^r$ contains a sunflower with p petals.*

The bound of Theorem 4.1 is roughly of the form r^r. After a lot of partial progress there was a major breakthrough for this problem which brought the bound to roughly $(\log r)^r$. This breakthrough relies on a rather general probabilistic tool which we discuss now.

For a set system $\mathcal{F}$ and a set of vertices S, let $\deg(S)$ denote the number of edges of $\mathcal{F}$ which contain S. We say $\mathcal{F}$ is *q-spread* for some $0 \le q \le 1$ if $\deg(S) \le q^{|S|}|\mathcal{F}|$ for every set of vertices S. A set system $\mathcal{F}$ being spread implies its edges are not clumped together on a small set of vertices, with sets $S = \{v\}$ in particular showing that no vertex is in more than q-proportion of the edges of $\mathcal{F}$.

Theorem 4.4 (Spreadness Theorem)**.** *There exists some absolute constant K_0 such that the following holds: if $\mathcal{F}$ is a q-spread set system with ground set V such that every edge of $\mathcal{F}$ has size at most r and if $C \ge K_0$ is real, then the random set $W \subseteq V$ chosen uniformly at random amongst subsets of V of size $Cq\log r \cdot |V|$ satisfies*

$$\Pr[W \textit{ contains an edge of } \mathcal{F}] \ge 1 - 8C^{-1}.$$

That is, for spread set systems a random set of not too large a size likely contains an edge of $\mathcal{F}$. This theorem was proven by Frankston, Kahn, Narayanan, and Park building on ideas of Alweiss, Lovett, Wu, and Zhang who proved the first sunflower bound of the form $(\log r)^r$. A more optimized version of their argument was observed by Bell, Chueluecha, and Warkne, giving the following result which remains the best known bound for the problem.

Theorem 4.5. *There exists an absolute constant $C > 0$ such that if $\mathcal{F}$ is an r-graph with at least $(Cp\log r)^r$ edges, then $\mathcal{F}$ contains a sunflower with p petals.*

Proof. We prove this by induction on r, the $r = 1$ case being trivial. Let $\mathcal{F}$ be an r-graph with at least $(Cp\log r)^r$ edges. If $\mathcal{F}$ is not q-spread with $q = (Cp\log r)^{-1}$, then there exists some $S \subseteq V$ such that

$$\deg(S) > q^{|S|}|\mathcal{F}| \ge (Cp\log r)^{r-|S|}.$$

We note that this first inequality can only hold if S is non-empty. This inequality means that the link hypergraph $\mathcal{F}_S = \{F \setminus S : F \in \mathcal{F}, S \subseteq F\}$ has size at least $(Cp\log r)^{r-|S|} \ge (Cp\log(r-|S|))^{r-|S|}$. Since $\mathcal{F}_S$ is an $(r-|S|)$-uniform hypergraph, by induction $\mathcal{F}_S$ contains a sunflower with p edges $F_1 \setminus S, \ldots, F_p \setminus S \in \mathcal{F}_S$. It is not difficult to check that $F_1, \ldots, F_k \in \mathcal{F}$ forms a sunflower in $\mathcal{F}$. We conclude that any $\mathcal{F}$ with at least $(Cp\log r)^r$ edges which is not q-spread contains a sunflower with p petals, so from now on we may assume $\mathcal{F}$ is q-spread.

Possibly by adding isolated vertices to $\mathcal{F}$, we can assume that the size of the ground set V of $\mathcal{F}$ is a multiple of $2p$. Let $V_1, \ldots, V_{2p}$ be a uniform random partition of V such that each $V_i \subseteq V$ has size $(2p)^{-1}|V|$. This means that each V_i is a uniformly chosen set of V of size $(2p)^{-1}|V| = \frac{1}{2}C(\log r)q|V|$. Let 1_i be the indicator variable for the event that V_i contains an edge of $\mathcal{F}$. By Theorem 4.4, we have $\Pr[1_i = 1] \ge \frac{1}{2}$ provided C is sufficiently large. This means $\mathbb{E}[\sum 1_i] \ge p$, and hence there exists some partition $V_1, \ldots, V_{2p}$ such that $\sum 1_i \ge p$, which in particular means there exist p disjoint edges of $\mathcal{F}$. This is a sunflower in $\mathcal{F}$ with p edges, proving the result. $\square$

The Spreadness Theorem has some very short (but not so easy) proofs along with a wide number of applications. In order to keep our narrative focused we defer further discussion of this theorem to Section 15.

4.2 The Delta-System Method

It's maybe better to use eg e to denote edges of $\mathcal{F}$ and capital letters to denote subsets and/or edges of $\mathcal{J}$ to avoid confusion/overloading on F'''.

The Erdős-Rado sunflower lemma shows that sunflowers are ubiquitous in large set systems. This ubiquity can be exploited in a general method known as the Delta-system method, where the term "Delta-system" refers to the original name used by Erdős and Rado to refer to sunflowers but which has since fallen out of use. In particular, we will use this method to prove the Erdős-Ko-Rado Theorem for t-intersecting families, which we recall are families with $|F \cap F'| \geq t$ for all $F, F' \in \mathcal{F}$.

Theorem 4.6. *If $\mathcal{F} \subseteq \binom{[n]}{r}$ is t-intersecting and if n is sufficiently large in terms of r, then $|\mathcal{F}| \leq \binom{n-t}{r-t}$ with equality holding if and only if $\mathcal{F}$ consists of every edge containing some fixed set T of size t.*

Roughly speaking, the Delta-system method can be interpreted as any proof which uses sunflowers. A basic observation which often finds use in such proofs is the following due to Deza, Erdős, and Frankl [**?**].

Lemma 4.7. *If $\mathcal{F}$ is an r-uniform set system which contains a sunflower with $r+1$ petals and kernel K, then for every set of r vertices F there exists some $F' \in \mathcal{F}$ with $F \cap F' \subseteq K$.*

Proof. Let $S_1, \ldots, S_{r+1}$ be the edges of the sunflower. Since each of the sets $S_1 \setminus K, \ldots, S_{r+1} \setminus K$ are non-empty disjoint sets, some $S_i \setminus K$ set must be disjoint from F. Taking $F' = S_i$ gives the result Insert picture. □

An effective tool used in conjunction with this observation is Füredi's intersection semilattice lemma, which itself is a result proven using the Delta-system method. Its full statement is a little intimidating, so we start with a basic version of it.

Lemma 4.8 (Weak Intersection Semilattice Lemma)**.** *For all integers $r, p \geq 1$ there exists $c = c(r,p) > 0$ such that for every r-uniform set system $\mathcal{F}$, there exists a subgraph $\mathcal{F}' \subseteq \mathcal{F}$ with $|\mathcal{F}'| \geq c|\mathcal{F}|$ such that for all $F, F' \in \mathcal{F}'$, the set $F \cap F'$ is the kernel of a sunflower with p petals.*

That is, we can approximate $\mathcal{F}$ by a hypergraph $\mathcal{F}'$ such that any two edges of $\mathcal{F}'$ are petals of a large sunflower. This together with the t-intersecting Erdős-Ko-Rado Theorem quickly gives the following result due to Mubayi and Zhao [**?**].

Corollary 4.9 ([**?**])**.** *For all $r, p, t \geq 1$ there exists a constant $C = C(r,p)$ such that if $\mathcal{F} \subseteq \binom{[n]}{r}$ has $|\mathcal{F}| \geq Cn^{r-t-1}$, then $\mathcal{F}$ contains a sunflower with p petals and kernel K satisfying $|K| \leq t$.*

Note that when $t = r-1$ this states that $|\mathcal{F}| \geq C$ implies $\mathcal{F}$ contains a sunflower with p petals (with the additional kernel condition $|K| \leq r-1$ being trivially satisfied), and as such this recovers the Erdős-Rado sunflower lemma up to the exact value of C.

Proof. We may assume n is sufficiently large in terms of r, as otherwise one can trivially find a sufficiently large C for which this holds. Let $C = c^{-1}$ with c the constant from Lemma 4.8. It follows that if $\mathcal{F}' \subseteq \mathcal{F}$ is the family guaranteed by Lemma 4.8 then

$$|\mathcal{F}'| \geq c|\mathcal{F}| \geq n^{r-t-1} > \binom{n-t-1}{r-t-1}.$$

By Theorem 4.6 applied to $(t+1)$-intersecting hypergraphs, we find that $\mathcal{F}'$ must contain two edges F, F' which intersect in less than $t+1$ vertices. By Lemma 4.8 we have that $F \cap F'$ is the kernel of a sunflower with at least p petals, proving the result. $\square$

We now state the full intersection semilattice lemma, which requires a bit of notation. If $\mathcal{F}$ is an r-uniform set system with ground set V, then we say that a partition $V_1, \ldots, V_r$ is an r-partition of $\mathcal{F}$ if every edge of $\mathcal{F}$ intersects each V_i in exactly 1 vertex, and we say $\mathcal{F}$ is r-partite if an r-partition exists for $\mathcal{F}$. Given an r-partition $V_1, \ldots, V_r$ for $\mathcal{F}$ and a set of vertices S we define $\mathrm{proj}(S) := \{i : S \cap V_i \neq \emptyset\}$. That is, $\mathrm{proj}(S)$ records which parts the vertices of S are in. Given a set system $\mathcal{F}$ and a set of vertices K, define $\deg^*_{\mathcal{F}}(K)$ to be the largest integer d such that there exist edges $F_1, \ldots, F_d \in \mathcal{F}$ with $F_i \cap F_j = K$ for all $i \neq j$. In other words, $\deg^*_{\mathcal{F}}(K)$ is the size of the largest sunflower which contains K as its kernel.

Theorem 4.10 (Intersection Semilattice Lemma)**.** *For all r, p, there exists $c = c(r,p) > 0$ such that for every r-uniform set system $\mathcal{F}$, there exists an r-partite subgraph $\mathcal{F}' \subseteq \mathcal{F}$ with $|\mathcal{F}'| \geq c|\mathcal{F}|$ and a hypergraph $\mathcal{J} \subseteq 2^{[r]} \setminus [r]$ such that:*

(1) $\mathcal{J}$ is intersection closed, i.e. $I, J \in \mathcal{J}$ implies $I \cap J \in \mathcal{J}$.

(2) For every $F \in \mathcal{F}'$, $\{\mathrm{proj}(F \cap F') : F' \in \mathcal{F}' \setminus \{F\}\} = \mathcal{J}$.

(3) $\deg^*_{\mathcal{F}'}(F \cap F') \geq p$ for all $F, F' \in \mathcal{F}'$.

To be clear, $\mathcal{J} \subseteq 2^{[r]} \setminus [r]$ means $\mathcal{J}$ is a hypergraph on $[r]$ which does not contain the edge of size r. Observe that if we ignore (1) and (2) we get back Lemma 4.8.

Roughly speaking, this result says that we can approximate a large chunk of $\mathcal{F}$, namely $\mathcal{F}'$, by a small hypergraph $\mathcal{J}$ in the sense that for any edge $F \in \mathcal{F}'$, the hypergraph $\mathcal{J}$ tells you by (2) exactly how the other edges of $\mathcal{F}'$ can intersect F, and moreover (3) guarantees that each possible intersection occurs at least p times. For convenience we say that an $\mathcal{F}'$ as in the conclusion of this theorem is $(p, \mathcal{J})$-homogeneous.

We postpone proving this result and look at some applications instead. For $\mathcal{J} \subseteq 2^{[r]} \setminus [r]$, define the rank

$$\mathrm{rank}(\mathcal{J}) = \min\{|T| : T \subseteq [r],\ T \not\subseteq I\ \forall I \in \mathcal{J}\},$$

that is, this is the smallest integer t such that there exists a t-set not contained in an edge of $\mathcal{J}$. For example, $\mathrm{rank}(\mathcal{J}) > 1$ if and only if every vertex of $[r]$ is contained in an edge of $\mathcal{J}$.

Lemma 4.11. *If $\mathcal{F}'$ is an n-vertex $(p, \mathcal{J})$-homogeneous r-uniform set system, then $|\mathcal{F}'| \leq \binom{n}{\mathrm{rank}(\mathcal{J})}$.*

Note that p does not appear in this bound. Before looking at the proof, the reader may want to try proving this result for themselves when $\operatorname{rank}(\mathcal{J}) = 1$ in order to get a sense for the definitions.

Proof. Let $T \subseteq [r]$ be a set such that $|T| = \operatorname{rank}(\mathcal{J})$ and such that $T \not\subseteq J$ for all $J \in \mathcal{J}$. Given an edge $F \in \mathcal{F}'$, let $\phi(F) = F \cap \bigcup_{i \in T} V_i$. We claim that ϕ is injective. Indeed, if $\phi(F) = \phi(F')$, then $T \subseteq \operatorname{proj}(F \cap F') \in \mathcal{J}$, a contradiction to our assumption on T. Since ϕ maps edges of $\mathcal{F}'$ injectively to sets of size $\operatorname{rank}(\mathcal{J})$ of the ground set of $\mathcal{F}'$ we conclude the desired bound on $|\mathcal{F}'|$. $\square$

We can use this to give a of the Erdős-Ko-Rado theorem for t-intersecting hypergraphs.

Proof of Theorem 4.6. Apply Lemma 4.10 with $p = r+1$ and let $\mathcal{F}', \mathcal{J}$ be the resulting hypergraphs with $\bigcup V_i$ the r-partition of $\mathcal{F}'$. First note that if $\operatorname{rank}(\mathcal{J}) < r - t$, then by Lemma 4.11 we have

$$|\mathcal{F}| \le c^{-1}|\mathcal{F}'| \le c^{-1}\binom{n}{r-t-1} < \binom{n-t}{r-t},$$

where this last step holds for n sufficiently large in terms of $c = c(r, r+1)$. Thus if $|\mathcal{F}| \ge \binom{n-t}{r-t}$, we must have $\operatorname{rank}(\mathcal{J}) \ge r - t$.

Let S be a set with $|S| = \operatorname{rank}(\mathcal{J})$ such that no edge of $\mathcal{J}$ contains S, and let $T = [r] \setminus S$. By the above we may assume $|S| \ge r - t$, and hence $|T| \le t$. By definition of $\operatorname{rank}(\mathcal{J}) = |S|$, for every $i \in S$, there exists an edge $J_i \in \mathcal{J}$ such that $S \setminus \{i\} \subseteq J_i$. Note that $i \notin J_i$ by assumption of S not being contained in any edge of $\mathcal{J}$, which means $J := \bigcap_{i \in S} J_i \subseteq T$. Because $\mathcal{J}$ is intersection closed, we have $J \in \mathcal{J}$.

We claim that $|J| \ge t$. Indeed, by definition of $J \in \mathcal{J}$, there exist two edges of $\mathcal{F}' \subseteq \mathcal{F}$ whose intersection is exactly J, and by the t-intersecting property we must have $|J| \ge t$. Because $|T| \le t$ and $J \subseteq T$, we conclude that $J = T \in \mathcal{J}$.

Now let $F \in \mathcal{F}'$ and $K = F \cap \bigcup_{i \in T} V_i$, noting that $|K| = t$. Lemma 4.10 (2) and (3) combine to guarantee that there is a sunflower of $\mathcal{F}' \subseteq \mathcal{F}$ with at least $r+1$ petals and K as its kernel. This implies by Lemma 4.7 that for every edge $F' \in \mathcal{F}$, there exists an edge $F'' \in \mathcal{F}'$ which contains K and which is disjoint from $F' \setminus K$. Because $|F' \cap F''| \ge t$ we must have $K \subseteq F'$. In other words, every edge of $\mathcal{F}$ must contain the t-set K. This implies the result. $\square$

A more careful look at this proof shows that it in fact gives the following stability result.

Corollary 4.12. *For all r, t there exists a constant $c' = c'(r)$ such that if $\mathcal{F} \subseteq \binom{[n]}{r}$ is t-intersecting with $|\mathcal{F}| > c'\binom{n-t}{r-t}$, then there exists a set of size t which is contained in every edge of $\mathcal{F}$.*

A similar argument as above works for more general intersection problems. Given a set $L \subseteq \{0, 1, \ldots, r-1\}$, we say that a hypergraph $\mathcal{F}$ is an (n, r, L)-system if it's an n-vertex r-uniform set system with $|F \cap F'| \in L$ for all distinct $F, F' \in \mathcal{F}$. For example, $(n, r, \{t, t+1, \ldots, r\})$-systems are t-intersecting set systems. Little is known about how large (n, r, L)-systems can be for general L, but one can get effective bounds in terms of ranks. To this end, for any $L \subseteq \{0, 1, \ldots, r-1\}$, define

$$\operatorname{rank}(r, L) = \max_{\mathcal{J}} \operatorname{rank}(\mathcal{J}),$$

where the maximum ranges over all $\mathcal{J} \subseteq 2^{[r]} \setminus [r]$ which are intersection-closed with $|J| \in L$ for all $J \in \mathcal{J}$.

Theorem 4.13. *If $\mathcal{F}$ is an (n, r, L)-system, then*

$$|\mathcal{F}| = O(n^{\operatorname{rank}(r,L)}).$$

Proof. Let $\mathcal{F}', \mathcal{J}$ be as in Lemma 4.10. Observe that $\mathcal{J}$ is intersection closed and that $|J| \in L$ for all $J \in \mathcal{J}$ (since otherwise two edges of $\mathcal{F}' \subseteq \mathcal{F}$ would fail to have $|F \cap F'| \in L$). Thus letting c be the constant from Lemma 4.10, we have

$$|\mathcal{F}| \le c^{-1}|\mathcal{F}'| \le c^{-1}\binom{n}{\operatorname{rank}(\mathcal{J})} \le c^{-1}\binom{n}{\operatorname{rank}(r, L)},$$

where this second inequality used Lemma 4.11 and the last inequality used the definition of $\operatorname{rank}(r, L)$. We conclude the result. $\square$

It is conjectured by Frankl that for all r, L there exist (n, r, L)-systems of size $\omega(n^{\operatorname{rank}(r,L)-1})$. This is unknown in general, but see e.g. [**?**, Theorem 16.6] for a construction of size $\Omega(n^{1+1/(r-1)})$ whenever $\operatorname{rank}(r, L) \ge 2$.

One of the best general bounds on the size of (n, r, L)-systems is the Deza-Erdős-Frankl theorem, which states that such a system $\mathcal{F}$ satisfies

$$|\mathcal{F}| \le \prod_{\ell \in L} \frac{n-\ell}{r-\ell} = O(n^{|L|}),$$

provided n is sufficiently large. One can prove this using a variant of the Delta-system method, though we omit doing so here. Instead, we give an easy proof of the asymptotic result by utilizing the following.

Lemma 4.14. *Every $L \subseteq \{0, 1, \ldots, r-1\}$ satisfies* $\operatorname{rank}(r, L) \le |L|$.

Proof. We prove the result by induction on $|L|$, the case $|L| = 0$ being trivial. Assume we have proven the result up to some value of $|L|$. First consider the case $0 \in L$. Let $\mathcal{J}$ be an intersection closed hypergraph on $2^{[r]} \setminus [r]$ with $|J| \in L$ for all $J \in \mathcal{J}$ and $\operatorname{rank}(\mathcal{J}) = \operatorname{rank}(r, L)$. For any vertex $x \in [r]$, let $\mathcal{J}_x = \{J - x : x \in J \in \mathcal{J}\}$ be the link hypergraph. If $L' = \{\ell - 1 : \ell \in L,\ \ell \neq 0\}$, then we see that $\mathcal{J}_x$ is intersection closed with $|J| \in L'$ for all $J \in \mathcal{J}_x$. Because $|L'| = |L|-1$, our inductive hypothesis implies that $\operatorname{rank}(r-1, L') \le |L| - 1$, i.e. there exists some set T of size $|L| - 1$ in $[r] \setminus \{x\}$ which is not contained in any edge of $\mathcal{J}_x$, which implies $T \cup \{x\}$ is a set of size $|L|$ not contained in any edge of $\mathcal{J}$. We conclude $\operatorname{rank}(r, L) = \operatorname{rank}(\mathcal{J}) \le |L|$.

Now assume $0 \notin L$, and again let $\mathcal{J}$ be an intersection closed hypergraph on $2^{[r]} \setminus [r]$ with $|J| \in L$ for all $J \in \mathcal{J}$ with $\operatorname{rank}(\mathcal{J}) = \operatorname{rank}(r, L)$. Let $I = \bigcap_{J \in \mathcal{J}} J$. Note that by definition I is contained in every edge of $\mathcal{J}$, and by the intersection closed property we have $I \in \mathcal{J}$, and hence

$|I| \in L$. Define $L' = \{\ell - |I| : \ell \geq |I|\}$, and note that the link hypergraph $\mathcal{J}_I = \{J \setminus I : J \in \mathcal{J}\}$ has all of its edge sizes lying in L'. Since $|L'| \leq |L|$ and $0 \in L'$, the previous case implies $\text{rank}(\mathcal{J}_I) \leq |L|$, which implies there exists some set $J \subseteq [r] \setminus I$ of size L not contained in an edge of $\mathcal{J}_I$, and hence this set continues to not be contained in an edge of $\mathcal{J}$ (since every edge of $\mathcal{J}$ is the union of an edge of $\mathcal{J}_I$ with I). We conclude $\text{rank}(r, L) = \text{rank}(\mathcal{J}) \leq |L|$, proving the result. □

This together with the previous theorem immediately gives the following.

Corollary 4.15. *If $\mathcal{F}$ is an (n, r, L)-system, then*

$$|\mathcal{F}| = O(n^{|L|}).$$

It is not difficult to show that this result is tight if $L = \{0, 1, \ldots, t-1\}$.

Before moving on, we note that while all of our applications here came from extremal set theory, the intersection semilattice lemma has application to other areas of extremal combinatorics as well, and for this we refer the reader to a survey by Kupavskii which covers the delta system in far greater detail.

4.3 Proof of the Semilattice Intersection Lemma

It is a basic exercise in probabilistic combinatorics that every r-uniform set system $\mathcal{F}$ contains an r-partite subgraph $\mathcal{F}' \subseteq \mathcal{F}$ with $|\mathcal{F}'| \geq \frac{r!}{r^r}|\mathcal{F}|$. Thus to prove the result, it suffices to prove that if $\mathcal{F}$ has r-partition $\bigcup V_i$ then one can find a subgraph $\mathcal{F}' \subseteq \mathcal{F}$ and $\mathcal{J} \subseteq 2^{[r]} \setminus [r]$ such that $|\mathcal{F}'| \geq c(r, p)|\mathcal{F}|$ and

(1) $\mathcal{J}$ is intersection closed, i.e. $I, J \in \mathcal{J}$ implies $I \cap J \in \mathcal{J}$.

(2) For every $F \in \mathcal{F}'$, $\{\text{proj}(F \cap F') : F' \in \mathcal{F}' \setminus \{F\}\} = \mathcal{J}$.

(3) $\deg^*_{\mathcal{F}'}(F \cap F') \geq p$ for all $F, F' \in \mathcal{F}'$.

Actually, only the latter two properties are relevant for the proof.

Lemma 4.16. *If there exists $\mathcal{F}', \mathcal{J}$ satisfying (2) and (3) with $p \geq r+1$, then they automatically satisfy (1).*

Proof. Let $J_1, J_2 \in \mathcal{J}$. By (2) and (3) this means that for any edge $F \in \mathcal{F}'$, there exist edges F_1, F_2 such that F, F_i intersect exactly in the coordinates of J_i and that these intersections are kernels of sunflowers in $\mathcal{F}'$ on at least $r + 1$ petals. This implies there exists an edge $F' \in \mathcal{F}'$ which contains $F \cap F_1$ and which is otherwise disjoint from F_2, namely by taking F' to be an appropriate edge of the sunflower with core $F \cap F_1$ (or equivalently by using Lemma 4.7). With this $\text{proj}(F' \cap F_2) = J_1 \cap J_2$, so necessarily $J_1 \cap J_2 \in \mathcal{J}$. Insert picture. □

With this claim in mind, we only have to find $\mathcal{F}', \mathcal{J}$ satisfying (2) and (3) (this is immediate if $p \geq r+1$, and for all other values of p we can take $c(r,p) = c(r, r+1)$ and apply the $s = r+1$ result). We will do this by iteratively partitioning $\mathcal{F}$ which do not satisfy (2) or (3) into a finite number of "better" set systems. For this and the rest of the subsection, we define for a set system $\mathcal{F}$ the intersection pattern

$$\mathcal{I}(\mathcal{F}) = \{\mathrm{proj}(F \cap F') : F, F' \in \mathcal{F},\ F \neq F'\}.$$

Lemma 4.17. *For any r-partite r-graph $\mathcal{F}$, one can decompose $\mathcal{F}$ as $\mathcal{F} = \mathcal{F}_0 \cup \bigcup_{I \in \mathcal{I}(\mathcal{F})} \mathcal{F}_I$ such that $\mathcal{F}_0$ satisfies (2) and (3) for some $\mathcal{J}$, and such that for all I, there does not exist a set K with $\mathrm{proj}(K) = I$ and which is the kernel of a sunflower on at least p petals in $\mathcal{F}_I$.*

Proof. The idea of the proof is to start with $\mathcal{F}_0 = \mathcal{F}$ and then iteratively remove edges from this set until $\mathcal{F}_0$ satisfies (2) and (3), for which we observe the following.

Claim 4.18. *If a set system $\mathcal{F}'$ does not satisfy both (2) and (3) for any $\mathcal{J}$, then there exists an edge $F \in \mathcal{F}'$ and $K \subseteq F$ such that $\mathrm{proj}(K) \in \mathcal{I}(\mathcal{F}')$ but K is not the kernel of a sunflower with at least p petals in $\mathcal{F}'$.*

Proof. If (3) is not satisfied for some $F, F' \in \mathcal{F}$ then the claim follows by taking F and $K = F \cap F'$. If (2) is not satisfied for any $\mathcal{J}$ then it in particular is not satisfied for $\mathcal{J} = \mathcal{I}(\mathcal{F}')$. Because $\mathrm{proj}(F \cap F') \in \mathcal{J} = \mathcal{I}(\mathcal{F}')$ for all distinct F, F' by definition, the failure of (2) means there exists some $F \in \mathcal{F}'$ and $J \in \mathcal{J} = \mathcal{I}(\mathcal{F}')$ such that $\mathrm{proj}(F \cap F') \neq J$ for all $F' \in \mathcal{F}' \setminus \{F\}$. This means that if $K \subseteq F$ is the set with $\mathrm{proj}(K) = J$ then K is not the kernel of a sunflower of size at least 2, which in particular satisfies the conditions of the claim. $\square$

With this in mind, we consider the following procedure. Initially start with $\mathcal{F}_0 = \mathcal{F}$ and $\mathcal{F}_I = \emptyset$ for all $I \in \mathcal{I}(\mathcal{F})$. If at some point $\mathcal{F}_0$ satisfies (2) and (3) then we stop and output the current sets. Otherwise there exist K and $F \in \mathcal{F}_0$ as in the claim and we remove F from $\mathcal{F}_0$ and add it to $\mathcal{F}_{\mathrm{proj}(K)}$.

We claim that this procedure gives the desired result. Indeed, $\mathcal{F}_0$ satisfies (2) and (3) by construction. Assume for contradiction that there existed some I with $\mathcal{F}_I$ containing a sunflower on at least p petals with kernel K satisfying $\mathrm{proj}(K) = I$. Let F be the first edge of this sunflower that was added to $\mathcal{F}_I$ during the procedure. This implies that every edge of the sunflower was in $\mathcal{F}_0$ right before F was removed, i.e. that $\mathcal{F}_0$ contained a sunflower with at least p petals and kernel K. This contradicts us removing F from $\mathcal{F}_0$ and adding it to $\mathcal{F}_I$ at this step, so we conclude no such sunflower exists in any $\mathcal{F}_I$. $\square$

In a similar spirit to the above, we want to take any $\mathcal{F}_I$ from Lemma 4.17 and split this into "better" set systems.

Lemma 4.19. *Let $\mathcal{F}$ be an r-partite r-graph and I a set such that $I \neq \mathrm{proj}(K)$ for any K which is the kernel of a sunflower with at least p petals. Then one can decompose $\mathcal{F}$ as $\mathcal{F} = \mathcal{F}^1 \cup \cdots \mathcal{F}^{r(p-1)}$ such that $I \notin \mathcal{I}(\mathcal{F}^i)$ for all i.*

Proof. For each set of vertices K with $\mathrm{proj}(K) = I$, let $\mathcal{F}(K)$ be the link hypergraph defined by $\mathcal{F}(K) = \{F \setminus K : K \subseteq F \in H\}$.

Claim 4.20. *One can partition each $\mathcal{F}(K)$ into at most $r(p-1)$ intersecting families.*

Proof. Let $M_1, \ldots, M_t$ be a largest matching of $\mathcal{F}(K)$. Note that $t < p$, as otherwise $M_i \cup K$ would be edges of $\mathcal{F}$ forming a sunflower with p petals with kernel K. Because $M_1, \ldots, M_t$ is a largest matching, every edge of $\mathcal{F}(K)$ must contain one of the $(r - |K|)t \le r(p-1)$ vertices in $\bigcup M_i$. Partitioning $\mathcal{F}(K)$ based on which vertex each edge contains gives the claim. $\square$

Let $\mathcal{F}^1(K), \ldots, \mathcal{F}^{r(p-1)}(K)$ be the intersecting families guaranteed by this claim. Let $\mathcal{F}^i[K] = \{F' \cup K : F' \in \mathcal{F}^i(K)\}$ and let $\mathcal{F}^i = \bigcup_{K:\mathrm{proj}(K)=I} \mathcal{F}^i[K]$. Note that the $\mathcal{F}^i$ families decompose $\mathcal{F}$ since for each $F \in \mathcal{F}$ there exists a unique $K \subseteq F$ with $\mathrm{proj}(K)$, and hence $F \in \bigcup_i \mathcal{F}^i[K]$. Further note that if $F, F' \in \mathcal{F}^i$ then $K = F \cap F'$ can not have $\mathrm{proj}(K) = I$, as otherwise this would imply $F \setminus K, F' \setminus K \in \mathcal{F}^i(K)$ and hence that F, F' intersect in a vertex outside of K since $\mathcal{F}^i(K)$ is intersecting. This implies $I \notin \mathcal{I}(\mathcal{F}^i)$, giving the result. $\square$

We now put these results together to prove the semilattice lemma.

Sketch of Proof of Intersection Semilattice Lemma. As noted at the start of the subsection it suffices to prove the result for r-partite $\mathcal{F}$. First take the partition $\mathcal{F}_0 \cup \bigcup_{I \in \mathcal{I}(\mathcal{F})} \mathcal{F}_I$ from Lemma 4.17. If $\mathcal{F}_0$ is the largest set in this partition then $\mathcal{F}' = \mathcal{F}$ satisfies all the conditions with $|\mathcal{F}'| \ge 2^{-r}|\mathcal{F}|$ and we are done. Thus we can assume there is some $I \in \mathcal{I}(\mathcal{F})$ with $|\mathcal{F}_I| \ge 2^{-r}|\mathcal{F}|$. By Lemma 4.19 we can partition $\mathcal{F}_I$ into sets $\mathcal{F}^1, \ldots, \mathcal{F}^{r(p-1)}$, and by the pigeonhole principle we find $|\mathcal{F}^i| \ge \frac{1}{r(p-1)}|\mathcal{F}_I| \ge \frac{1}{rp}2^{-r}|\mathcal{F}|$ and that $\mathcal{I}(\mathcal{F}^i) \subsetneq \mathcal{I}(\mathcal{F})$. We now iteratively repeat this exact same procedure with $\mathcal{F}^i$ replaced by $\mathcal{F}$ and because the relationship $\mathcal{I}(\mathcal{F}^i) \subsetneq \mathcal{I}(\mathcal{F})$ can happen at most 2^r times we find that this must stop after a bounded number of steps. More formally one can claim that for any $\mathcal{F}$ we can find $\mathcal{F}'$ with $|\mathcal{F}'| \ge (rp2^r)^{|\mathcal{I}(\mathcal{F})|}$ and the approach above establishes the inductive step on $|\mathcal{I}(\mathcal{F})|$. $\square$

4.4 Other Hypergraph Approximations

The semi-lattice intersection lemma gives a way of "approximating" set systems $\mathcal{F}$ by the much simpler set system $\mathcal{J}$. There are a few other methods in the literature which achieve a similar goal.

Perhaps the most similar is the method of spread approximations developed by Kupavskii and Zhakaraov. While the Delta-system method gives that the links of each $F \in \mathcal{F}'$ contains large sunflowers, the spread approximation method gives the stronger condition that a random partition of the vertex set of links is likely to contain a large sunflower (which is a more robust condition). Moreover, the "error term" $\mathcal{F} \setminus \mathcal{F}'$ for spread approximations is typically much smaller than that of Theorem 4.10 (which only approximates a constant proportion of $\mathcal{F}$, which is in some sense necessary since it can only perfectly approximate r-partite r-graphs). On the other hand, the approximating hypergraph for spread approximations is more "complex", in the sense that it is not just a hypergraph on $[r]$ but all of $V(\mathcal{F})$, and moreover you end up losing the homogeneity property of Theorem 4.10 which says that every two edges have the same intersection pattern.

Another famous hypergraph approximation is the Junta method, which was developed by Dinur and Friedgut [?]. This roughly says that if $\mathcal{F}$ is an intersecting hypergraph, then there exists a hypergraph $\mathcal{J}$ on a small set of vertices J of $\mathcal{F}$ such that almost every edge $F \in \mathcal{F}$ has $F \cap J \in \mathcal{J}$. We omit going into this in detail and refer the reader to [?].

Finally, we emphasize a fundamental weakness in all of these approaches, which is that they only work when n is quite large. In many cases this is okay, but for some results like the t-intersecting Erdős-Ko-Rado theorem, it is of interest in nailing down the exact dependency on n. In cases such as these a more careful argument is needed.

4.5 Exercises

1. It is straightforward to see that any two sets form a sunflower with two petals. Prove that there exist an infinite set system $\mathcal{F}$ which contains no sunflower with three petals (Hint: by the Erdős-Rado sunflower lemma we know such an $\mathcal{F}$ can not be uniform) [1+].

2. Prove for all $r, p \geq 1$ that there exists an r-uniform set system $\mathcal{F}$ with $|\mathcal{F}| = (p-1)^r$ which does not contain a sunflower with p petals [2-].

3. Prove that if there exists a constant $C = C(p) \geq 1$ such that every r-uniform C^{-1}-spread set system with at least C^r edges contains a sunflower with p petals, then every r-uniform spread set system with at least C^r edges contains a sunflower with p petals. In particular, it suffices to prove the Erdős-Rado sunflower conjecture for spread set systems [2-].

4. A fundamental result of Deza's says that if $\mathcal{F}$ is an r-uniform set system with $|\mathcal{F}| \geq r^2 - r + 2$ and if ℓ is an integer such that $|F \cap F'| = \ell$ for all distinct $F, F' \in \mathcal{F}$, then $\mathcal{F}$ is a sunflower.

   (a) Prove this when $\ell = 1$ (Hint: prove that if $\mathcal{F}$ contains a sufficiently large sunflower, then $\mathcal{F}$ must be a sunflower) [2].

   (b) Prove that there exist infinitely many r such that there exist r-uniform $\mathcal{F}$ which are not sunflowers with $|\mathcal{F}| = r^2 - r + 1$ and $|F \cap F'| = 1$ for all distinct $F, F' \in \mathcal{F}$ [2+].

5. For each integer $n \geq 1$, let $\mathcal{F}_n$ denote the set system which encodes perfect matchings in the complete graph K_{2n}; that is, the ground set of $\mathcal{F}$ is $E(K_{2n})$ and some n elements $e_1, \ldots, e_n \in E(K_{2n})$ form a hyperedge in $\mathcal{F}$ if and only if $e_1, \ldots, e_n$ form a perfect matching in K_n.

   (a) Prove that there exists some $C > 0$ such that $\mathcal{F}_n$ is Cn^{-1}-spread for all $n \geq C$ [2+].

   (b) Let $G_{2n,m}$ denote the random graph on $[2n]$ chosen uniformly at random from all such graphs with exactly m edges. Conclude that there exists some $C' > 0$ such that if $m \geq C' n \log n$ then

$$\Pr[G_{2n,m} \text{ contains a perfect matching}] \geq .99$$

[1+].

* * *

6. Refine Lemma 4.11 by proving that if $\mathcal{F}'$ is an n-vertex $(p, \mathcal{J})$-homogeneous r-uniform set system and if $s := |[r] - \bigcup_{J \in \mathcal{J}} J| \geq 1$, then $|\mathcal{F}'| \leq n/s$ [2-].

5 Linear Algebra and Polynomial Methods

Roughly speaking, the *linear algebra method* in combinatorics works as follows:

1 Associate a "linear algebraic object" M to your problem (e.g. a matrix or a list of vectors).

2. Determine algebraic information about M (e.g. its rank, eigenvalues, eigenvectors),

3. Use this algebraic information to conclude something about your original problem.

The linear algebra method applies to a broad range of problems. We only scratch the surface here, and we refer the reader to books by Babai and Frankl [**?**] and by Matoušek [**?**] for a more thorough treatment of this versatile method.

5.1 Modular Intersections

We begin with a classical application of the linear algebra method: Oddtown.

Consider the following (somewhat whimsical) setup. The city of Oddtown has a number of clubs, each of which follows the following odd set of rules: each club must have an odd number of people, and every two distinct clubs must have an even number of people in common.

The main question now becomes: if Oddtown has n people, what's the maximum number of clubs it can have? Equivalently, if $\mathcal{F} = \{F_1, F_2, \ldots, F_m\} \subseteq 2^{[n]}$ is a set system such that $|F_i|$ is odd for all i and such that $|F_i \cap F_j|$ is even for all $i \neq j$, then what is the maximum size of $\mathcal{F}$?

A very simple construction is to take $F_i = \{i\}$ for all i, which trivially satisfies the stated conditions. However, it's far from the only construction. For example, if n is even one can also take each F_i to be either $\{i\}$ or $[n] \setminus \{i\}$, and there are many, many more constructions achieving a bound of n (in fact, there's close to 2^{n^2} non-isomorphic constructions due to Szegedy [**?**, Exercise 1.1.14]).

Given all of these constructions, it seems plausible that (1) the true answer is indeed n, and (2) proving this might be difficult (since we have to come up with an argument that somehow deals with all of these constructions in a unified way). Fortunately, the linear algebra method manages to give a unified approach for all of these constructions in an extremely elegant way. More generally, if a given problem has many distinct looking extremal constructions, then it is often the case that the linear algebra method can come in handy.

Theorem 5.1 (Oddtown). *Let $\mathcal{F} \subseteq 2^{[n]}$ be a set system such that $|F|$ is odd for all $F \in \mathcal{F}$ and such that $|F \cap F'|$ is even for all $F \neq F' \in \mathcal{F}$. Then $|\mathcal{F}| \leq n$.*

Proof. Given a set $F \subseteq [n]$, define its characteristic vector $\chi_F \in \mathbb{F}_2^n$ by having $(\chi_F)_i = 1$ if $i \in F$ and $(\chi_F)_i = 0$ otherwise. Note crucially that for any F, F', the dot product satisfies

$$\langle \chi_F, \chi_{F'} \rangle = |F \cap F'| \mod 2.$$

We claim that $\{\chi_F : F \in \mathcal{F}\}$ is a set of linearly independent vectors. Indeed, say we had

$$\sum_{F\in\mathcal{F}} \lambda_F \chi_F = 0.$$

Take any $F' \in \mathcal{F}$ and apply the dot product on both sides to get

$$\sum_{F\in\mathcal{F}} \lambda_F \langle \chi_F, \chi_{F'} \rangle = 0.$$

By the observation above and the hypothesis of the theorem, we see $\langle \chi_F, \chi_{F'} \rangle = 0$ if $F \neq F'$ and that $\langle \chi_{F'}, \chi_{F'} \rangle = 1$. Thus the above says $\lambda_{F'} = 0$, and as $F' \in \mathcal{F}$ was arbitrary, we conclude that these vectors are indeed linearly independent.

Since we have $|\mathcal{F}|$ linearly independent vectors in $\mathbb{F}_2^n$, we must have $|\mathcal{F}| \leq n$, giving the result.

□

Since that was so clean, let's give another (essentially equivalent) proof of this result using slightly different language, which in some situations might be simpler to think about.

Proof. Write $\mathcal{F} = \{F_1, \ldots, F_m\}$ and let M be the $m \times n$ matrix over $\mathbb{F}_2$ which has $M_{i,j} = 1$ if $j \in F_i$ and $M_{i,j} = 0$ otherwise. Let $L = MM^T$. We claim that L is the $m \times m$ identity matrix. Indeed, one can verify that $L_{i,j} \equiv |F_i \cap F_j| \mod 2$, so the hypothesis of the theorem gives this claim.

Using the general fact

$$\text{rank}(AB) \leq \text{rank}(A) \leq \#\text{columns of } A,$$

which is valid for any two matrices A, B for which AB makes sense; we see that we can apply this with $A = B^T = M$ to conclude that

$$m = \text{rank}(I_m) \leq \text{rank}(M) \leq n,$$

proving the result. □

While the proof of Oddtown is extremely nice, one might complain that the problem itself is rather ad hoc. Here we give a far reaching generalization of the Oddtown problem which, in addition to being nice on its own, has a number of important applications.

To this end, we adopt the following (non-standard) notation. For an integer p and a set $L \subseteq \{0, 1, \ldots, p-1\}$, we say that a family $\mathcal{F}$ is (p, L)-*modular intersecting* if $|F| \mod p \notin L$ for all $F \in \mathcal{F}$ and if $|F \cap F'| \mod p \in L$ for all $F \neq F' \in \mathcal{F}$.

For example, a family is $(2, \{0\})$-modular intersecting if and only if it follows the rules of Oddtown. By following a somewhat similar strategy as in Oddtown, we can prove the following.

Proposition 5.2. *For any prime p and $L \subseteq \{0, 1, \ldots, p-1\}$, if $\mathcal{F} \subseteq 2^{[n]}$ is (p, L)-modular intersecting then*

$$|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n+i-1}{i}.$$

Proof. Similar to before, we define the characteristic vector $\chi_F \in \mathbb{F}_p^n$ by having $(\chi_F)_i = 1$ if $i \in F$ and $(\chi_F)_i = 0$ otherwise, and we again observe that $\langle \chi_F, \chi_{F'} \rangle = |F \cap F'| \mod p$. Unfortunately we can't conclude that these vectors are linearly independent like we could in the Oddtown case, but we can do this if we generalize our vectors somewhat.

To this end, for each $F \in \mathcal{F}$, define the polynomial $p_F : \mathbb{F}_p^n \to \mathbb{F}_p$ by

$$p_F(\vec{x}) = \prod_{\ell \in L} (\langle \chi_F, \vec{x} \rangle - \ell).$$

Note that by hypothesis, we have $p_F(\chi_{F'}) = 0$ if $F \neq F' \in \mathcal{F}$ (since $\langle \chi_F, \chi_{F'} \rangle \equiv |F \cap F'| \equiv \ell \mod p$ for some $\ell \in L$) and that $p_F(\chi_F) \neq 0$ (since $|F| \not\equiv \ell \mod p$ for any $\ell \in L$).

We claim that the polynomials $\{p_F : F \in \mathcal{F}\}$ are linearly independent. Indeed, say we had

$$\sum_{F \in \mathcal{F}} \lambda_F p_F = 0.$$

By plugging in $\chi_{F'}$ for some $F' \in \mathcal{F}$ into both sides of this equality, the observation above implies $c_{F'} \lambda_{F'} = 0$ for some $c_{F'} \neq 0$, and hence $\lambda_{F'} = 0$, proving the claim.

Observe that each polynomial p_F has degree at most $|L|$. It is a basic exercise to show that the dimension of the space of polynomials in n variables with degree at most $|L|$ is equal to $\sum_{i=0}^{|L|} \binom{n+i-1}{i}$. Since we constructed a set of $|\mathcal{F}|$ linearly independent polynomials that lie in a space of dimension $\sum_{i=0}^{|L|} \binom{n+i-1}{i}$, we must have $|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n+i-1}{i}$, proving the result. $\square$

The bound of Proposition 5.2 is relatively close to tight. If $L = \{0, 1, \ldots, s-1\}$ then $\mathcal{F} = \binom{[n]}{s}$ is (p, L)-modular intersecting and has size $\binom{n}{s} \approx \sum_{i=0}^{s} \binom{n+i-1}{i}$. However, one can do better by refining our argument (though we emphasize that the present weaker bound of Proposition 5.2 is typically all one needs for applications). For this there are two somewhat standard approaches.

5.1.1 Refinement #1: Multilinearization

Towards refining this proof/bound, one might first turn to the simplest case of $p = 2$, $L = \{0\}$. Since this is just Oddtown, we know the answer here should be n, but the bound we get is $n+1$. By carefully analyzing the proof, one realizes that the one place our argument isn't sharp is when we argue about the dimension of the space spanned by the p_F polynomials. Indeed, we naively said that they lived in the space of degree at most 1 polynomials, but in fact they all lie in the smaller vector space spanned by just the monomials $\{x_i\}$, and this gives the optimal bound.

We are thus left with the problem of trying to find a smaller subspace for us to work in while achieving the same conclusion, and there are a couple of semi-standard ways of doing this. One way is to try and find polynomials which are more "efficient", i.e. of lower degree, for which our conclusion still holds. And indeed, if we look back through the proof of Proposition 5.2, we see that although our polynomials were defined to have codomain $\mathbb{F}_p^n$, we only fed them 0-1 vectors to show linear independence. As such, it would suffice to look at polynomials $\bar{p}_F$ which agree with the p_F polynomials on 0-1 vectors, and this can be done simply by replacing each

x_i^α in the expansion of p_F by x_i (i.e. by replacing p_F by a multilinear function). For example, if $p_F(\vec{x}) = 3x_1x_2^2 + 2x_1^3x_3^2$ then we would want to look at $\bar{p}_F(\vec{x}) = 3x_1x_2 + 2x_1x_3$. By using this we can prove the following.

Theorem 5.3. *For any prime p and $L \subseteq \{0, 1, \ldots, p-1\}$, if $\mathcal{F} \subseteq 2^{[n]}$ is (p, L)-modular intersecting then*

$$|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n}{i}.$$

This theorem was originally proven by Deza, Frankl, and Singhi [?], with the following simpler proof due to Alon, Babai, and Suzuki [?].

Proof. Define p_F as before, and define the multilinear polynomial $\bar{p}_F$ by replacing each x_i^α in the expansion of p_F by x_i. Note that $p_F(\chi_{F'}) = \bar{p}_F(\chi_{F'})$ for all F', so by the same proof as before we conclude that the $\bar{p}_F$ polynomials are linearly independent. Moreover, these linearly independent polynomials all lie in the space of multilinear polynomials in n variables with degree at most $|L|$, which is a space of dimension $\sum_{i=0}^{|L|} \binom{n}{i}$. We thus must have $|\mathcal{F}| \leq \sum_{i=0}^{|L|} \binom{n}{i}$, proving the result. $\square$

5.1.2 Refinement #1.5: Shrinking the Vector Space

The multilinearization refinement essentially makes use of the fact that the set of vectors p_F which are naturally defined to live in a given vector space V in fact live in a smaller (but somewhat less natural) vector space $V' \subseteq V$. While mulitlinearization is perhaps the most standard way to obtain such a V', there are other problems for which this idea can be used. To demonstrate this we consider an application to discrete geometry, which is another area where linear algebraic approaches can be quite useful.

We say that a set of points $S \subseteq \mathbb{R}^n$ is an s-distance set if

$$|\{\|\vec{x} - \vec{y}\| : \vec{x}, \vec{y} \in S,\ \vec{x} \neq \vec{y}\}| \leq s;$$

that is, if there at most s distinct distances between the points in S. For example, it is not too difficult to show that if $S \subseteq \mathbb{R}^n$ is a 1-distance set then $|S| \leq n+1$ with this being best possible by considering a regular simplex (e.g. a line in $\mathbb{R}$, a triangle in $\mathbb{R}^2$, and a tetrahedron in $\mathbb{R}^3$). However, even for $s = 2$ the answer is unknown. We use a linear algebra approach of Larman, Rogers, and Seidel to get a pretty good bound for this.

Theorem 5.4. *If $S \subseteq \mathbb{R}^n$ is a 2-distance set then $|S| \leq \frac{1}{2}(n+4)(n+1)$.*

It is an exercise to show that there exist 2-distance sets with $|S| = \binom{n+1}{2}$, so asymptotically this bound is best possible.

Proof. Let S be a 2-distance set, say with its two possible distances being d_1, d_2. If either of these distances are 0 then S is a 1-distance set and hence $|S| \leq n+1$, so we may assume $d_1, d_2 > 0$.

Similar to our set system setup; for each $\vec{y} \in S$, we want to construct a small polynomimal $p_{\vec{y}}(\vec{x})$ which is non-zero when $\vec{x} = \vec{y}$ and is 0 when $\vec{x} \in S \setminus \{\vec{y}\}$. After some thought around the hypothesis of the theorem, one is perhaps led to define

$$p_{\vec{y}}(\vec{x}) = (\|\vec{x}-\vec{y}\|^2 - d_1^2)(\|\vec{x}-\vec{y}\|^2 - d_2^2) = (\sum(x_i-y_i)^2 - d_1^2)(\sum(x_i-y_i)^2 - d_2^2).$$

It is straightforward to check that $p_{\vec{y}}(\vec{y}) = d_1^2 d_2^2 \neq 0$ and $p_{\vec{y}}(\vec{x}) = 0$ for any other $\vec{x} \in S$, and as such it is straightforward to check that these polynomials are linearly independent. It follows then that $|S|$ is upper bounded by the dimension of the span of the $p_{\vec{y}}$ vectors.

Observe that each $p_{\vec{y}}(\vec{x})$ is a polynomial of degree at most 4 in n variables, and the vector space V of such polynomials has $\dim(V) = O(n^4)$ which is far larger than what we were hoping for. However, a closer inspection shows that every $p_{\vec{y}}(\vec{x})$ polynomial has the exact same term of degree 4, namely $(\sum x_i^2)^2$. As such, the $p_{\vec{y}}(\vec{x})$ vectors live in the smaller subspace $V' \subseteq v$ consisting of the span of all degree at most 3 polynomials in n variables together with $(\sum x_i^2)^2$, giving the much better bound $\dim(V') = O(n^3)$.

A more careful analysis in this direction shows in fact that each $p_{\vec{y}}(\vec{x})$ is the span of the following types of polynomials: (i) $(\sum x_i^2)^2$, (ii) $(\sum x_i^2)x_j$, (iii) $x_i x_j$ for i, j possibly equal, (iv) x_i, and (v) 1. The total dimension of the vector space V'' spanned by these 5 types of polynomials is $1 + n + \binom{n+1}{2} + n + 1 = \frac{1}{2}(n+4)(n+1)$. Because the $|S|$ polynomials $p_{\vec{y}}(\vec{x})$ are linearly independent and lie in V'', we find that $|S| \leq \dim(V'') = \frac{1}{2}(n+4)(n+1)$, proving the result. $\square$

The current best bound for the 2-distance problem is $\binom{n+2}{2}$ due to Blokhuis. They use the exact same proof approach we used here together with the additional refinement of adding in more polynomials.

5.1.3 Refinement #2: More Polynomials

The reader might complain that, although we have improved our bound for (p, L)-modular intersecting families through Theorem 5.3, it still fails to be tight for Oddtown. As far as we are aware, it is still an open problem as to whether the bound of Theorem 5.3 can be replaced by $|\mathcal{F}| \leq \binom{n}{|L|}$ for all L, which would be tight whenever $L = \{0, 1, \ldots, s-1\}$. However, such a general result is known to hold when $\mathcal{F}$ is uniform.

Theorem 5.5 (Frankl-Wilson [**?**]). *For any prime p and $L \subseteq \{0, 1, \ldots, p-1\}$, if $\mathcal{F} \subseteq \binom{[n]}{r}$ is (p, L)-modular intersecting, then*

$$|\mathcal{F}| \leq \binom{n}{|L|}.$$

We omit the proof of this result, which utilizes the powerful tool of incidence matrices, and instead refer the reader to [**?**, Chapter 7] for a more thorough treatment. Instead, we will prove a weaker version of this result which holds in the non-modular setting.

For this, given a set of integers L, we say that a set system $\mathcal{F}$ is *L-intersecting* if $|F \cap F'| \in L$ for all distinct $F, F' \in \mathcal{F}$.

Theorem 5.6 (Ray-Chaudhuri-Wilson [?]). *If L is a set of integers and $\mathcal{F} \subseteq \binom{[n]}{r}$ is L-intersecting, then*

$$|\mathcal{F}| \leq \binom{n}{|L|}.$$

Proof. Note that we may assume $L \subseteq \{0, 1, \ldots, r-1\}$ since these are the only intersections that can actually occur, and in particular we may assume $|L| \leq r$.

Similar to before, we define polynomials $p_F : \mathbb{R}^n \to \mathbb{R}$ by $p_F(\vec{x}) = \prod_{\ell \in L}(\langle \chi_F, \vec{x}\rangle - \ell)$ and we define $\bar{p}_F$ to be their multilinearizations. As before, we find that these $\bar{p}_F$ polynomials are linearly independent, giving a bound of $\sum_{i=0}^{|L|} \binom{n}{i}$ since these polynomials lie in the vector space V of multilinear polynomials of degree at most $|L|$.

Observe that the bound above fails to be sharp if and only if these $\bar{p}_F$ do not span all of V. In particular, to prove the result, it suffices to find a set of $\sum_{i=0}^{|L|-1} \binom{n}{i}$ polynomials of V which are linearly independent with the $\bar{p}_F$ vectors.

Let's say we were to write these supposed polynomials as q_I for $I \subseteq [n]$ with $|I| \leq |L| - 1$; what would be a good choice for these polynomials? To mimic the previous proof, we would like to have $q_I(\chi_F) = 0$ for all $F \in \mathcal{F}$. Since all we know about the individual elements of $\mathcal{F}$ is that they each of size r, perhaps a reasonable condition to demand is

$$q_I(\vec{x}) = q'_I(\vec{x}) \left(\sum_{j=1}^{n} x_j - r\right),$$

where q'_I is some appropriate polynomial of degree at most $|L| - 1$. Indeed, if we do this and if we have

$$\sum_F \lambda_F \bar{p}_F + \sum_I \lambda_I q_I = 0,$$

then by plugging in $\chi_{F'}$ into both sides of this equality we see that $\lambda_F = 0$ for all $F \in \mathcal{F}$, which is a promising start to the argument. The next thing we want to do is choose these q'_I polynomials such that the q_I vectors themselves are linearly independent. One possibility is to set

$$q_I(\vec{x}) = \prod_{i \in I} x_i \cdot \left(\sum_{j=1}^{n} x_j - r\right).$$

Note that with this, for any set $J \subseteq [n]$ with $|J| \leq |L| - 1 < r$, we have $q_I(\chi_J) = 0$ if $I \not\subseteq J$, and we have $q_I(\chi_J) \neq 0$ if $I \subseteq J$ (here we implicitly use $|J| < r$ to ensure $\sum_{j=1}^{n} (\chi_J)_j - r \neq 0$).

With this in mind, say we had

$$\sum_I \lambda_I q_I = 0,$$

and let J be a smallest set such that $\lambda_J \neq 0$. By plugging χ_J into both sides of this expression, the observation above implies that we get $c_J \lambda_J = 0$ for some $c_J \neq 0$, and hence $\lambda_J = 0$, a contradiction. We conclude that the q_I polynomials are linearly independent, and hence all of the polynomials $\{\bar{p}_F\}_F \cup \{q_I\}_I$ are linearly independent vectors. Unfortunately the q_I polynomials aren't in V, i.e. aren't multilinear, but this can be easily remedied by considering $\{\bar{p}_F\}_F \cup \{\bar{q}_I\}_I$, and again we conclude that these vectors (in V) are linearly independent, giving the desired bound on $|\mathcal{F}|$. $\square$

We note that the exact proof as written *almost* goes through if instead of demanding $|F| = r$ for $F \in \mathcal{F}$, we only demand $|F| \equiv r \mod p$. In particular, the current proof will go through for this variant of the problem if we further assume $r \notin [|L|-1] \mod p$ (which holds if e.g. $|L| \le r$), but without this assumption we can't assume $q_I(\chi_J) \neq 0$ whenever $I \subseteq J$. Nevertheless, it does turn out that this bound continues to hold under the weaker hypothesis $|F| \equiv r \mod p$, but the proof of this ends up being somewhat more involved and we refer the interested reader to [**?**, Theorem 5.37].

5.1.4 Refinement #2.5: Even More Polynomials and Stability

Need to change the uniformity notation at some point to match the rest of the text.

It has recently been observed that even more polynomials can be added to the proof of the Ray-Chaudhuri-Wilson Theorem above, giving both stronger bound as well as effective "stability" results.

To motivate this refinement, let us look back at the current argument. There we introduced multilinear polynomials $\bar{p}_F$ of degree $|L|$ corresponding to each $F \in \mathcal{F}$ along with polynomials q_I of degree at most $|L|$ corresponding to each $I \subseteq [n]$ of size at most $|L| - 1$. To refine this argument further, we would need to introduce some additional multilinear polynomials of degree at most $|L|$, and naively it perhaps makes the most sense to have these polynomials be associated to subsets of $[n]$ like our current set of polynomials. Since these polynomials must have size at most $|L|$ and since every set of size at most $|L| - 1$ has already been accounted for by the q_I, the most natural choice would be for our new polynomials to correspond to sets $J \in \binom{[n]}{|L|}$ of size exactly $|L|$, and given this, seemingly the only reasonable polynomial for us to use would be

$$r_J(\hat{x}) := \prod_{j \in J} x_j,$$

since it was essentially this same polynomial that we used to encode each set $I \subseteq [n]$ of size less than $|L|$. However, we now have an issue: the polynomials q_I had an additional factor of $\sum x_j - k$ to guarantee that $q_I(\chi_F) = 0$, but we can no longer add this term to the r_J since we need these polynomials to have degree at most $|L|$. As such, we will not have $r_J(\chi_F) = 0$ for every $J \in \binom{[n]}{J}$ and $F \in \mathcal{F}$. The crucial observation, however, is that this will hold for *some* choices of J, and as such these polynomials can be added in to improve the bound of Ray-Chaudhuri-Wilson theorem. In particular, we will have $r_J(\chi_F) = 0$ for every $F \in \mathcal{F}$ precisely if J is not a subset of any $F \in \mathcal{F}$.

Motivated by the above, given a set system $\mathcal{F}$ and an integer s, we define the *s-th shadow $\partial^s \mathcal{F}$* to be the s-uniform hypergraph where a set S of size s is in $\partial^s \mathcal{F}$ if and only if there exists some $F \in \mathcal{F}$ which contains S as a subset. Formalizing our logic above gives the following refinement of Ray-Chaudhuri-Wilson, which is essentially due to Gao, Liu, and Xu [**?**].

Theorem 5.7 ([**?**])**.** *If L is a set of integers and $\mathcal{F} \subseteq \binom{[n]}{k}$ is L-intersecting, then*

$$|\mathcal{F}| \le \binom{n}{|L|} - \left| \binom{[n]}{|L|} \setminus \partial^{|L|} \mathcal{F} \right|.$$

Sketch of Proof. Define the polynomials $\bar{p}_F$ for $F \in \mathcal{F}$ and q_I for $I \in \binom{[n]}{<|L|}$ exactly as we do in the proof of Ray-Chaudhuri-Wilson. Further, for each non-shadow $J \in \binom{[n]}{|L|} \setminus \partial^{|L|}\mathcal{F}$ we define $r_J(\hat{x}) := \prod_{j \in J} x_j$. One can check exactly as before that these polynomials will be linearly independent of each other, giving the stated bound since we added an additional $|\binom{[n]}{|L|} \setminus \partial^{|L|}\mathcal{F}|$ polynomials compared to our previous argument. $\square$

From this we immediately get the following result, which can be seen as an L-intersecting generalization of a special case of Katona's intersecting shadow theorem [?].

Corollary 5.8. *If L is a set of integers and $\mathcal{F} \subseteq 2^{[n]}$ is L-intersecting and k-uniform, then*

$$|\partial^{|L|}\mathcal{F}| \geq |\mathcal{F}|.$$

In addition to Corollary 5.8, Theorem 5.7 has the advantage of reducing the problem of bounding the size of a largest k-uniform L-intersecting family to the combinatorial problem of bounding how large the non-shadow of such a family can be, and in some cases this new problem can be solved more effectively than the original. In particular, this can be done in the simplest case when $L = \{t, t+1, \ldots, k-1\}$, i.e. when $\mathcal{F}$ is t-intersecting. Moreover, this sort of proof naturally yields the following stability result, implying that every "near-extremal" t-intersecting family must behave quite similarly to the unique extremal construction.

Theorem 5.9 ([?])**.** *If $\mathcal{F} \subseteq \binom{[n]}{k}$ is t-intersecting, then either*

$$|\mathcal{F}| \leq \binom{n}{k-t} - \binom{n-3k}{k-t} = O(n^{k-t-1}),$$

or there exists a set of t vertices $T \subseteq [n]$ such that every edge of $\mathcal{F}$ contains T. In particular, for n sufficiently large we have $|\mathcal{F}| \leq \binom{n-t}{k-t}$.

Proof. Let $\mathcal{J} := \binom{[n]}{k-t} \setminus \partial^{k-t}\mathcal{F}$ be the non-shadows of $\mathcal{F}$. If $|\mathcal{J}| \geq \binom{n-3k}{k-t}$ then by Theorem 5.7 we have $|\mathcal{F}| \leq \binom{n}{k-t} - \binom{n-3k}{k-t}$, and as such we may assume from now on that $|\mathcal{J}| < \binom{n-3k}{k-t}$.

Let F_1 be an arbitrary element of $\mathcal{F}$. Observe that $|\binom{[n]\setminus F_1}{k-t}| \geq \binom{n-3k}{k-t} > |\mathcal{J}|$, and hence there exists some $(k-t)$-set $S_2 \subseteq [n] \setminus F_1$ which is not in $\mathcal{J}$, i.e. which is contained in $\partial^{k-t}\mathcal{F}$. By definition this means there exists some $F_2 \in \mathcal{F}$ which consists of S_2 together with some set of t vertices T. Because $\mathcal{F}$ is t-intersecting and S_2 is disjoint from F_1, we must have $F_1 \cap F_2 = T$. It remains now to show that every edge of $\mathcal{F}$ contains this set T.

To this end, let $F_3 \in \mathcal{F}$ be an arbitrary edge. Again we have that $|\binom{[n]\setminus(F_1 \cup F_2 \cup F_3)}{k-t}| \geq \binom{n-3k}{k-t} > |\mathcal{J}|$, so there exists some $(k-t)$-set $S_4 \subseteq [n] \setminus (F_1 \cup F_2 \cup F_3)$ in the shadow of $\mathcal{F}$, meaning there is some edge $F_4 \in \mathcal{F}$ consisting of S_4 together with some set T_4 of size exactly t. Because $\mathcal{F}$ is t-intersecting and S_4 is disjoint from F_1, F_2, F_3, we must have $F_i \cap F_4 = T_4$ for all $i = 1, 2, 3$. In particular this implies $T_4 \subseteq F_1 \cap F_2 = T$, and since both T_4, T are sets of size t we conclude that $T_4 = T$, proving that every edge F_3 contains T as desired. $\square$

With a slightly more careful analysis one can easily prove the same result with $\binom{n-3k}{k-t}$ replaced by $\binom{n-3k+2t}{k-t}$. More generally, it is natural to ask just how large $|\mathcal{F}|$ can be if it does not have every edge containing a common t-set. In particular, the $t = 1$ case is covered by the following famous result of Hilton and Milner [?].

Theorem 5.10 (Hilton-Milner)**.** *If $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting and $n > 2k$, then either there exists an element $v \in [n]$ contained in every edge of $\mathcal{F}$, or*

$$|\mathcal{F}| \le \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1.$$

This upper bound is best possible, as can be seen by considering the set system $\mathcal{F}$ consisting of the edge $F = \{2, 3, \ldots, k+1\}$ together with every k-set that contains both 1 and an element from F. There by now a wide array of proofs of the Hilton-Milner theorem. Here we will briefly outline an argument in the same spirit as that of Theorem 5.9 due to Ge, Xu, and Zhao [**?**] who in fact proved far more than just the Hilton-Milner theorem using this approach. As the exact details are a little complex, we will only give a very hih-level sketch of the underlying details.

Sketch of Proof. Let $\mathcal{F}$ be a k-uniform intersecting family on $[n]$, say without loss of generality that $1 \in [n]$ has the maximum degree in $\mathcal{F}$. The key idea is to partition $\mathcal{F}$ into two sets $\mathcal{F}_0, \mathcal{F}_1$ with $\mathcal{F}_1$ consisting of all the elements containing 1 and $\mathcal{F}_0$ containing the rest of the elements. Letting $\mathcal{J}_1$ consist of the $(k-1)$-sets which are not shadows of $\mathcal{F}_1$, we have from the exact same argument as above that

$$|\mathcal{F}_1| \le \binom{n}{k-1} - |\mathcal{J}_1|.$$

In fact, we can do slightly better than this by using the fact that $\mathcal{F}_1$ is essentially a set system on an $n-1$ element subset (since every element of $\mathcal{F}_1$ is guaranteed to contain 1). Thus by using a slightly more involved argument and letting $\mathcal{J}_1' \subseteq \mathcal{J}_1$ be the set of non-shadows not containing 1, one can show that

$$|\mathcal{F}_1| \le \binom{n-1}{k-1} - |\mathcal{J}_1'|,$$

which in total implies that

$$|\mathcal{F}| \le \binom{n-1}{k-1} - |\mathcal{J}_1'| + |\mathcal{F}_0|.$$

Similar to before, we automatically get an effective bound on $|\mathcal{F}|$ now if $|\mathcal{J}_1'| - |\mathcal{F}_0|$ is large, and otherwise we can perform a structural analysis implying that $\mathcal{F}$ is contained in a star. For this structural analysis, it is important that we are working with the modified quantity $|\mathcal{J}_1'| - |\mathcal{F}_0|$ rather than just the set of non-shadows of $\mathcal{F}$ directly, as the functional $|\mathcal{J}_1'| - |\mathcal{F}_0|$ allows us to play the two quantities $|\mathcal{J}_1'|$ and $|\mathcal{F}_0|$ off of each other. $\square$

5.2 Slice Rank

In the previous subsection we considered proofs using linear algebra. Here we go a step further and use *multilinear* algebra.

To this end, given a set S and an integer k, we let S^k denote the set of all k-tuples of elements of S. Given a field $\mathbb{F}$, we will say that any function of the form $f : S^k \to \mathbb{F}$ is a *k-tensor.* Note that when $k = 2$, we can express f as a matrix whose rows and columns are indexed by S, and as such we can think of k-tensors as "higher order" matrices. Our main goal here is to use "ranks" of tensors to obtain bounds for combinatorial problems, analogous to what we did in our second proof of Oddtown.

There are various non-equivalent ways one can generalize the notion of rank from matrices to tensors. One way is through *slice rank*. For this, we say that a k-tensor f is a *slice* if

$$f(x_1, \ldots, x_k) = g(x_i)h(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_k),$$

where i is some integer, g is a 1-tensor, and h is a $(k-1)$-tensor. Note that when $k = 2$, slices are exactly rank 1 matrices, i.e. the outer product of two vectors g and h. We define the slice rank $sr(f)$ of a k-tensor f to be the smallest integer r such that f can be written as the sum of r slices. Again note that in the case $k = 2$ this exactly corresponds to the usual notion of rank.

Although not every property of matrix rank carries over to the setting of slice rank, one important property that does is the fact that the slice rank of a "diagonal tensor" equals the number of non-zero entries it has. To this end, we say that a tensor is *proper diagonal* if $f(\vec{x}) \neq 0$ if and only if $x_i = x_j$ for all i, j.

Lemma 5.11. *If $f : S^k \to \mathbb{F}$ is a proper diagonal k-tensor, then $sr(f) = |S|$.*

Proof Sketch. We first show $sr(f) \leq |S|$. For each $a \in S$, let $g_a(x)$ be the 1-tensor with $g_a(a) = 1$ and $g_a(b) = 0$ otherwise. We observe

$$f(\vec{x}) = \sum_{a \in S} g_a(x_1) f(a, x_2, \ldots, x_k),$$

proving that f has slice rank at most $|S|$ as desired.

The proof of the lower bound $sr(f) \geq |S|$ is a somewhat fiddly induction argument that takes about two pages to do; we refer the reader to a nice writeup by Martinez [**?**, Lemma 2.2.9] for the full details. ☐

Our general approach using Lemma 5.11 will be as follows:

(1) Start with a set S which has some desired properties.

(2) Using the properties of S, we construct a proper diagonal k-tensor $f : S^k \to \mathbb{F}$ which is "simple" (e.g. a polynomial of low degree).

(3) Using the fact that f is "simple", we find some explicit way to write f as the sum of r slices.

(4) In total, steps (2) and (3) imply

$$|S| = sr(f) \leq r,$$

giving an effective bound on sets S with our desired properties.

This sort of approach was first used by Croot, Lev, and Pach [?] and was then adapted by Ellenberg and Gijswitj [?] before being systemetized by Tao [?].

We will use the general approach outlined above to give very short proofs of two problems which were previously thought to be incredibly difficult. Throughout this section, we suggest before the reader goes through each proof that they first sit down and try to construct an f as in the framework outlined above in order to solve the problem.

5.2.1 The Capset Problem

Given an abelian group G, we say that a triple of elements $(x, y, z) \in G^3$ forms a 3-term arithmetic progression (or 3AP for short) if $y - x = z - y$, and we say this is a non-trivial 3AP if we do not have $x = y = z$. A fundamental question in additive combinatorics is to determine how large a subset $S \subseteq G$ can be if it contains no non-trivial 3AP. For example, Roth's theorem famously says that when $G = \mathbb{Z}$ such a set must have density 0.

Another natural case to consider is $\mathbb{F}_p^n$ for prime p. The case $p = 2$ is trivial (if $S \subseteq \mathbb{F}_2^n$ contains two distinct elements x, y, then (x, y, x) forms a non-trivial 3AP). As such, the first interesting case is to look at $\mathbb{F}_3^n$, and in this setting sets $S \subseteq \mathbb{F}_3^n$ without non-trivial 3AP's are referred to as cap sets. Using a difficult Fourier analytic argument, Bateman and Katz [?] showed that capsets can have size at most $O(3^n/n^{1+\varepsilon})$ and Edel [?] gave the best known lower bound of about 2.2^n. At one point Tao mentioned that this capset problem was perhaps his favorite open problem and that he thought the answer was probably $(3 - o(1))^n$. It was thus a major shock when Ellenberg and Gijswijt [?] gave a remarkably short proof showing an upper bound of $(3 - \varepsilon)^n$.

Theorem 5.12 (Ellenberg-Gijswijt [?])**.** *For all primes p, there exists a real number $\varepsilon > 0$ such that any set $S \subseteq \mathbb{F}_p^n$ which contains no non-trivial 3AP has $|S| \leq (p - \varepsilon)^n$.*

Again, the reader is encouraged to try and come up with a tensor f which might work to give the result before going through the details of this proof.

Proof. For slight ease of notation we will only deal with the case $p = 3$, the proof for general p being almost identical. As outlined above, our goal will be to use S to construct a proper diagonal tensor f of low degree. Given the inputs of the problem, perhaps the most natural kind of tensor to consider is one of the form $f : S^3 \to \mathbb{F}_3$, where again we want to ensure that if $x, y, z \in S$ then $f(x, y, z) \neq 0$ if and only if $x = y = z$. Critically, because of our hypothesis that S contains no non-trivial 3AP, we see that we want $f(x, y, z) \neq 0$ if and only if x, y, z is a 3AP, and as such our goal is essentially equivalent to constructing an f which is the indicator function for 3AP's!

By definition, (x, y, z) is a 3AP if and only if $y - x = z - y$, and rearranging gives $x - 2y + z = 0$. As such, we're essentially left with the problem of constructing an indicator function for $x - 2y + z \neq 0$, or equivalently that $x_i - 2y_i + z_i \neq 0$ for all i. And this is somewhat easy: simply take

$$f(x, y, z) = \prod_{i=1}^{n} (1 - (x_i - 2y_i + z_i)^2),$$

which one can readily check is an indicator function for 3AP's in S, and hence is a proper diagonal tensor.

It remains to estimate the slice rank of f. That is, we want to show that f can be written as the sum of a small number of functions of the form e.g. $g(x)h(y,z)$. To this end, we make the observation that by the definition of f given above, each of its monomials has degree at most $2n$, and as such each of its monomials has one of its x-degree, y-degree, or z-degree is at most $2n/3$. The idea now is to group each of these monomials with the same low degree variable and then bound the slice rank of each of these groups by using the low degree variable.

To be somewhat more precise, given a vector $\alpha \in \{0,1,2\}^n$, define $x^\alpha = \prod x_i^{\alpha_i}$, and similarly define y^β and z^γ. Let $|\alpha| := \sum \alpha_i$, and let $\mathcal{M}$ denote the set of all triples (α,β,γ) with $\alpha,\beta,\gamma \in \{0,1,2\}^n$ and $|\alpha|+|\beta|+|\gamma| \le 2n$. Observe that each monomial of f is of the form $x^\alpha y^\beta z^\gamma$ for some $(\alpha,\beta,\gamma) \in \mathcal{M}$.

Let $\mathcal{M}_x \subseteq \mathcal{M}$ be the set of triples with $|\alpha| \le 2n/3$, let $\mathcal{M}_y \subseteq \mathcal{M}\backslash\mathcal{M}_x$ be those with $|\beta| \le 2n/3$, and let $\mathcal{M}_z \subseteq \mathcal{M} \setminus (\mathcal{M}_x \cup \mathcal{M}_y)$ be those with $|\gamma| \le 2n/3$. As noted above, $\mathcal{M}_x \cup \mathcal{M}_y \cup \mathcal{M}_z$ partition $\mathcal{M}$. Given α with $|\alpha| \le 2n/3$, define the polynomial

$$h_\alpha(y,z) = \sum_{\beta,\gamma:(\alpha,\beta,\gamma)\in\mathcal{M}_x} c_{\alpha,\beta,\gamma} y^\beta z^\gamma,$$

where $c_{\alpha,\beta,\gamma}$ is the (possibly 0) coefficient of $x^\alpha y^\beta z^\gamma$ in f. Similarly define $h_\beta(x,z)$ and $h_\gamma(x,y)$. By definition and the fact that $\mathcal{M}_x \cup \mathcal{M}_y \cup \mathcal{M}_z$ partition $\mathcal{M}$, we have

$$f(x,y,z) = \sum_{\alpha:|\alpha|\le 2n/3} x^\alpha h_\alpha(y,z) + \sum_{\beta:|\beta|\le 2n/3} y^\beta h_\beta(x,z) + \sum_{\gamma:|\gamma|\le 2n/3} z^\gamma h_\gamma(x,y).$$

With this, we conclude that the slice rank of f is at most 3 times the number of terms in eahc sum, which is equal to the number of vectors $\alpha \in \{0,1,2\}^n$ with $|\alpha| \le 2n/3$. It is not convince oneself that the number of such vectors is at most $(3-\varepsilon)^n$ for some ε (essentially because a random $\alpha \in \{0,1,2\}^n$ behaves like the sum of two bionomial random variables with total expectation n), proving the result. □

In the simplest non-trivial case $p = 3$ one can go through the analysis above and get an upper bound of roughly $|S| = O(2.756^n)$ for sets $S \subseteq \mathbb{F}_3^n$ not containing a capset. It turns out that this upper bound continues to hold for a slightly more general problem: we say that three sets $X, Y, Z \subseteq \mathbb{F}_3^n$ contain no rainbow 3AP if the only 3AP's $(x,y,z) \in X \times Y \times Z$ are trivial. One can show that the above proof yields that if X, Y, Z contains no rainbow 3AP then $|X|+|Y|+|Z| = O(2.756^n)$, and moreover there exist constructions that essentially match this upper bound. Thus any improvement to the bound for capsets has to somehow utilize that we only allow triples from S^3 and not from three distinct sets. We refer to the interested reader to the nice survey article by Grochow [?] for more on this topic.

5.2.2 Nonuniform Sunflowers

In the setting of uniform hypergraphs, the famous Erdős-Rado sunflower conjecture says that there exists a constant $C = C(p)$ such that every r-uniform hypergraph with at least C^r

hyperedges contains a sunflower with p petals. Here we consider an analogous conjecture in the non-uniform setting due to Erdős and Szemerédi [?]: there exists a constant $\varepsilon = \varepsilon(p) > 0$ such that for all $n \geq \varepsilon^{-1}$, every $\mathcal{F} \subseteq 2^{[n]}$ with at least $(2-\varepsilon)^n$ edges contains a sunflower with p petals. This conjecture turns out to be weaker than the Erdős-Rado sunflower conjecture [?].

The slice rank method allows us to give an easy proof of this non-uniform sunflower conjecture when $p = 3$.

Theorem 5.13 (Naslund-Sawin [?]). *If $\mathcal{F} \subseteq 2^{[n]}$ does not have a sunflower with 3 petals, then*

$$|\mathcal{F}| \leq (n+1) \sum_{t \leq n/3} \binom{n}{t} \leq 1.89^n.$$

Proof. Let S be the set of characteristic vectors corresponding to the edges of $\mathcal{F}$. As a first attempt for this problem, one would probably try to work with the set S directly to construct a tensor f, but this causes some technical issues pop up.

To get around these issues, we let $S_r \subseteq S$ be those characteristic vectors with r 1's. The insight here is that $\max_r |S_r| \leq |S| \leq (n+1) \cdot \max_r |S_r|$, so up to some negligible factor it suffices to bound the size of each S_r set. This will allow us to get around the previously mentioned issues.

With this in mind, our goal is to construct a 3-tensor f with domain S_r^3 such that $f(x,y,z) \neq 0$ if and only if $x = y = z$. The crucial insight is that if we do not have $x = y = z$, then there exists some i such that $x_i + y_i + z_i = 2$. Indeed, if this were not the case and x, y, z were all pairwise distinct from each other, then this would imply that every i is either contained in 0, 1, or 3 of the sets x, y, z, which would imply these distinct edges form a sunflower with 3 edges. If, say $x = y \neq z$, then the non-existence of such an i would imply $x \subseteq z$, a contradiction to $x \neq z$ both being in S_r, i.e. both having the same number of elements.

With this observation in mind, we see that our desired f is essentially the indicator function for [no coordinate i having $x_i + y_i + z_i = 2$]. Thus we can take $f : S_r^3 \to \mathbb{F}_3$ by defining

$$f(x,y,z) = \prod_{i=1}^{n} (x_i + y_i + z_i - 2),$$

and one can easily check that f is a proper diagonal tensor given that $S_r \subseteq S$ contains no 3-sunflowers, so it remains to bound the slice rank of f. Similar to before, we observe that each monomial of f has either x, y or z degree at most $n/3$, so by a similar trick as before we conclude

$$\max_r |S_r| \leq sr(f) \leq \sum_{t \leq n/3} \binom{n}{t} \leq 2^{nH(1/3)},$$

where $H(p) = -p \log_2(p) - (1-p)\log_2(1-p)$ is the binary entropy function. Evaluating this (and multiplying by $n+1$) gives the desired upper bound. □

Before moving on, we note that slice rank fits into a broader set of methods known as polynomial methods, with another famous technique in this area being the combinatorial Nullstellensatz. We omit talking about this further, but see eg my Methods in Extremal Combinatorics notes for more.

5.3 Exercises

1. Show that the vector space of polynomials in n variables of degree at most d has dimension $\sum_{i=0}^{d} \binom{n+i-1}{i}$ [1+].

2. Many examples of the linear algebra method require one to show that a set of functions $p_1, \ldots, p_m$ from a set S to a field $\mathbb{F}$ are linearly independent over $\mathbb{F}$. We discuss some standard ways to check this.

   (a) (Diagonal Criterion) Prove that if for all j there exists some $s_j \in S$ with $p_i(s_j) = 0$ for $i \neq j$ and $p_j(s_j) \neq 0$ then $p_1, \ldots, p_m$ are linearly independent over $\mathbb{F}$ [1+].

   (b) (Triangular Criterion) Prove that if for all j there exists some $s_j \in S$ with $p_i(s_j) = 0$ for $i < j$ and $p_j(s_j) \neq 0$ then $p_1, \ldots, p_m$ are linearly independent over $\mathbb{F}$ [1+].

3. The Oddtown problem considers families with $|F| \equiv 1 \mod 2$ and $|F \cap F'| \equiv 0 \mod 2$. Here we consider the other three natural variants of this problem.

   (a) (Reverse Oddtown) Determine the largest size of a family $\mathcal{F} \subseteq 2^{[n]}$ satisfying $|F| \equiv 0 \mod 2$ for $F \in \mathcal{F}$ and $|F \cap F'| \equiv 1 \mod 2$ for distinct $F, F' \in \mathcal{F}$ [2+].

   (b) (Eventown) Determine the largest size of a family $\mathcal{F} \subseteq 2^{[n]}$ satisfying $|F| \equiv 0 \mod 2$ for $F \in \mathcal{F}$ and $|F \cap F'| \equiv 0 \mod 2$ for distinct $F, F' \in \mathcal{F}$ (Hint: let V denote the span of the characteristic vectors of $\mathcal{F}$ over $\mathbb{F}_2^n$ and show that $V \subseteq V^\perp$; what does that say about the dimension of V?) [2].

   (c) (Reverse Eventown) Determine the largest size of a family $\mathcal{F} \subseteq 2^{[n]}$ satisfying $|F| \equiv 1 \mod 2$ for $F \in \mathcal{F}$ and $|F \cap F'| \equiv 1 \mod 2$ for distinct $F, F' \in \mathcal{F}$ [2].

4. Let $\mathcal{F} \subseteq 2^{[n]}$ be a set system which is $\{\ell\}$-intersecting, i.e. which has $|F \cap F'| = \ell$ for all distinct $F, F' \in \mathcal{F}$.

   (a) Prove that if each $F \in \mathcal{F}$ has $|F| \geq \ell + 1$ then $|\mathcal{F}| \leq n$ (Hint: show that the characteristic vectors χ_F are linearly independent by showing $\langle \sum_F \lambda_F \chi_F, \sum_F \lambda_F \chi_F \rangle > 0$ whenever the λ_F are not all 0) [2+].

   (b) (Non-uniform Fischer inequality) conclude that all $\{\ell\}$-intersecting families $\mathcal{F} \subseteq 2^{[n]}$ with $\ell, n \geq 1$ have $|\mathcal{F}| \leq n$ [1].

5. In this exercise we construct large s-distance sets.

   (a) Prove that for all $s \geq 1$ there exists an s-distance set S with $|S| = \binom{n}{s}$ [1+].

   (b) Prove that for all $s \geq 1$ there exists an s-distance set S with $|S| = \binom{n+1}{s}$ (Hint: Most likely your construction for the previous part lies in an $(n-1)$-dimensional hyperplane) [2].

6. Prove that every 2-distance set $S \subseteq \mathbb{R}^n$ has $|S| \leq \binom{n+2}{2}$ (Hint: Show that one can add the polynomials $1, x_1, \ldots, x_n$ to the current proof while maintaining linear independence) [3-].

$* * *$

7. One can in general define k-tensors to be any function f of the form $f : S_1 \times \cdots \times S_k \to \mathbb{F}$ with $\mathbb{F}$ a field and each S_i a (possibly distinct) set, and we can analogously define the slice rank of such a tensor. Prove for such an f that $sr(f) \leq \min_i |S_i|$ [1+].

6 Transversals and VC Dimension

In this chapter we study two different measurements for how “complex” a set system can be: transversal numbers and VC-dimension.

6.1 Transversals

A *transversal* (sometimes also called a *cover* or *piercing set*) of a set system $\mathcal{F}$ is a set of vertices T with the property that $T \cap F \neq \emptyset$ for all $F \in \mathcal{F}$; in other words, every edge contains an element of T. We define the *transversal number* $\tau(\mathcal{F})$ to be the minimum size of a transversal of $\mathcal{F}$. The transversal number of a set system can be viewed as a measure of the complexity of $\mathcal{F}$ since $\tau(\mathcal{F})$ being small means there is a small number of vertices which are responsible for all of the edges of $\mathcal{F}$.

Results about transversals show up all over discrete mathematics. For example, many results from graph theory can be stated in terms of relationships between transversal numbers and matchings numbers. To this end we say that a set of edges $\{F_1, \ldots, F_m\} \subseteq \mathcal{F}$ is a *matching* if these sets are pairwise disjoint, and we define the *matching number* $\nu(\mathcal{F})$ to be the maximum size of a matching of $\mathcal{F}$. The parameter $\nu(\mathcal{F})$ turns out to be dual to $\tau(\mathcal{F})$ and they relate to each other as follows.

Lemma 6.1. *If $\mathcal{F}$ is a set system where every edge has size at least 1 and at most r, then*

$$\nu(\mathcal{F}) \leq \tau(\mathcal{F}) \leq r\nu(\mathcal{F}).$$

Proof. Let $M = \{F_1, \ldots, F_{\nu(\mathcal{F})}\}$ be a largest matching of $\mathcal{F}$. Observe that every transversal T must intersect each edge of M and that no vertex of T can intersect more than one edge of M since these edges are disjoint. As such every transversal satisfies

$$\nu(\mathcal{F}) = |M| \leq |T| \leq \tau(\mathcal{F}).$$

On the other hand, we claim that $T = \bigcup_i F_i$ is a transversal of $\mathcal{F}$. Indeed, if there were some edge $F \neq \emptyset$ disjoint from T then $M \cup \{F\}$ would be a larger matching, giving a contradiction. □

The upper bound of Lemma 6.1 is best possible for arbitrary set systems, and there is a great deal of interest in improving upon this bound for nice set systems.

One good source for nice set systems is that of graphs. For example, König's Theorem exactly says that $\nu(G) = \tau(G)$ whenever G is a bipartite graph, giving a wide class of examples where the trivial lower bound of Lemma 6.1 is tight. Another example of this sort of phenomenon is a variant of the Helly Theorem for trees.

Theorem 6.2. *Let T be a tree and $\mathcal{T}$ a collection of subtrees of T. If for every $T'_1, \ldots, T'_k \in \mathcal{T}$ there exist $i \neq j$ with $V(T'_i) \cap V(T'_j) \neq \emptyset$, then there exists a set $V \subseteq V(T)$ of size at most $k-1$ with $V \cap V(T') \neq \emptyset$ for all $T'_i \in \mathcal{T}$.*

Equivalently this says that if $\nu(\mathcal{T}) \leq k-1$ then $\tau(\mathcal{T}) \leq k-1$, and hence $\nu(\mathcal{T}) = \tau(\mathcal{T})$ for all set systems of this form. A spiritually similar result is the Erdős-Pósa Theorem.

Theorem 6.3 (Erdős-Pósa)**.** *For every graph G and integer k, either G contains k vertex disjoint cycles or there exists a set $T \subseteq V(G)$ of size $|T| = O(k \log k)$ such that $G - T$ contains no cycles.*

That is, if $\mathcal{F}$ denotes the set system whose ground set is $V(G)$ and where each $F \in \mathcal{F}$ is the vertex set of a cycle in G, then this says that either $\nu(\mathcal{F}) \le k$ or $\tau(\mathcal{F}) = O(k \log k)$, which can equivalently be interpreted as saying $\tau(\mathcal{F}) = O(\nu(\mathcal{F}) \log(\nu(\mathcal{F})))$. We note that this logarithmic term is necessary in general, and in many cases this bound significantly improves upon the trivial upper bound of Lemma 6.1 since $\mathcal{F}$ can have edges as large as size $|V(G)|$ whereas $\log(\nu(\mathcal{F})) \le \log(|V(G)|)$. In addition to these classic results, there are a wide number of open problems in graph theory that can be framed in the language of bounding transversal numbers; we mention two such conjectures here.

Conjecture 6.4 (Bollobás-Erdős-Tuza)**.** *For a graph G, let $\mathcal{F}_\alpha$ denote the set system with ground set $V(G)$ where a set of vertices I forms a hyperedge if I is an independent set of maximum size in G. If G is an n-vertex graph with $\alpha(G) = \Omega(n)$, then $\tau(\mathcal{F}_\alpha) = o(n)$.*

Conjecture 6.5 (Tuza)**.** *For a graph G, let $\mathcal{F}_\Delta$ be the set system with ground set $E(G)$ and where three edges form a hyperedge in $\mathcal{F}_\Delta$ if they form a triangle in G. Then $\tau(\mathcal{F}_\Delta) \le 2\nu(\mathcal{F}_\Delta)$.*

That is, if a graph has only k disjoint triangles, then it is possible to find at most $2k$ edges which intersect the edge set of every triangle. The bound is best possible for G which are disjoint unions of K_4 and K_5. By Lemma 6.1 we trivially have $\tau(\mathcal{F}_\Delta) \le 3\nu(\mathcal{F}_\Delta)$, and the only known improvement for this result for general graphs is due Haxell who showed $\tau(\mathcal{F}_\Delta) \le \frac{66}{23}\nu(\mathcal{F}_\Delta) \le 2.87\nu(\mathcal{F}_\Delta)$.

Another common source of inspiration for transversal problems come from geometric settings. The most famous geometric transversal result is Helly's Theorem, which says that if $\mathcal{C}$ is a collection of convex sets in $\mathbb{R}^d$ with the property that any $\mathcal{C}' \subseteq \mathcal{C}$ of size at most $d+1$ has a point of common intersection, then there exists a point in the common intersection of $\mathcal{C}$. Equivalently, if we view $\mathcal{C}$ as a set system on $\mathbb{R}^d$ then Helly's Theorem says that if $\tau(\mathcal{C}') = 1$ for all $\mathcal{C}' \subseteq \mathcal{C}$ of size at most $d+1$, then $\tau(\mathcal{C}) = 1$. This gives rise to a new set of transversal type problems of the form: when is it the case that if every small subsystem $\mathcal{F}' \subseteq \mathcal{F}$ has small transversal number then the same holds for $\mathcal{F}$? For example, we have the following discrete version of Helly's Theorem.

Proposition 6.6. *If $\mathcal{F}$ is an r-uniform set system and if every $\mathcal{F}' \subseteq \mathcal{F}$ with $|\mathcal{F}'| \le r+1$ has $\tau(\mathcal{F}') = 1$, then $\tau(\mathcal{F}) = 1$.*

The bound $r+1$ is best possible for this result to hold, since $\binom{[r+1]}{r}$ has every r of its edges intersecting but no vertex in its common intersection.

Proof. Let $F = \{x_1, \ldots, x_r\} \in \mathcal{F}$ be arbitrary. If $T = \{x_i\}$ is a transversal for $\mathcal{F}$ for any choice of i then we are done, so we can assume for all i that there exists some $F_i \in \mathcal{F}$ disjoint from x_i. Observe that $\{F, F_1, \ldots, F_r\}$ is a set of at most $r+1$ edges of $\mathcal{F}$ which does not contain a vertex in their common intersection (any such vertex would need to be of the form x_i to be in F, but then $x_i \notin F_i$). This contradicts the hypothesis of the proposition, giving the result. □

There are two natural ways to try and generalize Proposition 6.6. The first is to weaken the assumption $\tau(\mathcal{F}')=1$ for $|\mathcal{F}'|\le r+1$ by replacing $r+1$ with a smaller number and asking to what extent we can bound $\tau(\mathcal{F})$ in this case.

Theorem 6.7 (Lovász)**.** *If $\mathcal{F}$ is an r-uniform set system and if every $\mathcal{F}'\subseteq\mathcal{F}$ with $|\mathcal{F}'|\le k$ has $\tau(\mathcal{F}')=1$, then $\tau(\mathcal{F})\le\frac{r+k-2}{k-1}$.*

Proof. We will inductively construct sets $F_1,\dots,F_k\in\mathcal{F}$ with the property that for all $j\le k$,

$$|F_1\cap F_2\cap\cdots\cap F_j|\le r-(j-1)(\tau(\mathcal{F})-1).$$

Note that any choice of $F_1\in\mathcal{F}$ satisfies this for $j=1$. Assume we have constructed $F_1,\dots,F_j$ for some $j<k$ satisfying this condition and let $T=F_1\cap\cdots\cap F_j$. Observe that T is a transversal of $\mathcal{F}$ since for any edge F we have $F\cap F=F\cap F_1\cap\cdots\cap F_j\neq\emptyset$ by hypothesis. It follows that $|T|\ge\tau(\mathcal{F})$, and hence there exists some subset $S\subseteq T$ of size $\tau(\mathcal{F})-1$. By definition S can not be a transversal, and hence there must exist some $F_{j+1}\in\mathcal{F}$ which is disjoint from S, and hence

$$|F_1\cap\cdots\cap F_{j+1}|\le|F_1\cap F_2\cap\cdots\cap F_j|-|S|\le r-j(\tau(\mathcal{F})-1),$$

proving the inductive step of our argument.

In total we find that there exist sets $F_1,\dots,F_k$ with

$$1\le|F_1\cap\cdots\cap F_k|\le r-(k-1)(\tau(\mathcal{F})-1),$$

where the lower bound uses the hypothesis of the theorem. Rearranging this inequality gives the desired upper bound on $\tau(\mathcal{F})$. ☐

Another direction to take Proposition 6.6 is to replace the 1's by some larger number t and ask when having $\tau(\mathcal{F}')\le t$ for small subsystems $\mathcal{F}'$ imply $\tau(\mathcal{F})\le t$. Similar to before the set system $\binom{[r+t]}{r}$ is such that every $\binom{r+t}{r}-1$ subsystem has transversal number at most t but the system itself has transversal number $t+1$. As such we must look at all susbsystems of size at most $\binom{r+t}{r}$ and this turns out to be sufficient.

Theorem 6.8 (Bollobás)**.** *If $\mathcal{F}$ is an r-uniform set system and if every $\mathcal{F}'\subseteq\mathcal{F}$ with $|\mathcal{F}'|\le\binom{r+t}{r}$ has $\tau(\mathcal{F}')\le t$, then $\tau(\mathcal{F})\le t$.*

Proof. Assume for contradiction that there was some $\mathcal{F}$ satisfying the hypothesis of the theorem with $\tau(\mathcal{F})\ge t+1$. By iteratively removing elements of $\mathcal{F}$ we can find some subfamily $\mathcal{A}\subseteq\mathcal{F}$ with the property $\tau(\mathcal{A})=t+1$ but $\tau(\mathcal{A}\setminus\{A\})=t$ for any $A\in\mathcal{A}$. We aim to show $|\mathcal{A}|\le\binom{r+t}{r}$, which together with $\tau(\mathcal{A})=t+1$ will give a contradiction to the hypothesis of the theorem. We do this by applying the Bollobás set pairs inequality.

To this end, observe that for each $A_i\in\mathcal{A}$ there exists a transversal B_i of $\mathcal{A}\setminus\{A_i\}$ of size t by construction. This means B_i intersects each $A_j\in\mathcal{A}$ with $j\neq i$, and the fact that $\tau(\mathcal{A})=t+1$ implies that B_i must be disjoint from A_i as otherwise this would be a transversal of size t. We conclude that $\mathcal{A}$ and $\mathcal{B}=\{B_1,\dots\}$ are Bollobás intersecting with these set systems being r and t-uniform, respectively, so from the Bollobás set pairs inequality we find

$$|\mathcal{A}|\le\binom{r+t}{r}.$$

However, we assumed $\mathcal{F}$ had no subsystem of this size with transversal number more than t, a contradiction. $\square$

6.2 VC Dimension

We now study a somewhat more technical notion for measuring the complexity of a set system.

Definition 6. Given a set system $\mathcal{F}$ with ground set V, we say that a subset $X \subseteq V$ is *shattered* by $\mathcal{F}$ if for every subset $X' \subseteq X$ there exists some $F_{X'} \in \mathcal{F}$ such that $F_{X'} \cap X = X'$. We say that a set system $\mathcal{F}$ has *VC-dimension* d if there exists a set X of size d which is shattered by $\mathcal{F}$ and if no set of larger cardinality is shattered by $\mathcal{F}$.

Roughly speaking we think of set systems with few shattered sets as being "simple." This intuition can be formalized through the following.

Lemma 6.9. *The size $|\mathcal{F}|$ of a set system $\mathcal{F}$ is at most the number of sets shattered by $\mathcal{F}$.*

Proof. We prove this by induction on the size of the ground set of $\mathcal{F}$, and without loss of generality we may assume the ground set is $[n]$ for some integer n. The case $n = 0$ is trivial, so we assume we have proven the result up to some value $n \geq 1$.

As in our proofs using compression arguments, we want to represent $\mathcal{F}$ by set systems with ground set $[n-1]$ in order to apply our inductive hypothesis. The previous families we used $\mathcal{F}(\overline{n})$ and $\mathcal{F}_L(n)$ do not end up working here. Instead, we define

$$\mathcal{F}_1 = \{F \subseteq [n-1] : \text{either } F \in \mathcal{F} \text{ or } F \cup \{n\} \in \mathcal{F}\},$$

$$\mathcal{F}_2 = \{F \subseteq [n-1] : F \in \mathcal{F} \text{ and } F \cup \{n\} \in \mathcal{F}\}.$$

It is not difficult to see that $|\mathcal{F}_1| + |\mathcal{F}_2| = |\mathcal{F}|$, since for each $F \in \mathcal{F}_1$ at least 1 of the sets $F, F \cup \{n\}$ is in $\mathcal{F}$ with exactly two of these sets being in $\mathcal{F}$ if and only if $F \in \mathcal{F}_2$ as well. Let $\mathcal{S}_1, \mathcal{S}_2$ be the sets that are shattered by these respective families, noting that $|\mathcal{F}_i| \leq |\mathcal{S}_i|$ by our inductive hypothesis. Let $\mathcal{S}$ denote the set of shattered sets of $\mathcal{F}$.

We claim that $\mathcal{S}_1 \subseteq \mathcal{S}$. Indeed, consider some $X \in \mathcal{S}_1 \subseteq 2^{[n-1]}$, meaning that for each $X' \subseteq X$ there exists some $F_X \in \mathcal{F}_1 \subseteq 2^{[n-1]}$ such that $F_{X'} \cap X = X'$. By definition of $\mathcal{F}_1$ this means there exists some set $F'_{X'} \in \mathcal{F}$ (namely either $F_{X'}$ or $F_{X'} \cup \{n\}$) which contains $F_{X'}$ and no additional elements of $[n-1] \supseteq X$, and hence $F'_{X'} \cap X = X'$ as well, implying that X is shattered.

We next claim that if $X \in \mathcal{S}_2$, then $X \cup \{n\} \in \mathcal{S}$. Indeed, for each $X' \subseteq X$ there is some $F_{X'} \in \mathcal{F}_2$ with $F_{X'} \cap X = X'$, and by definition this means that $F_{X'}, F_{X'} \cup \{n\} \in \mathcal{F}$. Intersecting these two sets with $X \cup \{n\}$ give $X', X' \cup \{n\}$. As this holds for arbitrary $X' \subseteq X$ we conclude that $X \cup \{n\}$ is shattered in $\mathcal{S}$.

Observe that the two sets $\mathcal{S}_1, \mathcal{S}_2$ are disjoint, as one only uses sets avoiding n while the other always uses n. This combined with the claims above give

$$|\mathcal{S}| \geq |\mathcal{S}_1| + |\mathcal{S}_2| \geq |\mathcal{F}_1| + |\mathcal{F}_2| = |\mathcal{F}|,$$

proving the result. $\square$

Lemma 6.9 immediately implies the following bound for sets of bounded VC-dimension.

Theorem 6.10 (Sauer-Shelah Lemma)**.** *If $\mathcal{F}$ is a set system with ground set V and VC-dimension at most d, then*

$$|\mathcal{F}| \le \sum_{i=0}^{d} \binom{|V|}{i}.$$

Indeed, this follows since a set system with VC-dimension at most d can only shatter sets of size at most d. Theorem 6.10 as well as the notion of VC-dimension was discovered independently by a number of authors including Sauer [**?**], Shelah [**?**], and Vapnik and Chervonenkis [**?**], with this latter group being where the name "VC-dimension" comes from. The strengthened bound Lemma 6.9 is due to Pajor [**?**].

As mentioned, the VC-dimension of a set system roughly measures how "complex" it is, and we will motivate this intuition with a few examples. To start, given a set system $\mathcal{F}$ and a set of vertices S, we define the *trace* of S in $\mathcal{F}$ to be the collection of sets

$$\mathrm{Tr}_{\mathcal{F}}(S) := \{F \cap S : F \in \mathcal{F}\}.$$

We denote this simply by $\mathrm{Tr}(S)$ whenever $\mathcal{F}$ is clear from context.

Lemma 6.11. *If $\mathcal{F}$ is a set system with ground set V of VC-dimension at most d, then every set $S \subseteq V$ satisfies*

$$|\mathrm{Tr}(S)| \le \sum_{i=0}^{d} \binom{|S|}{i} \le \max\{1, 2|S|^d\}.$$

That is, if the VC-dimension of $\mathcal{F}$ is small, then the number of ways that the edges in $\mathcal{F}$ can intersect with a given set S is at most polynomial in the size of S. This is in sharp contrast with what would happen in a "random" set system where the number of such intersections is much closer to the maximum possible value of $2^{|S|}$.

Proof. The second inequality is straightforward, so we only prove the first. By the Sauer-Shelah Lemma, it suffices to show the set system $\mathrm{Tr}(S)$ has VC-dimension at most d.

Assume for contradiction there was some set $X \subseteq S$ of size greater than d shattered by $\mathrm{Tr}(S)$. By definition this means that for each $X' \subseteq S$ there exists a set $F_{X'} \in \mathcal{F}$ such that

$$X' = (S \cap F_{X'}) \cap X = F_{X'} \cap X.$$

These sets $F_{X'} \in \mathcal{F}$ show that X is in fact shattered in the original set system $\mathcal{F}$, a contradiction to this system having VC-dimension at most d. $\square$

Another way in which set systems of small VC-dimension set systems are "simple" is through them having small transversals.

Theorem 6.12 (Haussler-Welzl [**?**])**.** *If $\mathcal{F}$ is a set system on V which has VC-dimension at most d and $\min_{F\in\mathcal{F}} |F| \ge \delta|V|$ for some $0 < \delta < 1$, then*

$$\tau(\mathcal{F}) \le O\left(\frac{d}{\delta}\log\frac{1}{\delta}\right).$$

In particular, this bound is independent of the size of $\mathcal{F}$, which is not something one would expect in "typical" set systems. Something like our condition requiring each element of $\mathcal{F}$ to be large is necessary for this conclusion to hold, as otherwise as we discuss in the exercises $\mathcal{F}$ could contain a large number of sets of size 1, say, which will not increase the VC-dimension by much but which will force $\tau(\mathcal{F})$ to be large.

Proof. The bound is trivial if $\tau(\mathcal{F}) \le 1$, so we may assume $t := \tau(\mathcal{F}) - 1 \ge 1$. We will prove the result by bounding the number of pairs of sets (X_0, X_1) such that (i) X_0, X_1 are disjoint with $X_0 \cup X_1$ having size exactly t, and (ii) there exists some $F \in \mathcal{F}$ with $F \cap (X_0 \cup X_1) = F \cap X_1$. Let $\mathcal{X}$ denote the set of all such pairs.

To begin, we claim that

$$|\mathcal{X}| \ge \sum_{i=0}^{t} \binom{|V|}{i}\binom{\delta|V|}{t-i} = \binom{(1+\delta)|V|}{t}.$$

Indeed, consider the following process for constructing pairs in $\mathcal{X}$. Start with some integer $0 \le i \le t$ and pick an arbitrary set X_0 of size i. Because $i \le t < \tau(\mathcal{F})$, we by definition have that X_0 is not a transversal of $\mathcal{F}$, meaning that there exists some $F \in \mathcal{F}$ disjoint from X_0. We then take X_1 to be any subset of F of size $t - i$ (with no pair being produced if $|F| < t - i$). Observe that (X_0, X_1) is a pair of disjoint sets whose union has size t, and that the set F from this procedure witnesses $F \cap (X_0 \cup X_1) = F \cap X_1$. As such, every pair formed in this way lies in $\mathcal{X}$, and it is not difficult to see that these pair are all distinct. Moreover, for each integer i the number of choices for X_0 is exactly $\binom{|V|}{i}$, and given this the number of choices for X_1 is $\binom{|F|}{t-i} \ge \binom{\delta|V|}{i}$. Putting this all together gives the desired lower bound.

For the upper bound, we claim that

$$|\mathcal{X}| \le \binom{|V|}{t} \cdot 2t^d.$$

Indeed, we can first pick the set $X_0 \cup X_1$ in $\binom{|V|}{t}$ ways. We then observe that if $(X_0, X_1) \in \mathcal{X}$, then $X_1 \in \mathrm{Tr}(X_0 \cup X_1)$, which by Lemma 6.11 and the assumptions $t \ge 1$ implies that the number of choices for X_1 is at most $2t^d$. Because (X_0, X_1) is uniquely determined by $X_0 \cup X_1$ and X_1, we conclude that the number of pairs is at most the bound above.

Combining these two bounds and using basic inequalities for binomial coefficients implies

$$2t^d \ge (1+\delta)^t \approx e^{t\delta}.$$

This implies $t = O(\frac{d}{\delta}\log\frac{1}{\delta})$, which in turn implies the same bound for $\tau(\mathcal{F}) = t + 1$. □

6.3 VC-Dimension for Graphs

In this subsection we use VC-dimension for some problems in graph theory. To begin, given a graph G we say that a set of vertices T is a *hitting set* of G if T intersects every independent set of G of size $\alpha(G)$, i.e. every maximum independent set of G. Equivalently, T is hitting if $\alpha(G - T) < \alpha(G)$. Let $h(G)$ denote the smallest size of a hitting set of G.

It was asked by Hajebi, Li, and Spirkl [?] whether for all integers $t \geq 2$ there exists some function f_t such that $h(G) \leq f_t(\omega(G))$ (with $\omega(G)$ denoting the size of the largest clique of G) whenever G contains no induced matching of size t. This was answered in a strong form by Ai, Liu, Xu, and Zhou [?].

Theorem 6.13 ([?]). *If G does not have an induced matching of size t, then*

$$h(G) = O_t(\omega(G)^{3t-3} \log \omega(G)).$$

Proof. We begin by translating the problem into a statement about set systems by letting $\mathcal{F}$ denote the set system on $V(G)$ consisting of all maximum independent sets. With this we observe that a hitting set for G is the same as a transversal for $\mathcal{F}$, so $h(G) = \tau(\mathcal{F})$ and it suffices to upper bound the transversal number of this system. Given this interpretation, a natural thing to try is Haussler-Welzl. In particular, if G is an n-vertex graph and if $\mathcal{F}$ has VC-dimension d then Haussler-Welzl implies

$$h(G) = \tau(\mathcal{F}) = O\left(d\frac{n}{\alpha(G)} \log\left(\frac{n}{\alpha(G)}\right)\right), \tag{1}$$

where here we used that each element of $\mathcal{F}$ uses exactly an $\alpha(G)/n$ fraction of the ground set $V(G)$. To have any hope of using (1) to give our desired result, we must show that graphs G as in the hypothesis of our theorem have both d and $n/\alpha(G)$ bounded as a function of $\omega(G)$. We do this in two steps.

Claim 6.14. *The VC-dimension d of $\mathcal{F}$ satisfies $d = O_t(\omega(G)^{t-1})$.*

Proof. By definition there exist some set $X = \{x_1, \ldots, x_d\}$ which is shattered by $\mathcal{F}$. In particular, this means there exists some (maximum) independent set $I \subseteq V(G)$ with $I \cap X = X$, implying that X itself is an independent set. Our aim now is to use a large part of this independent set X as one side of an induced matching in G, implying that X must be small since G contains no large induced matchings.

Because X is shattered, we have for each $j \in [d]$ some maximum independent set $I_j \in \mathcal{F}$ with $I_j \cap X = X \setminus \{x_j\}$. Observe that x_j must be adjacent to some vertex $y_j \in I_j$, as otherwise $I_j \cup \{x_j\}$ would be a larger independent set, contradicting that I_j is maximum. Let $Y = \{y_1, \ldots, y_d\}$, noting that $y_i \neq y_j$ for $i \neq j$ since they have different neighbors and that $X \cap Y = \emptyset$ since X is independent and each vertex in Y has (exactly) one neighbor in X.

We claim that $G[Y]$ does not contain an independent set of size at least t. Indeed, if say $y_1, \ldots, y_t$ were independent then these vertices together with the (disjoint) independent set $x_1, \ldots, x_t$ would form an induced matching of size t. We also observe that $G[Y]$ does not contain a clique larger than $\omega(G)$. By standard bounds on Ramsey numbers then, we must have

$$d \leq R(\omega(G) + 1, t) \leq \binom{\omega(G) + t - 1}{t - 1} = O_t(\omega(G)^{t-1}).$$

$\square$

It remains to upper bound $n/\alpha(G)$. For this, one might recall the classical bound

$$n/\alpha(G) \leq \chi(G).$$

As such, it suffices to bound the chromatic number of G in terms of its clique number. While this is impossible to do for arbitrary graphs, a classic result of Wagon [**?**] establishes this for graphs without induced matchings, the proof of which we reproduce below.

Claim 6.15. *If G is a graph without a matching of size t, then $\chi(G) = O(\omega(G)^{2t-2})$.*

Proof. Define a sequence of functions $f_1, f_2, \ldots$ by having $f_1(\omega) = 1$ and inductively setting $f_t(\omega) = \binom{\omega}{2} f_{t-1}(\omega) + \omega$. We will show that G as in the claim satisfy $\chi(G) \le f_t(\omega(G))$, from which the result follows. This is trivial if $t = 1$, so assume we have proven the result up to some value t.

Let K be a maximum clique of G. For each pair of distinct vertices $x, y \in K$, define $S_{x,y}$ to be the set of vertices v which are not adjacent to either x or y. Observe that $G[S_{x,y}]$ contains no induced matching of size $t-1$, as otherwise such a matching together with xy would give an induced matching of size t in G. As such, we inductively have that $\chi(G[S_{x,y}]) \le f_{t-1}(\omega(G))$, and thus letting $S := \bigcup S_{x,y}$ we find that $\chi(G[S]) \le \binom{\omega(G)}{2} f_{t-1}(\omega(G))$.

It remains then to color the vertices $v \notin S$. We claim that every such vertex has exactly one vertex $x \in K$ which it is non-adjacent to. Indeed, a vertex $v \notin S$ has at most one non-neighbor in K by definition of S, and if v were adjacent to every vertex in K then $K \cup \{v\}$ would be a larger clique in G, a contradiction to K having maximum size. Let S_x be the set of vertices which are adjacent to every vertex in $K \setminus \{x\}$. Note that S_x must be an independent set, as any edge yz together with $K \setminus \{x\}$ would give a larger independent set. As such, we can partition the vertices of $V(G) \setminus S$ into $|K| = \omega(G)$ independent sets S_x, which combined with our analysis above implies that

$$\chi(G) \le \binom{\omega(G)}{2} f_{t-1}(\omega(G)) + \omega(G) = f_t(\omega(G)),$$

giving the result. $\square$

These claims together with (1) gives the desired result. $\square$

More generally, it is conjectured in [**?**] that if G avoids induced copies of some forest F, then there exists some function f depending on F such that $h(G) \le f(\omega(G))$. While this result verifies the conjecture whenever F is a matching, the current proof approach fundamentally can not prove the result in any other cases. Indeed, if G is a matching of size $n/2$, then the set system $\mathcal{F}$ consisting of the maximum independent sets of G will have VC-dimension $n/2$, which is arbitrarily large in terms of $\omega(G) = 2$ despite G avoiding every induced subgraph which is not a matching.

The previous example uses an auxiliary set system associated to a graph G, namely the set system of its maximum independent sets. While there are many different sorts of set systems one can associate to a given graph, the following is the most standard example of such a system within the context of VC-dimension.

Definition 7. Given a graph G, we define the *neighborhood set system*

$$\mathcal{F}_G := \{N(v) : v \in V(G)\}.$$

We say that a graph has VC-dimension d if $\mathcal{F}_G$ has VC-dimension d.

One reason one might be motivated to study this particular kind of set system is due to its transverals. Indeed, one can check that a set T is a transversal for $\mathcal{F}_G$ if and only if it is a set of vertices such that $\bigcup_{v\in T} N(v) = V(G)$. Haussler-Welzl then implies that if G is a graph with small VC-dimension and linear minimum degree, then it contains a small covering of $V(G)$ by neighborhoods.

The observation above might remind the reader of proper colorings which one can think of as coverings of $V(G)$ by independent sets. Although having small VC-dimension and large minimum degree is not enough to ensure small chromatic number due to K_n, this conclusion does hold if we put some additional hypothesis on G.

Lemma 6.16. *If G is a graph which has VC-dimension at most d, minimum degree at least $\delta|V(G)|$, and contains no C_ℓ for some ℓ, then '*

$$\chi(G) = O(\ell \cdot d\delta^{-1} \log(\delta^{-1})).$$

Proof. By Haussler-Welzl there exists a set $T \subseteq V(G)$ of size $O(d\delta^{-1} \log(\delta^{-1}))$ such that $V(G) = \bigcup_{v\in T} N(v)$. As such, it remains to show that $G[N(v)]$ has chromatic number at most $O(\ell)$ for each $v \in T$. And indeed, since G is C_ℓ-free, the graph $G[N(v)]$ can not contain a path on $\ell - 1$ vertices. Such graphs can easily be shown to have chromatic number at most $O(\ell)$ (by showing that they have degeneracy at most this, for example), proving the result. $\square$

Lemma 6.16 gives a strong conclusion about the graph G whenever G has small VC-dimension. While such results can be interesting on their own, we will primarily be interested in such results as a first step towards proving similar statements without the need of imposing any hypothesis on the VC-dimension. In particular, the above motivates studying the following parameter.

Definition 8. Given a graph F, we define the *chromatic threshold* $\delta_\chi(F)$ to be the infimum over all $\delta \geq 0$ such that every F-free graph G with $\delta(G) \geq \delta|V(G)|$ has $\chi(G) = O_\delta(1)$.

That is, what's the smallest minimum degree $\delta|V(G)|$ which forces an F-free graph to have bounded chromatic number? Lemma 6.16 naively suggests that $F = C_\ell$ might have $\delta_\chi(C_\ell) = 0$ since this is true if we restrict only to graphs with bounded VC-dimension. This is in fact immediately seen to be true if ℓ is even simply because C_ℓ-free graphs have at most $O(|V(G)|^{3/2})$ edges, and a nice result of Thomasson [?] further shows that $\delta_\chi(C_\ell) = 0$ for all $\ell \neq 3$. However, the behavior of δ_χ is surprisingly different for triangles.

Theorem 6.17. *We have $\delta_\chi(C_3) = 1/3$.*

The lower bound $\delta_\chi(C_3) \geq 1/3$ was proven by Hajnal (see [?]) and the upper bound $\delta_\chi(C_3) \leq 1/3$ is due to Thomassen [?]. Here we give an alternative proof of this upper bound due to Łuczak and Thomassé [?] based in the framework of VC-dimension. They key result we need for this is the following.

Theorem 6.18. *If G is a maximal triangle-free graph with $\delta(G) > |V(G)|/3$, then G has VC-dimension at most 3.*

Proof. The only place we will use our minimum degree condition is through the following observation.

Claim 6.19. *For any set $K \subseteq V(G)$ of $3k$ vertices, there exists a vertex z with at least $k+1$ neighbors in K.*

Indeed, by hypothesis we have $\sum_{v \in K} \deg(v) > k|V(G)|$, so by the pigeonhole principle there exist some vertex z incident to at least $k+1$ edges incident to K. This in turn implies the following result of Brandt [**?**].

Claim 6.20. *The graph G does not contain an induced cube, i.e. a $K_{4,4}$ minus a perfect matching.*

Proof. Assume for contradiction that G contained an induced cube, say with $x_1, \ldots, x_4, y_1, \ldots, y_4$ its vertices such that $x_i y_j \in E(G)$ if and only if $i \neq j$. Because G is maximal triangle-free, for each i there must exist a vertex z_i adjacent to the non-adjacent vertices x_i, y_i. Note that we must have $z_i \neq z_j$ for $i \neq j$, as otherwise $x_i, y_j, z_i = z_j$ would form a triangle. Let K denote the set of these 12 vertices.

By Claim 6.19, there exists a vertex z' adjacent to at least 5 vertices of K. A little case analysis shows that up to relabeling the vertices, the neighbors of z'_1 in K must be of the form x_1, y_1, z_2, z_3, z_4. Similarly if we look at the 9 vertices $K - \{x_1, y_1, z_1\}$ we conclude from Claim 6.19 that there exists some z'_4 adjacent to at least 4 vertices of $K - \{x_1, y_1, z_1\}$, which again up to relabeling must be x_4, y_4, z_2, z_3. Observe that $z'_1 \neq z'_4$ since they have distinct neighborhoods into K.

Consider now the set of 12 vertices $K' := (K \setminus \{z_1, z_4\}) \cup \{z'_1, z'_4\}$. One can check that the graph induced by K' has no independent set of size 5, but Claim 6.19 implies that there exists a vertex adjacent to at least 5 vertices of K', contradicting that G is triangle-free. We conclude that G contains no induced cube, proving the result. □

Now assume for contradiction that G shatters some set of four vertices $X = \{x_1, \ldots, x_4\}$. By definition of shattering there exists some vertex z adjacent to all of these vertices, and since G is triangle-free this implies X is an independent set. Again by definition of shattering we have for each i that there exists a vertex y_i such that $y_i \sim x_j$ if and only if $i \neq j$. Note that $y_i \notin X$ for any i since X is an independent set and y_i has neighbors in X, and also observe that the y_i vertices form an independent set since they pairwise have common neighbors. We conclude that X together with the y_i vertices induce a cube in G, contradicting our claim. We conclude that G has VC-dimension at most 3 as desired. □

This now quickly gives the upper bound of Theorem 6.17.

Proof of $\delta_\chi(C_3) \leq 1/3$. Let G be a triangle-free graph with $\delta(G) > |V(G)|/3$ and let G' be any maximal triangle-free graph containing G. By Theorem 6.18 G' has VC-dimension at most 3, which by Lemma 6.16 implies that

$$\chi(G) \leq \chi(G') = O(1),$$

proving the result. □

Our proof above not only shows that $\delta_\chi(C_3) \le 1/3$ but in fact shows the stronger fact that any triangle-free graph with $\delta(G) > |V(G)|/3$ has bounded chromatic number. One can push this argument further to show that triangle-free graphs with $\delta(G) \ge |V(G)|/3 - O(1)$ have bounded chromatic number, which was first done by Łuczak and Thomassé [?] using this VC-dimension based approach. In this same work, [?] introduced a variant of the classical notion of VC-dimension which was utilized by Allen, Böttcher, Griffiths, Kohayakawa, and Morris [?] to fully determine $\delta_\chi(F)$ for *every* graph F.

Before moving on, we briefly consider a variant of the chromatic threshold motivated by Theorem 6.18 originally introduced by Huang, Liu, Rong, and Xu [?].

Definition 9. Given a graph F, we define the *VC threshold* $\delta_{VC}(F)$ to be the infimum over the set of all $\delta \ge 0$ such that every maximal F-free graph G with $\delta(G) \ge \delta|V(G)|$ has VC-dimension at most $O_\delta(1)$.

We emphasize that this definition requires G to be maximal, which is needed to rule out G being a dense random bipartite graph. Note that Lemma 6.16 implies $\delta_{VC}(F) \ge \delta_\chi(F)$ and that Theorem 6.18 implies $\delta_{VC}(C_3) \le 1/3$. Again one trivially has $\delta_{VC}(F) = 0$ whenever F is bipartite, and it is known that $\delta_{VC}(K_r) = \frac{2r-5}{2r-3}$ for all r [?]. An argument spiritually similar to that of Theorem 6.18 was made by [?] to show $\delta_{VC}(C_\ell) \le \frac{1}{\ell}$ for all (odd) ℓ, with the authors using this result to prove tight bounds on another variant of the chromatic threshold. It remains open as to whether or not this upper bound on $\delta_{VC}(C_\ell)$ is tight for odd cycles in general.

6.4 Fractional and Geometric Transversals

Possibly put here, possibly put in geometric section.

6.5 Exercises

1. For all integers $r, \nu \ge 1$ construct an r-uniform $\mathcal{F}$ with $\nu(\mathcal{F}) = \nu$ and $\tau(\mathcal{F}) = r\nu$ [2-].

2. Prove that if $\mathcal{F}$ is an intersecting family then $\tau(\mathcal{F}) \le \min_{F\in\mathcal{F}} |F|$ [1+].

3. (Gyárfás) Prove that if $\mathcal{F}$ is a set system where every edge has size at most r, then for every set of vertices S the number of transversals T of $\mathcal{F}$ with $S \subseteq T$ and $|T| = \tau(\mathcal{F})$ is at most $r^{\tau(\mathcal{F})-|S|}$ [2].

4. (Erdős-Lovász) Prove that if $\mathcal{F}$ is an r-uniform intersecting family with $\tau(\mathcal{F}) = r$ then $|\mathcal{F}| \le r^r$ [2].

* * *

5. Recall that given a family $\mathcal{F} \subseteq 2^{[n]}$ and an element $i \in [n]$, we define for each $F \in \mathcal{F}$ the squashed edge
$$s_{-i}(F;\mathcal{F}) = \begin{cases} F \setminus \{i\} & i \in F,\ F \setminus \{i\} \notin \mathcal{F}, \\ F & \textit{otherwise}, \end{cases}$$

and the squashed family

$$s_{-i}(\mathcal{F}) = \{s_{-i}(F;\mathcal{F}) : F \in \mathcal{F}\}.$$

(a) Prove for all i and sets of vertices X that $|\mathrm{Tr}_{s_{-i}(\mathcal{F})}(X)| \le |\mathrm{Tr}_{\mathcal{F}}(X)|$ [2].

(b) Use this to prove the Sauer-Shelah lemma (Hint: use the exercises from Section 3 to compress $\mathcal{F}$ to a downset) [2-].

(c) Construct a family $\mathcal{F} \subseteq 2^{[n]}$ such that for all i, there exists $X \subseteq [n]$ with $\mathrm{Tr}_{s_{-i}(\mathcal{F})}(X) \not\subseteq \mathrm{Tr}_{\mathcal{F}}(X)$; this shows that (a) can not be strengthened to a subset relation like we had with shifting and shadows [1].

6. The Sauer-Shelah Lemma upper bounds the number of sets that $\mathcal{F} \subseteq 2^{[n]}$ can have before it contains a set S of size d with $|\mathrm{Tr}(S)| = 2^d$, and it is natural to ask what bounds one obtains when we replace 2^d with a smaller value.

   To this end, prove that if $\mathcal{F} \subseteq 2^{[n]}$ has $|\mathcal{F}| > 1 + n + n^2/4$, then there exists some $S \subseteq [n]$ of size 3 with $|\mathrm{Tr}(S)| \ge 7$ (Hint: reduce to $\mathcal{F}$ being a downset as in the previous problem and use Mantel's Theorem from graph theory) [2-].

7. Let $\mathcal{F}$ be a set system with VC dimension d.

   (a) Prove that if $\mathcal{F}'$ is a set system obtained by adding some number of edges of size at most k to $\mathcal{F}$, then $\mathcal{F}'$ has VC dimension at most $d + k + 1$ [2].

   (b) For all k, construct an $\mathcal{F}'$ as above with VC dimension equal to $d + k + 1$ [1+]

8. Let $\mathcal{F}, \mathcal{F}'$ be two set systems on the same ground set V such that each set system has VC-dimension at most d. We consider various operations on this family which play nicely with respect to VC-dimension.

   (a) Show that the complement family $\overline{\mathcal{F}} := \{V \setminus F : F \in \mathcal{F}\}$ has VC-dimension at most d [1].

   (b) Show that any subfamily $\tilde{\mathcal{F}} \subseteq \mathcal{F}$ has VC-dimension at most d [1].

   (c) Show that the intersection set system $\mathcal{I}_{\mathcal{F},\mathcal{F}'} := \{F \cap F' : F \in \mathcal{F},\ F' \in \mathcal{F}'\}$ has VC-dimension at most $O(d)$ (Hint: prove that for any set X we have $|\mathrm{Tr}_{\mathcal{I}_{\mathcal{F},\mathcal{F}'}}(X)| \le |\mathrm{Tr}_{\mathcal{F}}(X)| \cdot |\mathrm{Tr}_{\mathcal{F}'}(X)|$) [2].

   (d) Define the dual set system $\mathcal{F}^*$ to be the set system with ground set $\mathcal{F}$ together with $|V|$ edges of the form $F_v^* := \{F : v \in F\}$ for each $v \in V$. That is, this is the set system obtained by interchanging the roles of vertices and edges, and it can also be thought of as the set system whose incidence matrix is the transpose of $\mathcal{F}$. Prove that $\mathcal{F}^*$ has VC-dimension at most $2^{d+1} - 1$ (Hint: observe that it suffices to prove that if $\mathcal{F}^*$ has VC-dimension at most 2^d then $\mathcal{F}$ has VC-dimension at most d) [2+].

9. For a set system $\mathcal{F}$ with ground set V, we say that a map $f : V \to \mathbb{R}_{\geq 0}$ is a *fractional transversal* if $\sum_{v \in F} f(v) \geq 1$ for all $F \in \mathcal{F}$, and we define the *fractional transversal number* $\tau^*(\mathcal{F})$ to be the smallest value of $\sum_v f(v)$ amongst all fractional transversals f. Prove that if $\mathcal{F}$ is a set system with VC-dimension at most d then $\tau(\mathcal{F}) = O(d\tau^*(\mathcal{F}) \log \tau^*(\mathcal{F}))$ [2+].

10. (Packing Lemma) Prove that if $\delta > 0$ and if $\mathcal{F}$ is a set system with ground set V and VC-dimension at most d such that every pair of distinct sets $F, F' \in \mathcal{F}$ has symmetric difference $|F \Delta F'| \geq \delta |V|$, then $|\mathcal{F}| = O(\delta^{-d})$ [3].

11. Prove that if $\mathcal{F}$ is a set system with ground set V and VC-dimension at most d, then for all $\delta > 0$ there exists a partition $\mathcal{F}_1 \cup \cdots \cup \mathcal{F}_m$ with $m = O(\delta^{-d})$ such that for all i and distinct $F, F' \in \mathcal{F}_i$, we have $|F \Delta F'| \leq \delta |V|$ [2+].

Part II

Discrete Geometry

This part of the text largely follows Matoušek's book "Lectures on Discrete Geometry" chapters 1, 8, 9 and 10.

7 Convex Geometry

7.1 Basic Definitions and Fundamental Results

Convex geometry is the study of convex sets C, which are subsets of Euclidean space $\mathbb{R}^d$ with the property that for every two points $x, y \in C$, the set C contains the entire line segment from x to y.

A common way to construct convex sets is by taking the convex hull of a set of points. For $S = \{v_1, \ldots, v_n\} \subseteq \mathbb{R}^d$, we define the *convex hull* $\mathrm{conv}(S)$ to be the set of convex combinations of these points, i.e. all points of the form $\sum \lambda_i v_i$ where $\lambda_i \geq 0$ for all i and $\sum \lambda_i = 1$. Equivalently, it can be checked that $\mathrm{conv}(S)$ is the smallest convex set containing all of S. One of the most useful results concerning convex hulls is Radon's Theorem.

Theorem 7.1 (Radon's Theorem)**.** *If $S \subseteq \mathbb{R}^d$ is a set of at least $d+2$ points, then one can partition S into two subsets S_1, S_2 such that $\mathrm{conv}(S_1) \cap \mathrm{conv}(S_2) \neq \emptyset$.*

For example, if $d = 2$ and $|S| = 4$ then there are essentially two non-trivial cases to consider: either $\mathrm{conv}(S)$ is a quadrilateral (in which case taking S_1 to be two opposite points of the quadrilateral and S_2 the other points makes $\mathrm{conv}(S_1), \mathrm{conv}(S_2)$ two intersecting lines), or $\mathrm{conv}(S)$ is a triangle (in which case taking S_1 to be the vertices of the triangle and S_2 to be the remaining point inside the triangle works). The bound is best possible by considering $S = \{v_1, \ldots, v_{d+1}\}$ with v_i a standard basis vector for $1 \leq i \leq d$ and $v_{d+1} = 0$.

Here and in many other problems in convex geometry it can be useful to work in the language of affine geometry. We say that points $v_1, \ldots, v_n \in \mathbb{R}^d$ are *affinely dependent* if there exists real numbers α_i not all 0 with $\sum \alpha_i v_i = 0$ and $\sum \alpha_i = 0$, and otherwise we say these points are *affinely independent*. For example, any two distinct points are affinely independent, any three points not on a common line are affiniely independent, and any four points not on a common plane are affinely independent. We leave the following as exercises to the reader.

Lemma 7.2. *Let $v_1, \ldots, v_n \in \mathbb{R}^d$.*

(a) These points are affiniely independent if and only if $v_1 - v_n, \ldots, v_{n-1} - v_n$ are linearly independent.

(b) If $n \geq d+2$, then $v_1, \ldots, v_n$ are affinely dependent.

This last part of the lemma is exactly what we need to prove Radon's Theorem.

Proof of Radon's Theorem. Let $S = \{v_1, \ldots, v_n\}$ with $n \geq d+2$. By the previous lemma we know these points are affiniely dependent, meaning there exist α_i not all 0 with $\sum \alpha_i = 0$ and $\sum \alpha_i v_i = 0$. We aim to prove the result with $S_1 = \{v_i : \alpha_i > 0\}$ and $S_2 = \{v_i : \alpha_i \leq 0\}$. By hypothesis we have $\sum_{v_i \in S_1} \alpha_i v_i = \sum_{v_j \in S_2} (-\alpha_j) v_j$ where the real coefficients on both sides are non-negative but which may not sum to 1. To remedy this, we let $A = \sum_{i:\alpha_i > 0} \alpha_i = \sum_{i:\alpha_i \leq 0} -\alpha_i > 0$ since not all the α_i are 0. Observe then that $x := \sum_{v_i \in S_1} \frac{\alpha_i}{A} v_i = \sum_{v_j \in S_2} \frac{-\alpha_j}{A} v_j$

and that these are both convex combinations of S_1 and S_2, so $x \in \text{conv}(S_1) \cap \text{conv}(S_2)$, proving the result. $\square$

Radon's Theorem allows one to prove many other foundational results in convex geometry. For example, we observe that as written, a convex combination allows for every $v_i \in S$ to be given positive weight by λ_i. It turns out however that one only needs to put weight on a small number of elements.

Theorem 7.3 (Carathéodory's Theorem)**.** *If $S \subseteq \mathbb{R}^d$ is a finite set of points and $x \in \text{conv}(S)$, then there exist $v_1, \ldots, v_{d+1} \in S$ with $x \in \text{conv}(\{v_1, \ldots, v_{d+1}\})$.*

The bound of $d+1$ is best possible by considering $S = \{v_1, \ldots, v_{d+1}\}$ with v_i the ith standard basis vector in $\mathbb{R}^d$, $v_{d+1} = 0$, and $x = (1/2d, \ldots, 1/2d)$.

Proof. By hypothesis of $x \in \text{conv}(S)$ we have $x = \sum_{i=1}^n \lambda_i v_i$ for some $v_i \in S$ and $\lambda_i \geq 0$ with $\sum \lambda_i$. Choose such a sum with n as small as possible. If $n \leq d+1$ then we are done, so we may assume $n \geq d+2$. The following can maybe be made clearer with some sort of intermediate claim somewhere in the analysis.

By Radon's Theorem, there exist α_i not all 0 with $\sum \alpha_i v_i = 0$. By hypothesis there exist $\lambda_i \geq 0$ with $\sum \lambda_i = 1$ and $x = \sum \lambda_i v_i$. Then for any real ε we also have

$$x = \sum \lambda_i v_i + 0 = \sum \lambda_i v_i + \varepsilon \sum \alpha_i v_i = \sum (\lambda_i + \varepsilon \alpha_i) v_i.$$

Now take i to be an index with $\frac{\lambda_i}{|\alpha_i|}$ as small as possible amongst indices with $\alpha_i \neq 0$ and let $\varepsilon = -\frac{\lambda_i}{\alpha_i}$. By definition, $\lambda_j + \varepsilon \alpha_j \geq 0$ for all j since $\lambda_j \geq 0$ and $\lambda_i + \varepsilon \alpha_i = 0$. We also have that $\sum \lambda_j + \varepsilon \alpha_j = \sum \lambda_j = 1$, so in total we find that $\sum_{j \neq i} (\lambda_j + \varepsilon \alpha_j) v_j$ is a convex combination equal to x with one fewer non-zero term, a contradiction to our choice of n. We conclude then that we must have $n \leq d+1$, proving the result. $\square$

We can also use Radon's Theorem to prove what is perhaps the most famous result from convex geometry.

Theorem 7.4 (Helly's Theorem)**.** *If $\mathcal{C}$ is a finite collection of convex subsets of $\mathbb{R}^d$ such that every $\mathcal{C}' \subseteq \mathcal{C}$ with $|\mathcal{C}'| \leq d+1$ has $\bigcap_{C \in \mathcal{C}'} C \neq \emptyset$ then $\bigcap_{C \in \mathcal{C}} C \neq \emptyset$.*

Again $d+1$ is best possible: if one takes $S = \{v_1, \ldots, v_{d+1}\}$ with v_i a standard basis vector and $v_{d+1} = 0$ and if $C_i = \text{conv}(S \setminus \{v_i\})$ then one can check that any d of these C_i sets intersect (at some v_j vertex in particular) but that there exists no vertex in their common intersection.

Proof. We prove the result by induction on $|\mathcal{C}|$, the case $|\mathcal{C}| \leq d+1$ being trivial by hypothesis. Assume then that $\mathcal{C} = \{C_1, \ldots, C_n\}$ with $n \geq d+2$ is as in the hypothesis that we have proven the result for all smaller collections of convex sets. In particular, for each i the collection $\mathcal{C} \setminus \{C_i\}$ satisfies the condition of the hypothesis, so inductively we know there exists some $v_i \in \bigcap_{C \in \mathcal{C} \setminus \{C_i\}} C$ for all i And these must be distinct else a v_i is just in every C_i and we win. By Radon's Theorem we can partition the v_i points into two sets S_1, S_2 such that there exists some $x \in \text{conv}(S_1) \cap \text{conv}(S_2)$. We aim to show that $x \in C_i$ for all i.

To this end, say without loss of generality that $v_i \notin S_1$. Since $v_j \in C_i$ for all $j \neq i$, we have in particular that $v_j \in C_i$ for all $v_j \in S_1$. It follows maybe not obviously that C_i contains the convex hull of S_1 and hence contains x, finishing the proof. See if Matousek has a cleaner way of stating all this. $\square$

We note that Helly's Theorem is false in general if one allows $|\mathcal{C}|$ to be infinite, but this can be remedied if one requires each element of $\mathcal{C}$ to be compact.

7.2 Variants

In the previous subsection we studied three of the most fundamental results in convex geometry, namely Carathéodory, Radon, and Helly's Theorems. Here we study one variant for each of these results. We begin with a "colorful" or "rainbow" version of Carathéodory.

Theorem 7.5 (Colorful Carathéodory Theorem)**.** *If $S_1, \ldots, S_{d+1} \subseteq \mathbb{R}^d$ are finite sets of points and if $x \in \operatorname{conv}(S_i)$ for all i, then there exists $v_i \in S_i$ for all i such that $x \in \operatorname{conv}(\{v_1, \ldots, v_{d+1}\})$.*

The term "colorful" refers to us thinking of the points of each S_i set being given color i, and in this language the theorem says that if x is in the convex hull of each colored set, then x is in the convex hull of a rainbow point set (i.e. a point set which uses one vertex of each color). Note that when $S_i = S$ for all i this recovers the original Carathéodory Theorem.

Proof. We say that a set of points $R = \{v_1, \ldots, v_{d+1}\}$ is rainbow if $v_i \in S_i$ for all i. Choose a rainbow point set such that the distance from x to $\operatorname{conv}(R)$ is minimal (this distance is well defined since $\operatorname{conv}(R)$ is compact, and the minimum is well defined since the point sets are finite). If $x \in \operatorname{conv}(R)$ then we are done, so we can assume this is not the case. Let $y \in \operatorname{conv}(R)$ be a point of minimal distance to x. Let h denote the hyperplane containing y which is perpendicular to the line $\vec{xy}$ and let H denote the closed halfspace containing h and not x Insert picture.

Observe that H must contain all of R. Indeed, if there were some $y' \in R \setminus H$, then $\operatorname{conv}(R)$ would contain the line segment yy' since $\operatorname{conv}(R)$ is convex, and taking a point along this line sufficiently close to y would give a closer point to x, a contradiction to how y was chosen.

The observation above implies that $\operatorname{conv}(R) \cap H = \operatorname{conv}(R \cap H)$ Maybe this requires more justification and that $\operatorname{conv}(R \cap H)$ lies in a $(d-1)$-dimensional subspace. It follows from Carathéodory's Theorem applied to this smaller subspace that $y \in \operatorname{conv}(R) \cap H = \operatorname{conv}(R \cap H)$ can be written as the convex combination of only d points, say $v_1, \ldots, v_d$ without loss of generality. Now by definition, $x \in \operatorname{conv}(S_{d+1})$, and this means we can not have $S_{d+1} \subseteq H$ since $x \notin H$. Therefore, there must exist some $v'_{d+1} \in S_{d+1} \setminus H$. Because $y \in \operatorname{conv}(\{v_1, \ldots, v_d\})$, the set $R' = \{v_1, \ldots, v_d, v'_{d+1}\}$ contains the line segment from y to $v'_{d+1} \notin H$, and as before taking a point along this line close to y gives a point closer to x. It follows that x is closer to $\operatorname{conv}(R')$ than $\operatorname{conv}(R)$, a contradiction to how we chose R. It thus must be that $x \in \operatorname{conv}(R)$, proving the result. $\square$

Maybe exposiate on why colorful notions are useful/how they're studied a lot in eg latin transversals and the like. Let me maybe figure out Tverberg first and then come back to this.

We next generalize Radon's Theorem. This result says that we can partition any large point set S into two sets S_1, S_2 with $\text{conv}(S_1) \cap \text{conv}(S_2) \neq \emptyset$, and it is natural then to ask about partitioning S into r sets $S_1, \ldots, S_r$ with $\bigcap \text{conv}(S_i) \neq \emptyset$. It is not difficult to show that this is possible to do if $|S|$ is very large, but getting the tight bound as in Radon's Theorem is more difficult.

Theorem 7.6 (Tverberg's Theorem). *If $S \subseteq \mathbb{R}^d$ has $|S| \geq (d+1)(r-1)+1$, then one can partition S into sets $S_1, \ldots, S_r$ with $\bigcap_i \text{conv}(S_i) \neq \emptyset$.*

The bound $(d+1)(r-1)+1$ is best possible explain why.

We prove this result in an alternative setting which is somewhat easier to work with. To this end, for a finite set of points S we define the cone

$$\text{cone}(S) = \{\sum_{v \in S} \alpha_v v : \alpha_v \geq 0\}.$$

That is, this is the same as $\text{conv}(S)$ except we no longer require $\sum \alpha_v = 1$. In particular, $\text{conv}(S) \subseteq \text{cone}(S)$, and more generally $\text{cone}(S)$ consists of all points which lie on a ray from the origin to a point in $\text{conv}(S)$.

Theorem 7.7 (Tverberg's Theorem for Cones). *If $S \subseteq \mathbb{R}^{d+1}$ has $|S| \geq (d+1)(r-1)+1$ and $0 \notin \text{conv}(S)$, then one can partition S into sets $S_1, \ldots, S_r$ with $\bigcap_i \text{cone}(S_i) \neq \{0\}$.*

We emphasize that our set S lies in $\mathbb{R}^{d+1}$, not $\mathbb{R}^d$ like before.

Proof of Tverberg Assuming Tverberg for Cones. Given $S \subseteq \mathbb{R}^d$ as in the hypothesis, we define a new point set $S' \subseteq \mathbb{R}^{d+1}$ by replacing each $v = (v_1, \ldots, v_d) \in S$ with the point $v' = (v_1, \ldots, v_d, 1) \in \mathbb{R}^{d+1}$. We have $0 \notin \text{conv}(S')$ so Tverberg's Theorem for cones guarantees a partition $S_1', \ldots, S_r'$ with some $x \neq 0$ in $\bigcap_i \text{cone}(S_i')$. This implies that each $\text{cone}(S_i')$ contains the entire ray from 0 to x, and in particular it contains the point y on this ray with $y_{d+1} = 1$. One can check that $y \in \bigcap_i \text{conv}(S_i')$ which in turn implies that $(y_1, \ldots, y_d) \in \bigcap \text{conv}(S_i)$ where S_i is defined by projecting S_i' to $\mathbb{R}^d$. $\square$

It remains to prove the result for cones.

Proof of Tverberg's Theorem for Cones. The high level idea of the proof is as follows. We will define a map Φ which takes points v in $\mathbb{R}^d$ and maps them to r-subsets $\{v^{(1)}, \ldots, v^{(r)}\}$ of $\mathbb{R}^N$ for some appropriate choice of N. In this way, we can think of picking an r-partition of $S = \{v_1, \ldots, v_n\}$ as picking for each i some copy $v_i^{(j)}$, and we will design our map Φ so that an r-partition with a point in the intersection of the convex hulls corresponds to our choice of points $v_i^{(j)}$ having some appropriate properties. Moreover, we will view each set $V_i = \{v_i^{(1)}, \ldots, v_i^{(r)}\}$ as sets of points being given color i, allowing us to make use of the colorful Carathéodory Theorem.

So with the above in mind, what sort of conditions do we want at the end? Colorful says that any point in the common hull of each thing can be achieved by a rainbow thing. So okay we have this rainbow thing and point in the convex hull, how does that translate to a point in the convex hull of the base things? Or rather, in the cone of the base things. □

7.3 Applications

eg from your methods notes: Radon for balanced Sperner stuff, colorful Carathedory for Drisko

8 The (p, q)-Theorem

Include subsection on fractional and geometric transversals.

9 Geometric Constructions and Extremal Combinatorics

Do Borsuk's construction and maybe also Ramsey-Turan for K_4 showing the back and forth between extremal and geometry giving back to each other.

Part III

Topological Methods

10 The Borsuk Ulam Theorem

10.1 Colorings

Given that this isn't led directly into by extremal set theory I maybe should include some lead in like "we begin our study of more involved topological techniques by looking at a problem from extremal set theory which motivated much of this field."

Define the Kneser graph $K_{n;r}$ to be the graph with vertex set $\binom{[n]}{r}$ where two $F, F' \in \binom{[n]}{r}$ are adjacent in $K_{n;r}$ if and only if $F \cap F' = \emptyset$. In this language, intersecting families $\mathcal{F} \subseteq \binom{[n]}{r}$ are exactly independent sets of $K_{n;r}$, and in particular the Erdős-Ko-Rado Theorem can be phrased as saying the independence number $\alpha(K_{n;r})$ equals $\binom{n-1}{r-1}$ for $n \geq 2r$. A closely related parameter to the independence number of a graph is the chromatic number, and as such we can ask: what is $\chi(K_{n;r})$? In other words, what is the smallest size of a partition of $\binom{[n]}{r}$ into intersecting families?

To get started, we recall that for any graph G we have $\chi(G) \geq \frac{v(G)}{\alpha(G)}$, and hence for $n \geq 2r$ we have

$$\chi(K_{n;r}) \geq \frac{\binom{n}{r}}{\binom{n-1}{r-1}} = \frac{n}{r}.$$

What about upper bounds for the chromatic number, that is, proper colorings of $K_{n;r}$. A very simple idea is to color each set $F \in \binom{[n]}{r}$ by its smallest element. In this way, the set $\mathcal{F}_i$ of elements colored i all contain i and hence are intersecting/independent in $K_{n;r}$, proving that $\chi(K_{n;r}) \leq n$. Actually, our analysis of this coloring is suboptimal since the minimal element of each set in $\binom{[n]}{r}$ is always at least $n-r+1$ (with the extremal case being $F' = \{n-r+1, n-r+2, \ldots, n\}$), so really this coloring shows $\chi(K_{n;r}) \leq n-r+1$ and this is in fact optimal for $r = 1$. For $r \geq 2$ though, one might realize that $F' = \{n-r+1, \ldots, n\}$ is the only set being given color $n-r+1$ which seems rather inefficient, and actually if we color F' with color $n-r$ instead then this set will intersect every other set with color $n-r$ since such sets must contain $r-1 \geq 1$ elements greater than $n-r$ and hence intersects F', so this lets us save 1 additional color. Pondering this even further, we realize that any two sets F, F' with $\min(F), \min(F') \geq n-2r+2$ must intersect by the pigeonhole principle. In total then, if we color each $F \in \binom{[n]}{r}$ by color $\min(F)$ if $\min(F) \leq n-2r+1$ and otherwise giving it color $n-2r+2$, then this is a proper coloring of $K_{n;r}$ (i.e. any two sets given the same color intersect), and hence

$$\chi(K_{n;r}) \leq n-2r+2.$$

We now have two rather far apart bounds for $\chi(K_{n;r})$, both of which are derived from somewhat natural considerations. However, some thought leads one to believe that the lower bound is probably incorrect. For one, we know that for $n > 2r$ the only independent sets of $K_{n;r}$ of size exactly $\binom{n-1}{r-1}$ are stars, and as such the bound $n/r = v(K_{n;r})/\alpha(K_{n;r})$ could only be tight if we could find exactly n/r stars covering all of $\binom{[n]}{r}$ which one can quickly check is impossible. Another way to see this is by looking at small examples. For instance, $K_{5;2}$ is exactly the Petersen graph which has chromatic number $3 = 5 - 2 \cdot 2 + 2 > 5/2$, and this suggests that the upper bound should perhaps be the truth. And indeed, in groundbreaking work Lovász showed this to be the case.

Theorem 10.1 (Lovász-Kneser Theorem). *If $n \geq 2r-1$ then $\chi(K_{n;r}) = n - 2r + 2$.*

The result itself is quite nice because it gives a family of graphs whose chromatic numbers are arbitrarily large compared to their fractional chromatic number (of which very few such families are known) Maybe expand on this Also state the combinatorial interpretation of this statement.

More importantly than the result itself is how it is proven. Notably, this is one of the first ever combinatorial results proven using topological methods, and since then many more such results of this form have been found. In fact, every known proof of Theorem 10.1 uses ideas inspired by topology, specifically by some version of the Borsuk-Ulam Theorem. To state this fundamental result, we let S^d be the d-dimensional sphere, i.e. the topological space with point set $\{\vec{x} \in \mathbb{R}^{d+1} : \sum \vec{x}_i^2 = 1\}$ which is given the subspace topology from $\mathbb{R}^{d+1}$. Thus for example S^1 is the unit circle in the plane, and S^0 is just the two points ± 1 in $\mathbb{R}$.

Theorem 10.2 (Borsuk-Ulam). *The following statements are all equivalent and true for all $d \geq 0$:*

(i) For every continuous map $f : S^d \to \mathbb{R}^d$ there exists a point $\vec{x} \in S^d$ with $f(\vec{x}) = f(-\vec{x})$.

- *Probably insert more, will see what I end up needing; also can insert things as exercises.*
- *If $C_1, \ldots, C_{d+1}$ are closed subsets of S^d with $S^d = \bigcup C_i$, then there is some point $\vec{x}$ with $\vec{x}, -\vec{x} \in C_i$ for some i.*
- *If $U_1, \ldots, U_{d+1}$ are open subsets of S^d with $S^d = \bigcup U_i$, then there is some point $\vec{x}$ with $\vec{x}, -\vec{x} \in U_i$ for some i.*

Comment on what ends up staying: first one is classical, last few are some genralizeation by someone.

We will postpone talking about Lovász's original proof and instead give a much simpler proof found shortly after by Bárány. The idea is spiritually similar to the Katona circle argument: instead of putting the elements $[n]$ along a circle and then considering arcs which contain r-sets, we instead put the elements $[n]$ on a sphere S^d and then consider open hemispheres which contain r-sets. For the circle there is a natural way to place n points, namely equidistant along the circle. For larger d there is an analog of this "equidistant" construction.

Lemma 10.3 (Gale's Lemma). *For every $d \geq 0$ and $r \geq 1$ there exists points $\vec{x}_1, \ldots, \vec{x}_{2r+d} \in S^d$ such that every open hemisphere of S^d contains at least r of the $\vec{x}_j$ points.*

We omit the (not too hard and non-topological) proof of this geometric fact and show how to use this to prove Lovász-Kneser.

Proof I of Lovász-Kneser Theorem. The case $n = 2r-1$ is trivial, so we assume $n \geq 2r$ from now on. As mentioned above, we want to consider putting n points on some sphere S^d where it turns out we will take $d := n - 2r \geq 0$, and to this end we let $\vec{x}_1, \ldots, \vec{x}_n$ be the points guaranteed by Gale's lemma. For any point $\vec{x} \in S^d$, we define the open hemisphere $H(\vec{x})$ to be the open hemisphere of S^d centered at $\vec{x}$.

Assume for contradiction that there exists a proper coloring χ of $K_{n;r}$ which uses at most $n - 2r + 1 = d + 1$ colors. For each $1 \le i \le d + 1$, we define a set $A_i \subseteq S^d$ by having$\vec{x} \in A_i$ whenever there exist $\vec{x}_{j_1}, \ldots, \vec{x}_{j_r} \in H(\vec{x})$ with $\chi(\{j_1, \ldots, j_r\}) = i$; that is, A_i records which hemispheres contain a set which is in the ith color class. We also define $A_{d+1} = S^d \setminus \bigcup_{i=1}^d A_i$.

It is not difficult to see that each A_i set is open, since if $\vec{x}_j$ is in the open hemisphere $H(\vec{x})$ then $\vec{x}_j \in H(\vec{x}')$ for any $\vec{x}'$ which is sufficiently close to $\vec{x}$. Moreover, by construction of the $\vec{x}_j$ points, every open hemisphere $H(\vec{x})$ must contain some r points and hence each $\vec{x} \in S^d$ belongs to some A_i (namely the i which is how this r-set of points is colored). Thus $A_1, \ldots, A_{d+1}$ are open sets which cover S^d, so it follows from Borsuk-Ulam that there exists some $\vec{x}, -\vec{x}$ lying in some common A_i. By definition this means there is some $\{\vec{x}_{j_1}, \ldots, \vec{x}_{j_r}\} \in H(\vec{x})$ and some $\{\vec{x}_{j_1'}, \ldots, \vec{x}_{j_r'}\} \in H(-\vec{x})$ given the same color i, but the fact that these are antipodal hemispheres means $\{j_1, \ldots, j_r\} \cap \{j_1', \ldots, j_r'\} = \emptyset$, a contradiction to these two sets being given the same color. □

In fact, one can generalize this result further to prove that $\chi(G) \ge n - 2r + 2$ for some induced graph G of $K_{n;r}$. For example, the Petersen graph $K_{5;2}$ has chromatic number 3, but to prove $\chi(K_{5;2}) \ge 3$ it suffices to only look at a 5-cycle $C_5 \subseteq K_{5;2}$, say the one with vertex set $\{1, 3\}, \{2, 4\}, \{3, 5\}, \{1, 4\}, \{2, 5\}$. Motivated by this example, we say that a set $F \in \binom{[n]}{r}$ is 1-stable if no two elements of F are within cyclic distance 1 of each other (so if $i \in F$ then $i + 1, i - 1 \mod n \notin F$), and we define the Schrijver graph $S_{n;r}$ to be the graph whose vertex set consists of all 1-stable sets of $\binom{[n]}{r}$ where two sets are adjacent if they are disjoint. That is, $S_{n;r}$ is some induced subgraph of $K_{n;r}$. As such, the following is an even stronger version of the Lovász-Kneser Theorem.

Theorem 10.4 (Schrijver's Theorem)**.** *For all $n \ge 2r$ we have $\chi(S_{n;r}) = \chi(K_{n;r}) = n - 2r + 2$.*

Indeed, this result follows immediately from our previous proof together with the following slightly stronger version of Gale's lemma proven by Schrijver.

Lemma 10.5. *For every $d \ge 0$ and $r \ge 1$ there exists points $\vec{x}_1, \ldots, \vec{x}_{2r+d} \in S^d$ such that every open hemisphere of S^d contains at least r points $x_{j_1}, \ldots, x_{j_r}$ such that $\{j_1, \ldots, j_r\}$ is 1-stable.*

There are two natural follow ups to Schrijver's Theorem. The first is if one can prove $\chi(G) = \chi(K_{n;r})$ for an even smaller subgraph G, but it turns out this is impossible: $S_{n;r}$ is vertex-critical, meaning deleting any vertex decreases its chromatic number. Another direction is to consider s-stable sets, that is sets where every two elements are at cyclic distance at least s from each other. It is conjectured that the corresponding graph $K_{n;r,s}$ has chromatic number $n - (s + 1)(r - 1)$ for $s \ge 1$ and n sufficiently large, but this is only known in certain cases.

We proved Schrijver's Theorem by taking Bárány's proof and using a more sophisticated choice of point set. Another angle one could try and do is to take Bárány's proof and use a more sophisticated version of the Borsuk-Ulam Theorem. This approach was used by Greene to give what is generally considered to be the simplest known proof of the Lovász-Kneser Theorem which does not require the usage of Gale's lemma.

Proof II of Lovász-Lneser Theorem. Similar to before we put points on a sphere S^d though now we take $d = n - 2r + 1$ (which is one larger than our previous proof). We next let $\vec{x}_1, \ldots, \vec{x}_n$ be any set of points of S^d which are in general position, meaning that no $d+1$ of them lie on a common hyperplane through the origin. Note that a random choice of $\vec{x}_i$ will be in general position with probability 1, so it is very easy to find such points as compared to needing to use Gale's lemma.

Again we assume for contradiction that there exists some proper coloring of $K_{n;r}$ using $n - 2r + 1 = d$ colors and let A_i with $1 \le i \le d$ be as before defined to contain all $\vec{x}$ such that $H(\vec{x})$ contains points corresponding to a set given color i. Again these sets will be open but they may not cover all of S^d. To this end we introduce a new set $A_{d+1} = S^d \setminus \bigcup_{i=1}^d A_i$ which is closed since it is the complement of a union of open sets. We thus have that $A_1, \ldots, A_{d+1}$ are a bunch of open and closed sets whose union equals all of S^d. A variant of the Borsuk-Ulam Theorem proven by Greene? says that in this case there again exists some antipodal pair $\vec{x}, -\vec{x}$ contained in some A_i. Again having $1 \le i \le d$ would give a contradiction, so this antipodal pair must be in A_{d+1}. By definition of A_{d+1} this means $H(\vec{x}), H(-\vec{x})$ each contain at most $r-1$ of the $\vec{x}_j$ points (else they would contain some r-set being given some color). It follows then that at least $n - 2r + 2$ of the $\vec{x}_j$ points must lie on the equator $S^d - H(\vec{x}) - H(-\vec{x})$. But this means there is a hyperplane through the center of the sphere containing $n - 2r + 2 \ge d + 1$ points, a contradiction to how we chose our point set. $\square$

Can also talk about Dolnikov if desired/relevant to later discussion.

10.2 Mass Partitions

The Borsuk-Ulam Theorem can be used to prove other results beyond coloring statements. Probably start with classical ham sandwich then do necklace splitting.

11 The KKM Theorem

An important consequence of the Borsuk-Ulam Theorem is Brouwer's Fixed Point Theorem, which in turn is equivalent to something known as the KKM Theorem which has been used to solve a number of problems in combinatorics and discrete geometry. We motivate its statement in the following subsection, after which we look at some further applications.

Refer to Shira's survey/notes.

Maybe the right name is standard n-simplex.

11.1 Envy-Free Divisions

We begin by discussing a topic spiritually similar to mass partitions, namely that of envy-free divisions. As a toy version of this problem, say that you have a piece of cake which you want to cut in half and give to two kids Alice and Bob. If you're sloppy with your cutting then one slice will be larger than the other, making whichever kid does not receive that slice envious of the other. Even if you do manage to slice it perfectly in half, maybe one half has more icing or sprinkles than the other which again could make one of the kids envious. How then can you cut this cake while keeping both kids happy?

The practical solution to this problem is the "I cut you choose" method: you ask Alice to cut the cake so that she prefers both slices equally (which can be done using the intermediate value theorem assuming her preferences are "continuous"), and then Bob takes whichever of the two slices he prefers. By construction neither child will be envious of the other.

This cake-cutting problem with more children is complicated and requires non-trivial topological statements to resolve; this is perhaps not too surprising given that we seemingly needed the non-trivial Intermediate Value Theorem to solve. To discuss this further we need some formal definitions.

For simplicity we think of cutting a rectangular cake into n vertical slices, and in this way we can identify such a cutting by a sequence of numbers $(x_1, \ldots, x_n)$ with $\sum x_i = 1$ where we think of this tuple representing that the first slice uses the first x_1 proportion of the cake, the next uses the next x_2 proportion of the cake, and so on. We will denote[2] the set of all such tuples by Δ_{n-1} which is also called the standard $(n-1)$-simplex. Each child in our setup will have some (possibly distinct) preference functions $P : \Delta_{n-1} \to 2^{[n]}$ where we think of $P(\vec{x})$ for a given cut $\vec{x} = (x_1, \ldots, x_n) \in \Delta_{n-1}$ as defining the set of indices $i \in P(\vec{x})$ such that the child likes the ith cut the most. For example, one might define $P(\vec{x}) = \{i : x_i = \max_j x_j\}$ which means the child always prefers the largest possible slice of cake regardless of where its located Actually maybe I should think of the second slice as being the interval $[x_1, x_2]$; will have to figure this out. We will further require our preference functions to satisfy two additional properties, namely that $P(\vec{x})$ always contains some i with $x_i > 0$ for all $\vec{x}$ (i.e. every child prefers some cake over no cake), and that if $i \in P(\vec{x}_j)$ for some sequence of partitions $\vec{x}_1, \ldots$ which converges to some $\vec{x} \in \Delta_{n-1}$, then $i \in P(\vec{x})$.

[2]The strange indexing here comes from the fact that Δ_{n-1} is an $(n-1)$-dimensional object, e.g. Δ^2 defines a triangle and Δ^3 defines a tetrahedron.

Theorem 11.1 (Fair Division Theorem)**.** *If $P_1, \ldots, P_n$ are preference functions as defined above, then there exists some $\vec{x} \in \Delta_{n-1}$ and permutation $\pi : [n] \to [n]$ such that $\pi(j) \in P_j$ for all j.*

That is, regardless of what preferences the children have, there always exists some way of cutting the cake into n pieces such that each child can be given some piece which they think is best amongst all possible options. The Fair Division Theorem turns out to be exactly equivalent to a certain variant of Brouwer's Fixed Point Theorem which requires some additional definitions.

For each $I \subseteq [n]$, we define the face

$$\Delta_I = \{\vec{x} \in \Delta_{n-1} : x_i = 0 \ \forall i \notin I\}.$$

For example, Δ_2 (which is a triangle) has itself as a face, its three edges as faces, its three vertices as faces, and the empty set as a face. We say that a collection of subsets $A_1, \ldots, A_n$ of Δ_{n-1} is a *KKM cover* if the A_i sets are either all open or all closed, and if $\Delta_I \subseteq \bigcup_{i \in I} A_i$ for all faces Δ_I. In particular, $\Delta_{n-1} = \Delta_{[n]}$ This is really bad notation is contained in the union of the A_i, and also each A_i contains the vector $\vec{x}$ with a 1 in position i and 0's everywhere else. This notion of a cover was introduced by Knaster, Kuratowski, and Mazurkiewicz who proved the following. Draw a picture of a KKM cover for the triangle and note the point of intersection.

Theorem 11.2 (KKM Theorem)**.** *If $A_1, \ldots, A_n$ is a KKM cover of Δ_{n-1}, then $\bigcap A_i \neq \emptyset$.*

This theorem turns out to be equivalent to Brouwer's Fixed Point Theorem, though we omit discussing this further or maybe not idk. There are many different variantas of the KKM Theorem, and in particular we will want the following variant due to Gale.

Theorem 11.3 (Colorful KKM Theorem)**.** *Let $\mathcal{A}^1, \ldots, \mathcal{A}^n$ be collections of sets such that each $\mathcal{A}^j = \{A_1^j, \ldots, A_n^j\}$ is a KKM cover of Δ_{n-1}. Then there exists some permutation $\pi : [n] \to [n]$ such that $\bigcap_j A^j_{\pi(j)} \neq \emptyset$.*

Maybe change π to the bottom depending on the equivalence. We call this "colorful" because we can think of each set in $\mathcal{A}^j$ as being given some color j, with the conclusion being that we can find a set of each color which has a common point of intersection. This in turn implies the Fair Division Theorem.

Proof of Fair Division Theorem. If the jth child has preference function P_j, then we define the set $A_i^j \subseteq \Delta_{n-1}$ to be the set of $\vec{x}$ with $i \in P_j(\vec{x})$ (that is, these are the cuts of the cake such that child j is happy to have the ith slice). The sequence requirement for our preference functions implies each A_i^j set is closed, and the fact that $P_j(\vec{x})$ always contains some i with positive weight implies $\Delta_I \subseteq \bigcup_{i \in I} A_i^j$ for all I, j. It follows that each $A_1^j, \ldots, A_n^j$ is a KKM cover, and as such the Colorful KKM Theorem guarantees a permutation π and a point $\vec{x} \in \bigcap A^j_{\pi(j)}$. By definition this implies that $\pi(j) \in P_j(\vec{x})$ for all j, giving the result. $\square$

In fact this same sort of proof shows that the Fair Division Theorem is equivalent to the Colorful KKM Theorem in the case when all the sets are closed which itself can be used to prove the case when all the sets are open.

11.2 Piercing Geometric Sets

We next use the KKM Theorem to prove results about piercing geometric sets. Our approach follows a general framework detailed by McGinnis and Zerbib in their survey on using the KKM Theorem for showing the existence of some "nice object":

(i) Translate each possible "object" into a point in the simplex.

(ii) Show that if there exists no "nice" object, then one can define a KKM cover for the simplex.

(iii) Show that the point $\vec{x}$ guaranteed by the KKM Theorem gives a contradiction.

For example, in the Envy-Free Division Theorem our objects were cuts of the cake, with the nice objects of envy-free cuts being essentially equivalent to the existence of a KKM cover. Our next set of applications concerns piercing geometric objects, and for this we recall that given a set system $\mathcal{F}$, the matching number $\nu(\mathcal{F})$ is the largest size of a collection of pairwise disjoint sets in $\mathcal{F}$ and the transversal number $\tau(\mathcal{F})$ is the smallest size of a transversal, i.e. a set which intersects each element of $\mathcal{F}$.

Theorem 11.4 (Gallai's Theorem)**.** *If $\mathcal{F}$ is a finite set of compact intervals in $\mathbb{R}$, then $\nu(\mathcal{F}) = \tau(\mathcal{F})$.*

This result can be proven by elementary means, and in particular it follows from the more general Helly Theorem for trees stated somewhere; we give a KKM based proof of it to better understand the technique. In particular, we encourage the reader to try and think through each step of the KKM strategy outlined above before actually reading the details of how each step is done. Our exact proof will be somewhat verbose with us posing some motivating ideas for each step before summarizing the key facts that we need for the formal argument.

Proof. That $\nu(\mathcal{F}) \le \tau(\mathcal{F})$ is trivial, so it remains to show that $n := \nu(\mathcal{F}) \ge \tau(\mathcal{F})$, i.e. that there exists a set of at most n points which pierces all of $\mathcal{F}$. To use the KKM Theorem, we want to find a way of identifying potential transversals for $\mathcal{F}$ of size n as points $(x_1, \ldots, x_k)$ in the simplex for some integer k. With some thought, we realize that each point $\vec{x} \in \Delta_{k-1}$ can be identified as a set of k points in $[0,1]$, namely the points $x_1,\ x_1 + x_2,\ x_1 + x_2 + x_3$, and so on (this is also implicitily what we did in the Envy-Free division problem with these points corresponding to where the cake was cut), where we note that only the first $k-1$ of these points are interesting since $\sum x_i = 1$. Of course, for an arbitrary $\mathcal{F}$ such points in $[0,1]$ may be very far from a transversal. Crucially though, we observe that by rescaling and shifting the elements of $\mathcal{F}$ we can assume they all lie in $[0,1]$.

In summary: we will assume from now on that the elements of $\mathcal{F}$ all lie in $[0,1]$ (which we can do by appropriate scaling and shifting), and for each $\vec{x} = (x_1, \ldots, x_{n+1}) \in \Delta_n$ we define $T(\vec{x})$ to be the n points in $[0,1]$ of the form $\{\sum_{j=1}^{i} x_j : 1 \le i \le n\}$. We will also towards the next step of the KKM framework assume that there exists no nice object, i.e. that $T(\vec{x})$ is not a transversal of $\mathcal{F}$ for all $\vec{x}$.

Our goal now is to use the fact that $T(\vec{x})$ is not a transversal to construct a KKM cover, i.e. to assign $\vec{x}$ to at least one set A_i. That $T(\vec{x})$ is not a transversal means there exists some $F \in \mathcal{F}$ which $T(\vec{x})$ is disjoint from. Perhaps the most naive way to try using this information is to say something like "if $T(\vec{x})$ is disjoint from F_i then $\vec{x} \in A_i$", but this does not work since $\mathcal{F}$ may have far more than n elements. That is, we really need to encode $T(\vec{x})$ not being a transversal by some integer $1 \le i \le n+1$. Thinking further, we realize that if $T(\vec{x})$ is not a transveral, then there exists some $1 \le i \le n+1$ such that there exists some $F \in \mathcal{F}$ lying in $\left(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j\right)$.

Summarizing, we will define for each $1 \le i \le n+1$ the set A_i to consist of all $\vec{x} \in \Delta_n$ such that there exists some $F \in \mathcal{F}$ lying in $\left(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j\right)$. This definition together with our assumption $T(\vec{x})$ is not a transversal for all $\vec{x}$ implies

$$\Delta_n \subseteq \bigcup_i A_i,$$

which is the most basic property we need to apply the KKM Theorem. It remains to check the other properties.

That each A_i is open is relatively straightforward since e.g. if $F \subseteq \left(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j\right)$ then a slight perturbation of $\vec{x}$ continues to have F lying in this interval since F is closed. For the last property, we aim to show that for any $I \subseteq [n+1]$ and $\vec{x} \in \Delta_I$ that $\vec{x} \in \bigcup_{i \in I} A_i$. Because we already know $\vec{x} \in \Delta_n \subseteq \bigcup_i A_i$, it suffices to show that $\vec{x} \notin \bigcup_{i \notin I} A_i$. And indeed, for each $i \notin I$ we by definition of $\vec{x} \in \Delta_I$ have $x_i = 0$ and hence the interval $\left(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j\right)$ is empty. It follows that no F lies in this interval, and hence $\vec{x} \notin A_i$ as desired.

Putting all this together implies that the A_i form a KKM cover, so by the KKM Theorem there exists some $\vec{x} \in \bigcap_i A_i$. This means that for each $1 \le i \le n+1$, there exists some $F_i \in \mathcal{F}$ with $F_i \subseteq \left(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j\right)$. These F_i are thus all disjoint from each other, which implies $\nu(\mathcal{F}) \ge n+1 = \nu(\mathcal{F}) + 1$, a contradiction. □

The fact that our proof used the KKM Theorem means that the same exact argument can be used to give a colorful version of Gallai's Theorem simply by replacing the KKM Theorem with the colorful KKM Theorem.

Theorem 11.5 (Colorful Gallai Theorem)**.** *If $\mathcal{F}_1, \ldots, \mathcal{F}_{n+1}$ are finite sets of compact intervals of $\mathbb{R}$ such that $\tau(\mathcal{F}_i) \ge n+1$ for all i, then there exists a rainbow matching, i.e. $F_i \in \mathcal{F}_i$ for each i such that $F_i \cap F_j = \emptyset$ for all $i \ne j$.*

The two examples of the KKM Theorem we have seen so far have identified points of Δ_n with a set of points on $[0, 1]$, but there are other choices one can make. To state the coming result, for a collection of sets $\mathcal{F} \subseteq 2\mathbb{R}^2$ we say that a set of lines $\mathcal{L} \subseteq \mathbb{R}^2$ is a line transversal for $\mathcal{F}$ if for all $F \in \mathcal{F}$ there exists some $L \in \mathcal{L}$ with $F \cap L \ne \emptyset$, and we define $\tau_\ell(\mathcal{F})$ to be the smallest size of a line transversal for $\mathcal{F}$. We also recall that a set $C \subseteq \mathbb{R}^2$ is convex if C contains the line between any two points $p, q \in C$.

Theorem 11.6. *If $\mathcal{F}$ is a collection of compact convex sets in $\mathbb{R}^2$ such that every $\mathcal{F}' \subseteq \mathcal{F}$ of size at most 4 has $\tau_\ell(\mathcal{F}') \le 1$, then $\tau_\ell(\mathcal{F}) \le 2$.*

This is probably best possible, need to think about it.

Proof. The idea for the proof is that rather than identify points of the simplex by point sets on the unit interval, we instead identify them by point sets on the unit circle. To this end, we may assume by scaling that every set in $\mathcal{F}$ lies within the unit circle, which we recall can be parameterized by the set of points $(\cos(2\pi t), 2\pi \sin(2\pi t))$. For each $(x_1, x_2, x_3, x_4) \in \Delta_3$, consider the four points on the circle $y_i = (\cos(2\pi \sum_{j=1}^{i} x_j), \sin(2\pi \sum_{j=1}^{i} x_j))$ and let ℓ_1 be the line containing y_1, y_3 and ℓ_2 the line containing $y_2, y_4 = (1, 0)$. These two lines split the unit circle into four regions, and we let $R_i(\vec{x})$ denote the region which contains the arc from y_{i-1} to y_i. Insert picture.

Assume for contradiction that $\tau_\ell(\mathcal{F}) > 2$, which means that for any $\vec{x} \in \Delta_3$ the two lines ℓ_1, ℓ_2 fail to pierce some $F \in \mathcal{F}$. Because F is convex, this is only possible if $F \subseteq R_i(\vec{x})$ for some i. Motivated by this, we define $A_i \subseteq \Delta_3$ to consist of the points $\vec{x}$ such that there exists $F \in \mathcal{F}$ which lies in $R_i(\vec{x})$.

It is not difficult to see that A_i is open for all i since perturbing $\vec{x}$ slightly maintains the property of the compact sets F lying in $R_i(\vec{x})$. By assumption of $\tau_\ell(\mathcal{F}) > 2$ we have $\Delta_3 \subseteq \bigcup_i A_i$, so to prove this is a KKM cover it suffices to show for each $I \subseteq [4]$ and $\vec{x} \in \Delta_I$ that $\vec{x} \notin \bigcup_{i \notin I} A_i$. And indeed, if $i \notin I$ then $x_i = 0$ which implies that $y_{i-1} = y_i$ and hence $R_i(\vec{x}) = \emptyset$, so no element of $\mathcal{F}$ lies in this region and hence $\vec{x} \notin A_i$ as desired.

Given that the A_i form a KKM cover, it follows from the KKM Theorem that there exists some $\vec{x} \in \bigcap_i A_i$. This in turn implies that there exist $F_i \in \mathcal{F}$ with $F_i \subseteq R_i(\vec{x})$ for all i. It is not difficult to see that $\tau_\ell(\{F_1, \ldots, F_4\}) > 1$ since no line can pierce all four regions $R_i(\vec{x})$, a contradiction to our assumption on $\mathcal{F}$. We conclude that there does indeed exist a line transversal of $\mathcal{F}$ of size at most 2. $\square$

Again, an identical argument can be used to state a colorful version of this theorem, and a slightly more delicate version of the argument can be used to further guarantee that the line transversal of size 2 for $\mathcal{F}$ contains a vertical line; we omit the details of this and refer the reader to Survey. I might also consider doing the mass partition result with eg $k = 3$ to further demonstrate the ideas.

11.3 KKM Variants

A variant of Gallai's problem of piercing intervals in $\mathbb{R}$ is to replace intervals with d-intervals, which are sets $S \subseteq \mathbb{R}$ which are the union of at most d compact intervals. It is not difficult to show that collections of d-intervals with $d > 1$ can have $\nu(\mathcal{F}) < \tau(\mathcal{F})$, but a nice result obtained independently by Tardos and Kaiser shows that these two quantities can never be too far apart from each other.

Theorem 11.7. *If $\mathcal{F}$ is a family of d-intervals, then $\tau(\mathcal{F}) \leq (d^2 - d + 1)\nu(\mathcal{F})$.*

This bound is close to optimal, as Matousek constructed families with $\tau(\mathcal{F})/\nu(\mathcal{F}) = \Omega(d^2/\log d)$.

Naively, to prove this result we might again try to translate $\vec{x} \in \Delta_k$ into the set of points $T(\vec{x})$ of the form $\sum_{j=1}^{i} x_j$. However, now if some set $F \in \mathcal{F}$ is disjoint from $T(\vec{x})$ we can not say that this means F lies entirely within some interval $(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j)$ like before since F being a d-interval means it could lie within up to d such intervals. As such, these disjoint F no longer encode naturally by integers $1 \le i \le k+1$ like before, but rather by sets $I \subseteq [k+1]$ indicating which union of intervals contains F. As such, we will not want to consider a cover by sets of the form A_i but rather A_I.

Motivated by the above, we say that a collection of sets A_I with $I \subseteq [n+1]$ is a KKMS cover of Δ_n if the A_I sets are either all open or all closed, and if for every $J \subseteq [n+1]$ we have $\Delta_J \subseteq \bigcup_{I \subseteq J} A_I$. For example, if A_i is a KKM cover then by defining $A_{\{i\}} = A_i$ and $A_I = \emptyset$ we recover a KKMS cover.

The example above shows that we can not hope to have a point in the common intersection of a KKMS cover like we did with KKM covers. The right condition turns out to be the following strengthening of the KKM Theorem due to Shapley who represents the S in KKMS. For this we recall that a set system $\mathcal{F}$ on a ground set V has a fractional perfect matching if it has a fractional matching $\mu : \mathcal{F} \to \mathbb{R}$ with $\sum_{F \ni v} \mu(F) = 1$ for all $v \in V$.

Theorem 11.8 (KKMS Theorem)**.** *If $\{A_I : I \subseteq [n+1]\}$ is a KKMS cover of Δ_n, then there exist sets $I_1, \ldots, I_{n+1} \subseteq [n+1]$ such that (i) the set system $\{I_1, \ldots, I_{n+1}\}$ with ground set $[n+1]$ has a fractional perfect matching and (ii) we have $\bigcap_{j=1}^{n+1} A_{I_j} \neq \emptyset$.*

Because of how the KKMS Theorem is stated, its applications often require some additional results around fractional matchings, such as the following.

Theorem 11.9 (Füredi)**.** *If $\mathcal{F}$ is a set system where every edge has size at most d, then $\nu(\mathcal{F}) \ge \frac{\nu^*(\mathcal{F})}{d-1+1/d}$.*

We note that it is easy to prove the weaker bound $\nu(\mathcal{F}) \ge \nu^*(\mathcal{F})/d$ which in particular is enough to give a slightly weaker version of the Tardos-Kaiser Theorem. We also need the following which may or may not be done in the transversal section.

Lemma 11.10. *Let $\mathcal{F}$ be a set system with ground set V. If every edge of $\mathcal{F}$ has size at most d and if $\mathcal{F}$ has a fractional perfect matching and every edge has at most, then $\nu^*(\mathcal{F}) \ge |V|/r$.*

Combining these results allows us to turn our proof of Gallai's Theorem into a proof of the Tardos-Kaiser Theorem.

Proof of ??? Let $\mathcal{F}$ be a collection of d-intervals and let n be such that $\tau(\mathcal{F}) = n+1$. We aim to use this to construct a large matching of $\mathcal{F}$. To this end, we may assume by rescaling that each element of $\mathcal{F}$ lies in $[0,1]$, and for each $\vec{x} \in \Delta_n$ we let $T(\vec{x}) = \{\sum_{j=1}^{i} x_j : 1 \le i \le n\}$. Because $\tau(\mathcal{F}) > n$, each $T(\vec{x})$ set is not a transversal, so for each $\vec{x}$ there exists some $F \in \mathcal{F}$ disjoint from it. Motivated by this, for each $I \subseteq [n+1]$ we define A_I to be the set of $\vec{x}$ such that there exists an $F \in \mathcal{F}$ with $F \subseteq \bigcup_{i \in I} (\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j)$ and with $F \cap \bigcup_{i \notin I} (\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j) = \emptyset$.

We aim to show that the A_I sets form a KKMS cover. The fact that each of these sets is open is straightforward. For the other property, we note that $\Delta_n \subseteq \bigcup_I A_I$ by assumption of no $T(\vec{x})$ being a transversal of $\mathcal{F}$. As such, to prove that for each $J \subseteq [n+1]$ we have $\Delta_J \subseteq \bigcup_{I \subseteq J} A_I$ it suffices to show that for each $\vec{x} \in \Delta_J$ we have $\vec{x} \notin \bigcup_{I \not\subseteq J} A_I$. And indeed, if I contains some

$i \notin J$, then the interval $(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j)$ is empty and hence there can not exist an $F \in \mathcal{F}$ with $F \cap \bigcup_{i \notin I}(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j) = \emptyset$, meaning $\vec{x} \notin A_I$ as desired.

The above implies that the A_I sets form a KKMS cover, so by the KKMS Theorem there exists some set system $\mathcal{I} = \{I_1, \ldots, I_{n+1}\} \subseteq 2[n+1]$ which has a fractional perfect matching as well as some $\vec{x} \in \bigcap_j A_{I_j} \neq \emptyset$. This latter condition in particular implies $A_{I_j} \neq \emptyset$ for all j, and we crucially observe that this is only possible if $|I_j| \leq d$ since each $F \in \mathcal{F}$ is the union of at most d intervals and hence can be contained in at most d intervals of the form $(\sum_{j=1}^{i-1} x_j, \sum_{j=1}^{i} x_j)$. By our lemmas around fractional matchings, we find

$$\nu(\mathcal{I}) \geq \frac{\nu^*(\mathcal{I})}{d - 1 + 1/d} \geq \frac{n+1}{d^2 - d + 1},$$

and as such we can find some matching $\mathcal{M} \subseteq \mathcal{I}$ of at least this size. For each $I \in \mathcal{M}$ there is some $F_I \in \mathcal{F}$ which witnesses $\vec{x} \in A_I$ and it is not difficult to see that $\mathcal{M}$ being a matching implies $F_I \cap F_{I'} = \emptyset$ for distinct $I, I' \in \mathcal{M}$. It follows that $\nu(\mathcal{F}) \geq |\mathcal{M}| \geq \frac{n+1}{d^2-d+1} = \frac{\tau(\mathcal{F})}{d^2-d+1}$, proving the result. $\square$

There exist many other extensions and variants of the KKM Theorem. For example, there exist colorful versions of the KKMS Theorem which immediately yields a colorful version of the Tardos-Kaiser Theorem by using the same proof approach we have above. Another far reaching extension is Komiya's Theorem, which roughly allows one to replace the simplex Δ_n in KKMS with any polytope, which can be useful for problems which are not naturally encoded by the simplex.

Maybe talk more about Brouwer and Sperner here.

11.4 Exercises

1. Prove the Colorful Gallai Theorem [1+].

2. Prove that if $\mathcal{F}$ is a set system where every edge has size at most d then $\nu(\mathcal{F}) \geq \nu^*(\mathcal{F})/d$ [2].

12 Simplicial Complexes

The previous chapter demonstrated how certain combinatorial problems could be translated to looking at points on the simplex, whereupon topological results like the KKM Theorem could be used to solve the problem. In this chapter we study a more general geometric object known as a simplicial complex. Many combinatorial properties can be phrased in terms of simplicial complexes, whereupon a number of powerful topological tools such as the Borsuk-Ulam Theorem and more can be used to solve the problem at hand.

12.1 Simplicial Complex Basics

An *abstract simplicial complex* is simply a set system $\mathcal{F}$ with the property that if $F \in \mathcal{F}$ then $F' \in \mathcal{F}$ for all $F' \subseteq F$; in the extremal set theory community such set systems are more commonly called downsets or hereditary. Abstract simplicial complexes have a natural correspondence with certain topological spaces known as simplicial complexes, which requires a few basic definitions to set up.

Recall that a set of points $S = \{v_1, \ldots, v_n\}$ in $\mathbb{R}^d$ are affiniely independent if there do not exist $\alpha_i \in \mathbb{R}$ not all 0 with $\sum \alpha_i v_i = 0$ and $\sum \alpha_i = 0$. A *simplex* σ is the convex hull of a set S of affinely independent points in $\mathbb{R}^d$; we say that S is the set of vertices of σ and we define the dimension of the simplex by $\dim \sigma := |S| - 1$.

For example, a simplex of dimension 0 is just a point, of dimension 1 is a line, of dimension 2 is a triangle (since three points are affiniely independent only if they do not lie on a common line), and of dimension 3 is a tetrahedron (since four points are affinely independent only if they do not lie on a common plane). We will be interested in collections of simplices which are glued together in nice ways.

A *face* of a simplex is the convex hull of a subset of its vertices. In particular, the simplex and the empty set is always a face, and faces are always simplicies. A collection Δ of simplicies is a *geometric simplicial complex* if (i) every face of each $\sigma \in \Delta$ is a simplex of Δ and if (ii) for any $\sigma_1, \sigma_2 \in \Delta$ we have that $\sigma_1 \cap \sigma_2$ is a face of both σ_1 and σ_2. The union of all simplicies in Δ is called the polyhedron of Δ and will be denoted by $\|\Delta\|$.

Condition (i) of geometric simplicial complexes should remind the reader of the definition of abstract simplicial complexes, and indeed these two notions are essentially equivalent. For one direction, if Δ is a geometric simplicial complex, then we can define a set system $\mathcal{F}$ whose ground set is the set of all vertices used in a simplicial complex of Δ and whose sets $F \in \mathcal{F}$ are the vertex sets of each simplex in Δ which by (i) is an abstract simplicial complex Do we need (ii) here? Surely this is needed to say anything nice topologically. On the other hand, if $\mathcal{F}$ is an abstract simplicial complex on $[n]$, then one can consider n points affiniely independent points in $\mathbb{R}^n$ and then take $\Delta = \{\mathrm{conv}(V_F) : F \in \mathcal{F}\}$ where $V_F = \{v_i : i \in F\}$ which one can check is a geometric simplicial complex which we refer to as a *realization* of $\mathcal{F}$ More precisely we should say that Δ is a realization of $\mathcal{F}$ if this abstract complex associated to Δ is isomorphic to $\mathcal{F}$, and it turns out that all such Δ are homeomorphic which I guess isn't well defined yet..

One of the key ideas of topological combinatorics is that combinatorial properties of an abstract simplicial complex can be translated into topological properties of its geometric realizations which can be studied by using powerful tools from algebraic topology.

An important topological property which often ends up coming from this sort of translation is connectedness. To this end, we say that a topological space T is k-connected if for all $-1 \leq \ell \leq k$, every continuous map $f : S^\ell \to T$ from the ℓ-dimensional sphere can be extended to a continuous map $f' : B^{\ell+1} \to T$ from the $(\ell+1)$-dimensional ball. In other words, T has no "ℓ-dimensional holes" for all $\ell \leq k$. Here $S^{-1} = \emptyset$ and B^0 is a single-point, so (-1)-connected means T is non-empty.

In the remaining parts of this chapter, we associate simplicial complexes Δ to graphs and then study how the combinatorial properties of graphs translate to topological properties of Δ. This translations allows us to use topological tools on Δ to deduce combinatorial properties of our graph.

Okay I need to figure out the next thing to write (which may or may not be the next thing in the notes).

- Naively the simplest thing to do (both for next step and narrative flow) would be to skim through Borsuk-Ulam again and do these things since it's currently how I've written the language. After this I can do/figure out independence complexes and in particular somewhere around here I can put Sperner's lemma which is maybe naturally motivated by some of the ideas noted below.

  Yeah it's doubly good to do Borsuk-Ulam type stuff given that I opened with it.

12.2 Independence Complexes

Independent set complex is pretty natural object to associate to it.

- Can prove hypergraph Hall via Aharoni-Haxell and then use this to prove r=3 of Ryser

- Gabriel's notes prove some independent transversal result; I would probably start with the motivation in chapter 8 and then use this to reverse engineer the stuff in chapter 7 that we need for the proof

So okay, Gabriel defines a simplicial sphere basically to be a simplicial complex with no boundary and then defines connectedness with respect to the existence of simplicial mappings to such spheres. Hall's Theorem for hypergraph kind of nicely motivates the notion of independent transversals which in turn motivates the idea of a multicolored simplex. If all the independent transversal stuff really needs Sperner then maybe it's best to do this towards the end after noting what we can do with Borsuk-Ulam type things (though of course we just say this is true without proof).

12.3 ??TBD

- Using neighborhood complex to give lower bound on chromatic number of generalized Mycleskian which is only konwn proof
- Box complexes/kneser hypergraphs (see Lovasz or Matousek)
- Proving colored Tverberge using (probably just the statement of) Sarkaria for p-folded stuff could be neat (with me briefly mentioning that this uses Z_p actions rather than Z_2 actions).
- Partitioning a graph into connected pieces (Lovasz notes)
  - The Lovasz notes version seems a little involved for a result I don't care that much about, especially given that Gyori evidently proved this with non-topological methods. At most then I think I would sketch out the proof emphasizing the idea of what this particular simplicial complex gives you.
- Evasive properties (Lovasz notes, maybe also Bjorner)
  - This requires us building a simplicial complex so it would need to come after that stuff. It also while not terribly difficult does require a little bit of annoying group theory and definitions to state, so at the very least if I include this I would be inclined to do it towards the end of the course. It does have the advantage of just requiring the notion of contractability and no higher version of connectivity

Part IV
Bonus Topics

13 Designs

TODO

14 Harris Kleitman Inequality

This will be covered in the probabilistic method class and hence not be written up.

15 Spreadness and Spread Approximations

TODO

16 Incidence Problems

TODO